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THE IKKISH 21 QUANT MASTERY SERIES

500 ARITHMETIC WORD PROBLEMS

Target: SBI Clerk • IBPS Clerk • RRB Clerk & PO Exams

VOLUME 2 • ADVANCED & APPLIED CHAPTERS (Q251 - Q500)

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THE IKKISH 21 ARITHMETIC MASTERY (VOLUME 2: Q251 - Q500)

Time & Work, Pipes & Cisterns, Speed Time Distance, Trains, Boats & Mixtures

250 Practice Questions
With Step-by-Step Smart Solutions

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CHAPTER 01: AGES, AVERAGES & PARTNERSHIP DYNAMICS

38 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Partnership Profit Ratio = (Investment 1 × Time 1) : (Investment 2 × Time 2)
- Average = Total Sum of Quantities / Total Number of Quantities
- Weighted Average = (n1 × A1 + n2 × A2) / (n1 + n2)
- If a new person joins and average increases, Weight of New Person = Old Average + (Increase × New Total Count).

Q.251 Ages

The present ages of Ankit and Gaurav are in the ratio 4:7 respectively. Six years from now, the ratio of Gaurav's age to Ankit's age will be 5:3. Find the present age of Ankit.

(A) 42 (B) 48
(C) 54 (D) 36
(E) 60

Q.252 Ages

The population of a town is 500000, out of which 220000 are males. If 56% of the total population is literate and 42% of the males are literate, then what percentage of females are literate?

(A) 62 (B) 65
(C) 64 (D) 67
(E) 69

Q.253 Ages

In an Aptitude test, Yash and Divya scored an average of 80 marks. In the Reasoning test, Yash scored 20% more than his Aptitude marks, while Divya scored 10% less than her Aptitude marks. The total marks scored by both of them in Reasoning were 8 more than their total marks in Aptitude. Find the difference between their Reasoning marks.

(A) 18 (B) 20
(C) 28 (D) 24
(E) 30

Q.254 Ages

The average of 11 numbers is 42. The average of the first 6 numbers is 44 and the average of the last 6 numbers is 45. If the 6th number is excluded, then find the average of the remaining numbers.

(A) 35 (B) 37
(C) 41 (D) 43
(E) 39

Q.255 Ages

In a conference, every person shakes hands with every other person exactly once. If the total number of handshakes is 276, then find the number of people present in the conference.

(A) 20 (B) 24
(C) 21 (D) 23
(E) 22

Q.256 Ages

School X has 700 boys and School Y has 800 girls. If 60% of the boys from School X and 35% of the girls from School Y participated in a quiz competition, and there were no other participants, find the percentage of boys among the total participants.

(A) 55% (B) 60%
(C) 50% (D) 65%
(E) 45%

Q.257 Ages

The average age of Nidhi and Kritika is 50 years. If Rhea replaces Nidhi, the average age becomes 48 years and if Rhea replaces Kritika, the average age becomes 52 years. If the average age of Priya and Sona is 1.5 times the average age of Nidhi, Kritika and Rhea, then find the average age of all the five people.

(A) 55 years (B) 65 years
(C) 60 years (D) 62 years
(E) 58 years

Q.258 Ages

In a recruitment test, 68% of the candidates qualified in Reasoning and 72% qualified in Quant, while 15% of the candidates failed in both sections. If 1100 candidates qualified in both sections, find the total number of candidates who appeared in the test.

(A) 1600 (B) 1800
(C) 2000 (D) 2200
(E) 2500

Q.259 Ages

The ratio of the present ages of A and B is 4 : 5. Three years ago, C's age was 2/3 of A's age at that time. If B is 13 years older than C at present, find the ratio of B's present age to C's age after 5 years.

(A) 12 : 7 (B) 10 : 9
(C) 14 : 13 (D) 15 : 11
(E) 11 : 8

Q.260 Ages

An amount is divided among Anita, Bhumi and Charu in the ratio of 6:5:9, respectively. Divya separately receives an additional amount equal to 1/3 of the combined amount received by Anita, Bhumi and Charu. If the difference between the average amount of Anita and Charu and the average amount of Bhumi and Divya is Rs. 600, find the total amount received by all four persons.

(A) 8400 (B) 9600
(C) 7200 (D) 10800
(E) 12000

Q.261 Ages

Eight years ago, the ratio of the ages of Megha and Kunal was 7:6. Seven years hence, the ratio of their ages will be 10:9. Find the present age of Megha.

(A) 43 years (B) 37 years
(C) 45 years (D) 48 years
(E) 40 years

Q.262 Ages

The average age of a class, including the teacher, is 24 years. The average age of the boys is 30 years, and the number of boys and girls is equal. If the teacher's age is 6 years more than the average age of the girls, find the average age of the girls.

(A) 18 years (B) 16 years
(C) 20 years (D) 22 years
(E) 24 years

Q.263 Ages

Ishita's present age is 50% more than Payal's present age. The sum of their ages 7 years from now will be 89 years. Find the difference between their ages 3 years ago.

(A) 18 years (B) 9 years
(C) 12 years (D) 21 years
(E) 15 years

Q.264 Ages

Three years ago, the average age of Ira and Leena was 24 years. Six years from now, their ages will be in the ratio 5 : 6. Find the present age of the elder person.

(A) 28 years (B) 30 years
(C) 32 years (D) 34 years
(E) 36 years

Q.265 Ages

Six years ago, the ratio of the ages of Vikram and Sameer was 7 : 4. The ratio of Vikram's age four years from now to Sameer's age two years from now is 45 : 28. Find the present age of Sameer.

(A) 24 years (B) 26 years
(C) 28 years (D) 30 years
(E) 32 years

Q.266 Ages

The present age of Ananya is 38 years. Two years hence, the ratio of the ages of Rhea to Ananya will be 4 : 5. If the present age of Swati is 20% more than the present age of Rhea, then find Swati's age five years ago.

(A) 29 years (B) 33 years
(C) 31 years (D) 27 years
(E) 35 years

Q.267 Ages

The ratio of the present ages of Swati and Kritika is 5 : 8. Megha is 12 years older than Swati and 6 years younger than Kritika. Find Kritika's age 7 years ago.

(A) 37 years (B) 41 years
(C) 45 years (D) 39 years
(E) 48 years

Q.268 Ages

The ratio of the present age of B to that of A is 5 : 4, and the present age of C is 20% less than the present age of B. If the difference between the sum of the ages of B and C and the average age of A and B is 54 years, then find the average age of all three two years hence.

(A) 46 years (B) 50 years
(C) 52 years (D) 56 years
(E) 54 years

Q.269 Ages

Four years ago, Yash's age was equal to the average of the present ages of Yash, Sandeep and Ayush. Ayush is 2 years older than Sandeep and 5 years younger than Yash. If the average of their present ages is 29 years, then find the present age of Sandeep.

(A) 24 years (B) 26 years
(C) 28 years (D) 30 years
(E) 32 years

Q.270 Ages

t years hence, the ratio of the ages of Rhea and Shreya will be 4 : 5, and (t + 8) years hence, the ratio of their ages will be 5 : 6. If the present age of Rhea is 24 years, then find the present age of Shreya.

(A) 28 years (B) 30 years
(C) 32 years (D) 34 years
(E) 36 years

Q.271 Ages

Ayush had a certain amount. He invested 60% of it in a mutual fund, 15% in a fixed deposit, and 10% in gold. If the remaining amount with him is Rs. 7500, find the total amount he had initially.

(A) Rs. 45000 (B) Rs. 50000
(C) Rs. 60000 (D) Rs. 40000
(E) Rs. 55000

Q.272 Ages

Vikram's annual income is Rs. 120000. He spends 25% of his income on rent. From the remaining amount, x% is spent on insurance. The balance amount is invested in fixed deposits and gold in the ratio 4:5. If the amount invested in gold is Rs. 30000, find x.

(A) 40 (B) 35
(C) 45 (D) 30
(E) 50

Q.273 Ages

In a school, the ratio of boys to girls in a class is 3:5. The total number of students is 64. The average weight of the whole class is 45 kg, and the average weight of the boys is 50 kg. Find the average weight of the girls.

(A) 44 kg (B) 42 kg
(C) 40 kg (D) 46 kg
(E) 41 kg

Q.274 Ages

The average temperature of the first three days of a week is 40°C, and the average temperature of the last three days is 22°C. If the ratio of the average temperature of the whole week to the temperature on the fourth day is 5 : 4, find the temperature on the fourth day.

(A) 20°C (B) 28°C
(C) 24°C (D) 30°C
(E) 26°C

Q.275 Ages

The average weight of 45 players in a sports camp is 62 kg. When 15 new players join the camp, the average weight decreases by 3 kg. Find the average weight of the 15 new players.

(A) 50 kg (B) 48 kg
(C) 52 kg (D) 46 kg
(E) 54 kg

Q.276 Ages

The monthly salary of Suresh is 50% more than that of Avani. The average monthly expenditure of Suresh and Avani is Rs. 27,000. Avani's monthly saving is 40% less than Suresh's monthly saving, and Avani's monthly expenditure is Rs. 6,000 less than Suresh's monthly expenditure. Find the difference between their annual salaries. Assume both monthly salaries remain unchanged throughout the year.

(A) Rs. 3,60,000 (B) Rs. 3,00,000
(C) Rs. 4,20,000 (D) Rs. 2,40,000
(E) Rs. 4,80,000

Q.277 Ages

The average weight of all students in a batch is 48 kg. The average weight of boys is 52 kg, and the average weight of all 30 girls is 42 kg. Find the number of boys in the batch.

(A) 40 (B) 45
(C) 50 (D) 36
(E) 60

Q.278 Ages

The average salary of Pallavi and Megha is Rs. 4500. If Pallavi's salary is increased by 40% and Megha's salary is increased by 25%, then their total salary increases by 30%. By what percentage is Megha's initial salary more than Pallavi's initial salary?

(A) 50% (B) 75%
(C) 100% (D) 150%
(E) 200%

Q.279 Ages

The average monthly income of Keshav and Lalit is Rs. 5600. The average monthly income of Lalit and Kunal is Rs. 6200, and the average monthly income of Keshav and Kunal is Rs. 5900. Find Keshav's monthly income.

(A) Rs. 5100 (B) Rs. 5400
(C) Rs. 5200 (D) Rs. 5300
(E) Rs. 5500

Q.280 Ages

The average weight of all students in a class is 40 kg. The average weight of all the boys is 44 kg, and the average weight of all 30 girls is 34 kg. Find the number of boys in the class.

(A) 40 (B) 45
(C) 35 (D) 50
(E) 60

Q.281 Ages

The average weight of 10 students in a class is y kg. Two new students join the class with a total weight of 70 kg, and the average weight of the class decreases by y/12 kg. If the heavier student among the two new students weighs y − 28 kg, find the difference between the weights of the two new students.

(A) 10 kg (B) 12 kg
(C) 16 kg (D) 18 kg
(E) 14 kg

Q.282 Ages

The average weight of all the players in a club is 55 kg. When a new player weighing 76 kg joins the club, the average weight increases by 1 kg. Find the number of players in the club now.

(A) 18 (B) 24
(C) 21 (D) 20
(E) 22

Q.283 Ages

The average monthly expenditure of Rahul and Mayank is Rs. 5600, which is 20% less than the average monthly expenditure of Prateek and Hemant. Prateek spends Rs. 600 more than Mayank, and the average monthly expenditure of Mayank and Prateek is Rs. 5900. Find the average monthly expenditure of Rahul and Hemant.

(A) 6400 (B) 7000
(C) 6700 (D) 6500
(E) 6200

Q.284 Ages

The average daily numbers of visitors at a book fair during the first three days and the last three days of a week are 48 and 45, respectively. If the ratio of the average number of visitors for the whole week to the number of visitors on the fourth day is 5:4, then find the number of visitors on the fourth day.

(A) 42 (B) 30
(C) 40 (D) 36
(E) 32

Q.285 Ages

The average weight of Yash, Sohan, Keshav and Kunal is 24 kg, while the average weight of Sohan, Keshav and Kunal is 22 kg. If Nilesh's weight is 40% more than Yash's weight, find the average weight of Yash and Nilesh.

(A) 32 kg (B) 36 kg
(C) 42 kg (D) 30 kg
(E) 38 kg

Q.286 Ages

The average weight of Kabir, Aman and Aniket is 60 kg. If the average weight of Kabir and Aman is 55 kg and the average weight of Aman and Aniket is 58 kg, then find Aman's weight.

(A) 42 (B) 48
(C) 50 (D) 44
(E) 46

Q.287 Ages

The average temperature of the first three days of a week is 31°C, and the average temperature of the next three days is 28°C. If the average temperature of the entire week is 30°C, find the temperature of the last day of the week.

(A) 33°C (B) 30.5°C
(C) 36°C (D) 29°C
(E) 31.5°C

Q.288 Ages

In an exam, marks obtained by Riya, Shreya, Rhea and Kritika are 72, 58, 46 and 64, respectively. If the average marks of Riya, Shreya, Rhea, Kritika and Anjali in the same exam is 62, find the marks obtained by Anjali.

(A) 68 (B) 72
(C) 70 (D) 64
(E) 76

📖 Chapter 01 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.251 — **Correct Option: (B)**

• Let the present ages of Ankit and Gaurav be 4x and 7x respectively.
• After 6 years:
• Ankit's age = 4x + 6
• Gaurav's age = 7x + 6
• Given that after 6 years, the ratio of Gaurav's age to Ankit's age will be 5:3.
• So,
• (7x + 6) : (4x + 6) = 5 : 3
• Now cross-multiply:
• 3(7x + 6) = 5(4x + 6)
• 21x + 18 = 20x + 30
• x = 12
• Therefore, Ankit's present age = 4x = 4 × 12 = 48 years.

Solution Q.252 — **Correct Option: (D)**

• Total population = 500000
• Number of males = 220000
• So, number of females = 500000 - 220000 = 280000
• Total literate population = 56% of 500000
• = 500000 × 56 / 100
• = 280000
• Literate males = 42% of 220000
• = 220000 × 42 / 100
• = 92400
• Therefore, literate females = Total literates - Literate males
• = 280000 - 92400
• = 187600
• Now, percentage of literate females
• = 187600 × 100 / 280000
• = 67%

Solution Q.253 — **Correct Option: (D)**

• Average Aptitude marks of Yash and Divya = 80.
• So total Aptitude marks = 80 × 2 = 160.
• Let Yash's Aptitude marks be x.
• Then Divya's Aptitude marks = 160 - x.
• Yash's Reasoning marks = 120% of x = 1.2x.
• Divya's Reasoning marks = 90% of (160 - x).
• Total Reasoning marks are 8 more than total Aptitude marks.
• So total Reasoning marks = 160 + 8 = 168.
• Now,
• 1.2x + 0.9(160 - x) = 168
• 1.2x + 144 - 0.9x = 168
• 0.3x = 24
• x = 80
• So Divya's Aptitude marks = 160 - 80 = 80.
• Yash's Reasoning marks = 120% of 80 = 96.
• Divya's Reasoning marks = 90% of 80 = 72.
• Required difference = 96 - 72 = 24.

Solution Q.254 — **Correct Option: (E)**

• Average of 11 numbers = 42.
• Total of 11 numbers = 11 × 42 = 462.
• Average of first 6 numbers = 44.
• Sum of first 6 numbers = 6 × 44 = 264.
• Average of last 6 numbers = 45.
• Sum of last 6 numbers = 6 × 45 = 270.
• Here, the 6th number is common in both groups.
• 6th number = 264 + 270 - 462 = 72.
• Sum of remaining 10 numbers = 462 - 72 = 390.
• Average of remaining numbers = 390 / 10 = 39.

Solution Q.255 — **Correct Option: (B)**

• Let the number of people be n.
• Total handshakes = n × (n - 1) / 2.
• Given, total handshakes = 276.
• So, n × (n - 1) / 2 = 276.
• n × (n - 1) = 552.
• Now, 24 × 23 = 552.
• So, n = 24.

Solution Q.256 — **Correct Option: (B)**

• %
• Solution:
• Number of boys in School X = 700.
• Boys who participated = 60% of 700 = 700 × 60 / 100 = 420.
• Number of girls in School Y = 800. Girls who participated = 35% of 800 = 800 × 35 /
• 100 = 280.
• Total participants = 420 + 280 = 700.
• Required percentage of boys = 420 × 100 / 700 = 60%. Hence, boys are 60% of the
• total participants.

Solution Q.257 — **Correct Option: (C)**

• years
• Solution:
• Average age of Nidhi and Kritika = 50 years.
• So, Nidhi + Kritika = 50 × 2 = 100.
• If Rhea replaces Nidhi, average becomes 48 years.
• So, Rhea + Kritika = 48 × 2 = 96.
• If Rhea replaces Kritika, average becomes 52 years.
• So, Nidhi + Rhea = 52 × 2 = 104.
• Adding all three equations:
• (Nidhi + Kritika) + (Rhea + Kritika) + (Nidhi + Rhea) = 100 + 96 + 104.
• 2 × (Nidhi + Kritika + Rhea) = 300.
• Nidhi + Kritika + Rhea = 150.
• Average age of Nidhi, Kritika and Rhea = 150 / 3 = 50 years.
• Average age of Priya and Sona = 1.5 × 50 = 75 years.
• Sum of ages of Priya and Sona = 75 × 2 = 150.
• Sum of ages of all five people = 150 + 150 = 300.
• Average age of all five people = 300 / 5 = 60 years.

Solution Q.258 — **Correct Option: (C)**

• Candidates who failed in both sections = 15%.
• Candidates who qualified in at least one section = 100% - 15% = 85%.
• Qualified in Reasoning = 68%.
• Qualified in Quant = 72%.
• Qualified in both sections = 68% + 72% - 85% = 55%.
• 55% of total candidates = 1100.
• Total candidates = 1100 × 100 / 55 = 2000.

Solution Q.259 **Correct Option: (D)**

• : 11
• Solution:
• Let the present ages of A and B be 4x years and 5x years, respectively.
• Three years ago, A's age = 4x - 3.
• Three years ago, C's age = 2/3 of (4x - 3).
• So, C - 3 = 2/3 × (4x - 3).
• 3C - 9 = 8x - 6.
• 3C = 8x + 3.
• C = (8x + 3)/3.
• Given, B is 13 years older than C.
• 5x - (8x + 3)/3 = 13.
• 15x - 8x - 3 = 39.
• 7x = 42.
• x = 6.
• Present age of B = 5x = 30 years.
• Present age of C = (8 × 6 + 3)/3 = 51/3 = 17 years.
• C's age after 5 years = 17 + 5 = 22 years.
• Required ratio = 30 : 22 = 15 : 11.

Solution Q.260 **Correct Option: (B)**

• Let the amounts received by Anita, Bhumi and Charu be 6x, 5x and 9x respectively.
• Combined amount of Anita, Bhumi and Charu = 6x + 5x + 9x = 20x
• Divya's amount = 1/3 of 20x
• Divya's amount = 20x/3
• Average amount of Anita and Charu = (6x + 9x)/2
• Average amount of Anita and Charu = 15x/2
• Average amount of Bhumi and Divya = (5x + 20x/3)/2
• Average amount of Bhumi and Divya = (15x/3 + 20x/3)/2
• Average amount of Bhumi and Divya = (35x/3)/2
• Average amount of Bhumi and Divya = 35x/6
• Difference between the two averages = Rs. 600
• 15x/2 - 35x/6 = 600
• 45x/6 - 35x/6 = 600
• 10x/6 = 600
• 5x/3 = 600
• x = 600 × 3/5
• x = 360
• Total amount received by all four = 6x + 5x + 9x + 20x/3
• Total amount = 20x + 20x/3
• Total amount = 80x/3
• Total amount = 80 × 360/3
• Total amount = Rs. 9600

Solution Q.261 **Correct Option: (A)**

• years
• Solution:
• Eight years ago, the ratio of ages of Megha and Kunal was 7:6.
• Let their ages eight years ago be 7x and 6x.
• Present age of Megha = 7x + 8
• Present age of Kunal = 6x + 8
• Seven years hence:
• Megha's age = 7x + 8 + 7 = 7x + 15
• Kunal's age = 6x + 8 + 7 = 6x + 15
• Given future ratio = 10:9
• So, (7x + 15)/(6x + 15) = 10/9
• Cross multiply:
• 9 × (7x + 15) = 10 × (6x + 15)
• 63x + 135 = 60x + 150
• 63x - 60x = 150 - 135
• 3x = 15
• x = 15/3 = 5
• Present age of Megha = 7x + 8
• Present age of Megha = 7 × 5 + 8 = 35 + 8 = 43 years

Solution Q.262 **Correct Option: (A)**

• years
• Solution:
• Let the number of boys be n and the number of girls be n.
• Let the average age of girls = x years.
• Average age of boys = 30 years
• So, total age of boys = 30n
• Total age of girls = nx
• Teacher's age = x + 6
• Total number of persons = n + n + 1 = 2n + 1
• Average age of the class including teacher = 24 years
• So, total age of all persons = 24 × (2n + 1)
• Now,
• 30n + nx + x + 6 = 24 × (2n + 1)
• 30n + nx + x + 6 = 48n + 24
• nx + x = 48n - 30n + 24 - 6
• x(n + 1) = 18n + 18
• x(n + 1) = 18(n + 1)
• x = 18

Solution Q.263 **Correct Option: (E)**

• years
• Solution:
• Ishita's present age is 50% more than Payal's present age.
• So, Ishita's age = 150% of Payal's age = 3/2 of Payal's age
• Therefore, present age Ratio (Ishita : Payal) = 3 : 2
• The sum of their ages 7 years from now = 89 years
• In 7 years, both will gain 7 years each.
• Total increase in their ages = 7 + 7 = 14 years
• Present sum of their ages = 89 - 14 = 75 years
• Ratio sum = 3 + 2 = 5 parts
• Value of 1 part = 75/5 = 15 years
• Ishita's present age = 3 × 15 = 45 years
• Payal's present age = 2 × 15 = 30 years
• Present age difference = 45 - 30 = 15 years
• The difference between their ages is unchanged with time.
• Therefore, their age difference three years ago was also 15 years.

Solution Q.264 **Correct Option: (B)**

• years
• Solution:
• Three years ago, the average age of Ira and Leena was 24 years.
• So, their total age three years ago = 24 × 2 = 48 years.
• Present total age = 48 + 3 + 3 = 54 years.
• Six years from now, total age = 54 + 6 + 6 = 66 years.
• Six years from now, their age ratio will be 5 : 6.
• Total ratio parts = 5 + 6 = 11.
• One ratio part = 66/11 = 6.
• Future ages = 5 × 6 and 6 × 6.
• Future ages = 30 years and 36 years.
• Present ages = 30 - 6 and 36 - 6.
• Present ages = 24 years and 30 years.
• Therefore, the elder person's present age is 30 years.

Solution Q.265 **Correct Option: (B)**

• years
• Solution:
• Let the ages of Vikram and Sameer six years ago be 7x and 4x, respectively
• Present age of Vikram = 7x + 6
• Present age of Sameer = 4x + 6
• Vikram's age four years hence = 7x + 6 + 4
• Vikram's age four years hence = 7x + 10
• Sameer's age two years hence = 4x + 6 + 2
• Sameer's age two years hence = 4x + 8
• Given ratio = 45 : 28
• (7x + 10)/(4x + 8) = 45/28
• 28(7x + 10) = 45(4x + 8)
• 196x + 280 = 180x + 360
• 16x = 80
• x = 5
• Present age of Sameer = 4x + 6
• Present age of Sameer = 4 × 5 + 6
• Present age of Sameer = 26 years

Solution Q.266 **Correct Option: (C)**

• years
• Solution:
• Present age of Ananya = 38 years
• Ananya's age two years hence = 38 + 2
• Ananya's age two years hence = 40 years
• Two years hence, Rhea : Ananya = 4 : 5
• Rhea's age two years hence = 40 × 4/5
• Rhea's age two years hence = 32 years
• Present age of Rhea = 32 - 2
• Present age of Rhea = 30 years
• Swati's present age is 20% more than Rhea's present age
• Swati's present age = 120% of 30
• Swati's present age = 30 × 120/100
• Swati's present age = 36 years
• Swati's age five years ago = 36 - 5
• Swati's age five years ago = 31 years

Solution Q.267 **Correct Option: (B)**

• years
• Solution:
• Let the present ages of Swati and Kritika be 5x years and 8x years respectively.
• Megha is 12 years older than Swati.
• Megha's age = 5x + 12.
• Megha is 6 years younger than Kritika.
• Megha's age = 8x - 6.
• So, 5x + 12 = 8x - 6.
• 3x = 18.
• x = 6.
• Kritika's present age = 8 × 6 = 48 years.
• Kritika's age 7 years ago = 48 - 7 = 41 years.

Solution Q.268 **Correct Option: (E)**

• years
• Solution:
• Let the present ages of B and A be 5x and 4x years respectively.
• Present age of C = 20% less than B = 5x × 80/100 = 4x years.
• Sum of ages of B and C = 5x + 4x = 9x.
• Average age of A and B = (4x + 5x)/2 = 9x/2.
• Given difference = 54 years.
• 9x - 9x/2 = 54
• 9x/2 = 54
• 9x = 108
• x = 12
• Present age of A = 4 × 12 = 48 years.
• Present age of B = 5 × 12 = 60 years.
• Present age of C = 4 × 12 = 48 years.
• Ages after two years = 50 years, 62 years and 50 years.
• Average age after two years = (50 + 62 + 50)/3
• Average age after two years = 162/3 = 54 years.

Solution Q.269 **Correct Option: (B)**

• years
• Solution:
• Average of present ages of Yash, Sandeep and Ayush = 29 years
• Sum of present ages = 29 × 3
• Sum of present ages = 87 years
• Four years ago, Yash's age = 29 years
• Present age of Yash = 29 + 4
• Present age of Yash = 33 years
• Ayush is 5 years younger than Yash
• Present age of Ayush = 33 - 5
• Present age of Ayush = 28 years
• Ayush is 2 years older than Sandeep
• Present age of Sandeep = 28 - 2
• Present age of Sandeep = 26 years

Solution Q.270 **Correct Option: (C)**

• years
• Solution:
• Let the present age of Shreya = N years
• Present age of Rhea = 24 years
• t years hence,
• Age of Rhea = 24 + t
• Age of Shreya = N + t
• Ratio = 4 : 5
• So, (24 + t)/(N + t) = 4/5
• 5(24 + t) = 4(N + t)
• 120 + 5t = 4N + 4t
• t = 4N - 120
• (t + 8) years hence,
• Age of Rhea = 24 + t + 8 = 32 + t
• Age of Shreya = N + t + 8
• Ratio = 5 : 6
• So, (32 + t)/(N + t + 8) = 5/6
• 6(32 + t) = 5(N + t + 8)
• 192 + 6t = 5N + 5t + 40
• t = 5N - 152
• Now,
• 4N - 120 = 5N - 152
• N = 32
• Present age of Shreya = 32 years

Solution Q.271 **Correct Option: (B)**

• Rs. 50000
• Solution:
• Let the total amount be 100%.
• Amount invested in mutual fund = 60%
• Amount invested in fixed deposit = 15%
• Amount invested in gold = 10%
• Total invested amount = 60% + 15% + 10% = 85%
• Remaining amount = 100% - 85% = 15%
• Given remaining amount = Rs. 7500
• So, 15% of total amount = Rs. 7500
• Total amount = 7500 × 100/15
• Total amount = Rs. 50000

Solution Q.272 **Correct Option: (A)**

• Annual income of Vikram = Rs. 120000.
• He spends 25% on rent.
• Rent = 25% of 120000 = Rs. 30000.
• Remaining amount = 120000 - 30000 = Rs. 90000.
• The balance after insurance is invested in fixed deposits and gold in the ratio 4:5.
• Gold's share = 5 parts.
• Given, amount invested in gold = Rs. 30000.
• So, 5 parts = Rs. 30000.
• 1 part = 30000 / 5 = Rs. 6000.
• Total invested amount = 9 parts = 9 × 6000 = Rs. 54000.
• Amount spent on insurance = 90000 - 54000 = Rs. 36000.
• Insurance percentage of the remaining amount = 36000 / 90000 × 100 = 40%.
• So, x = 40.

Solution Q.273 **Correct Option: (B)**

• kg
• Solution:
• The ratio of boys to girls is 3:5.
• Total ratio parts = 3 + 5 = 8
• Total number of students = 64
• Number of boys = (64 × 3)/8
• = 24
• Number of girls = (64 × 5)/8
• = 40
• Average weight of the whole class = 45 kg
• So, total weight of the whole class = 64 × 45 = 2880 kg
• Average weight of boys = 50 kg
• So, total weight of boys = 24 × 50 = 1200 kg
• Total weight of girls = Total weight of class - Total weight of boys
• Total weight of girls = 2880 - 1200 = 1680 kg
• Average weight of girls = 1680/40
• = 42 kg

Solution Q.274 **Correct Option: (C)**

• °C
• Solution:
• Average temperature of first three days = 40°C
• So, total temperature of first three days = 40 × 3 = 120°C
• Average temperature of last three days = 22°C
• So, total temperature of last three days = 22 × 3 = 66°C
• Total temperature of six days = 120 + 66 = 186°C
• Let the temperature on the fourth day = X
• Total temperature of the whole week = 186 + X
• Average temperature of the whole week: (186 + X)/7
• Given ratio of weekly average temperature to fourth day temperature = 5 : 4
• Now,
• 4 × (186 + X) = 5 × 7 × X
• 744 + 4X = 35X
• 35X - 4X = 744
• 31X = 744
• Value of X: 744/31
• = 24

Solution Q.275 **Correct Option: (A)**

• kg
• Solution:
• Initial number of players = 45
• Initial average weight = 62 kg
• Initial total weight = 45 × 62 = 2790 kg
• 15 new players join the camp.
• Total number of players now = 45 + 15 = 60
• Average decreases by 3 kg.
• New average weight = 62 - 3 = 59 kg
• New total weight = 60 × 59 = 3540 kg
• Total weight of 15 new players = 3540 - 2790 = 750 kg
• Average weight of 15 new players = 750/15
• Average weight = 50 kg

Solution Q.276 **Correct Option: (A)**

• Rs. 3,60,000
• Solution:
• Suresh's monthly salary is 50% more than Avani's monthly salary.
• So, Suresh's salary : Avani's salary = 150 : 100 = 3 : 2
• Let Suresh's monthly salary = 3x and Avani's monthly salary = 2x.
• Average monthly expenditure = Rs. 27,000
• So, total monthly expenditure = 27000 × 2 = Rs. 54,000
• Avani's expenditure is Rs. 6,000 less than Suresh's expenditure.
• Let Suresh's expenditure = e.
• Then Avani's expenditure = e - 6000.
• So, e + (e - 6000) = 54000
• 2e - 6000 = 54000
• 2e = 60000
• e = 30000
• Suresh's monthly expenditure = Rs. 30,000
• Avani's monthly expenditure = 30000 - 6000 = Rs. 24,000
• Suresh's monthly saving = 3x - 30000
• Avani's monthly saving = 2x - 24000
• Avani's saving is 40% less than Suresh's saving.
• So, Avani's saving = 60% of Suresh's saving = 3/5 of Suresh's saving.
• 2x - 24000 = 3/5 × (3x - 30000)
• Cross multiply:
• 5 × (2x - 24000) = 3 × (3x - 30000)
• 10x - 120000 = 9x - 90000
• 10x - 9x = 120000 - 90000
• x = 30000
• Difference between monthly salaries = 3x - 2x = x = Rs. 30,000
• Difference between annual salaries = 30000 × 12
• Difference between annual salaries = Rs. 3,60,000

Solution Q.277 **Correct Option: (B)**

• Let the number of boys = x
• Average weight of boys = 52 kg
• Total weight of boys = 52x
• Number of girls = 30
• Average weight of girls = 42 kg
• Total weight of girls = 30 × 42 = 1260 kg
• Average weight of all students = 48 kg
• Total number of students = x + 30
• So, total weight of all students = 48 × (x + 30)
• Now,
• 52x + 1260 = 48 × (x + 30)
• 52x + 1260 = 48x + 1440
• 52x - 48x = 1440 - 1260
• 4x = 180
• x = 180/4 = 45

Solution Q.278 **Correct Option: (C)**

• %
• Solution:
• Average salary of Pallavi and Megha = Rs. 4500.
• Total initial salary = 4500 × 2 = Rs. 9000.
• Let Pallavi's initial salary be Rs. x.
• Then Megha's initial salary = Rs. (9000 - x).
• Total salary increases by 30%.
• Total increase = 30% of 9000.
• Total increase = 9000 × 30/100 = Rs. 2700.
• Increase in Pallavi's salary = 40% of x = 40x/100.
• Increase in Megha's salary = 25% of (9000 - x) = 25(9000 - x)/100.
• So,
• 40x/100 + 25(9000 - x)/100 = 2700
• 40x + 25(9000 - x) = 270000
• 40x + 225000 - 25x = 270000
• 15x = 270000 - 225000
• 15x = 45000
• x = 45000/15
• x = 3000
• Pallavi's initial salary = Rs. 3000.
• Megha's initial salary = 9000 - 3000 = Rs. 6000.
• Difference = 6000 - 3000 = Rs. 3000.
• Required percentage = 3000/3000 × 100 = 100%.

Solution Q.279 **Correct Option: (D)**

• Rs. 5300
• Solution:
• Average income of Keshav and Lalit = Rs. 5600
• So, Keshav + Lalit = 5600 × 2 = Rs. 11200
• Average income of Lalit and Kunal = Rs. 6200
• So, Lalit + Kunal = 6200 × 2 = Rs. 12400
• Average income of Keshav and Kunal = Rs. 5900
• So, Keshav + Kunal = 5900 × 2 = Rs. 11800
• Now:
• 2 × Keshav = (Keshav + Lalit) + (Keshav + Kunal) - (Lalit + Kunal)
• 2 × Keshav = 11200 + 11800 - 12400
• 2 × Keshav = 10600
• Keshav = 10600/2
• Keshav = Rs. 5300

Solution Q.280 **Correct Option: (B)**

• Let the number of boys = x
• Average weight of boys = 44 kg
• Total weight of boys = 44x kg
• Number of girls = 30
• Average weight of girls = 34 kg
• Total weight of girls = 30 × 34 = 1020 kg
• Average weight of all students = 40 kg
• Total number of students = x + 30
• Total weight of all students = 40(x + 30)
• Now,
• 44x + 1020 = 40(x + 30)
• 44x + 1020 = 40x + 1200
• 44x - 40x = 1200 - 1020
• 4x = 180
• x = 180/4 = 45

Solution Q.281 **Correct Option: (E)**

• kg
• Solution:
• The average weight of 10 students is y kg.
• So, total weight of 10 students = 10y kg.
• Two new students join the class with a total weight of 70 kg.
• So, total weight after two students join = 10y + 70.
• Now, total number of students = 10 + 2 = 12.
• The average weight decreases by y/12 kg.
• So, new average weight = y - y/12
• New average weight = 12y/12 - y/12
• New average weight = 11y/12
• Now, total weight of 12 students can also be written as:
• New average × Number of students
• Total weight of 12 students = 11y/12 × 12
• Total weight of 12 students = 11y.
• So,
• 11y = 10y + 70
• 11y - 10y = 70
• y = 70
• Now, the heavier student's weight = y - 28.
• Heavier student's weight = 70 - 28 = 42 kg.
• Total weight of the two new students = 70 kg.
• So, lighter student's weight = 70 - 42 = 28 kg.
• Required difference = 42 - 28 = 14 kg.

Solution Q.282 **Correct Option: (C)**

• Let the initial number of players = x
• Initial average weight = 55 kg.
• Initial total weight = 55x kg.
• After one player of 76 kg joins, number of players = x + 1.
• New average weight = 55 + 1 = 56 kg.
• New total weight = 56 × (x + 1).
• Now, 55x + 76 = 56 × (x + 1)
• 55x + 76 = 56x + 56
• 56x - 55x = 76 - 56
• x = 20
• Initial number of players = 20.
• Number of players now = 20 + 1 = 21.

Solution Q.283 **Correct Option: (C)**

• Average expenditure of Rahul and Mayank = Rs. 5600
• So, Rahul + Mayank = 5600 × 2 = Rs. 11200
• This average is 20% less than the average of Prateek and Hemant.
• So, Rs. 5600 = 80% of average of Prateek and Hemant
• Average of Prateek and Hemant = 5600 × 100/80 = Rs. 7000
• Prateek + Hemant = 7000 × 2 = Rs. 14000
• Average of Mayank and Prateek = Rs. 5900
• Mayank + Prateek = 5900 × 2 = Rs. 11800
• Prateek = Mayank + 600
• So, Mayank + Mayank + 600 = 11800
• 2 × Mayank = 11200
• Mayank = Rs. 5600
• Prateek = 5600 + 600 = Rs. 6200
• Rahul = 11200 - 5600 = Rs. 5600
• Hemant = 14000 - 6200 = Rs. 7800
• Average expenditure of Rahul and Hemant = (5600 + 7800)/2 = Rs. 6700

Solution Q.284 **Correct Option: (D)**

- Sum of visitors during the first three days = 48 × 3 = 144.
- Sum of visitors during the last three days = 45 × 3 = 135.
- Let the number of visitors on the fourth day = x
- Total visitors in the week = 144 + 135 + x = 279 + x.
- Average visitors for the whole week = (279 + x) / 7.
- Given ratio of weekly average to fourth day visitors = 5:4.
- So, weekly average = 5x/4.
- Now, (279 + x) / 7 = 5x/4.
- 4 × (279 + x) = 35x.
- 1116 + 4x = 35x.
- 31x = 1116.
- x = 36.
- Number of visitors on the fourth day = 36.

Solution Q.285 **Correct Option: (B)**

- kg
- Solution:
- Average weight of Yash, Sohan, Keshav and Kunal = 24 kg.
- Total weight of all four = 24 × 4 = 96 kg.
- Average weight of Sohan, Keshav and Kunal = 22 kg.
- Total weight of Sohan, Keshav and Kunal = 22 × 3 = 66 kg.
- Yash's weight = 96 - 66 = 30 kg.
- Nilesh's weight is 40% more than Yash's weight.
- Nilesh's weight = 30 × 140/100 = 42 kg.
- Average weight of Yash and Nilesh = (30 + 42) / 2
- Average weight = 72 / 2 = 36 kg.

Solution Q.286 **Correct Option: (E)**

- Average weight of Kabir, Aman and Aniket = 60 kg
- So, total weight of all three
- = 60 × 3
- = 180 kg
- Average weight of Kabir and Aman = 55 kg
- So, Kabir + Aman = 55 × 2 = 110 kg
- Average weight of Aman and Aniket = 58 kg
- So, Aman + Aniket = 58 × 2 = 116 kg
- Now add both pair sums:
- Kabir + Aman + Aman + Aniket = 110 + 116
- Kabir + Aniket + 2Nikhil = 226
- But Kabir + Aman + Aniket = 180
- So, subtracting total weight from 226 gives:
- Aman = 226 - 180
- = 46 kg
- Hence, Aman's weight is 46 kg.

Solution Q.287 **Correct Option: (A)**

- °C
- Solution:
- Average temperature of the first three days = 31°C.
- Total temperature of the first three days = 31 × 3 = 93°C.
- Average temperature of the next three days = 28°C.
- Total temperature of the next three days = 28 × 3 = 84°C.
- Average temperature of the entire week = 30°C.
- Total temperature of 7 days = 30 × 7 = 210°C.
- Temperature of the last day = 210 - 93 - 84 = 33°C.

Solution Q.288 **Correct Option: (C)**

- Average marks of 5 students = 62
- Total marks of 5 students = 62 × 5 = 310
- Total marks of Riya, Shreya, Rhea and Kritika = 72 + 58 + 46 + 64 = 240
- Marks obtained by Anjali = 310 - 240
- = 70

CHAPTER 02: TIME & WORK, WAGES & ALTERNATE DAYS

37 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Total Work = LCM of individual times (Work in units).
- Efficiency = Total Work Units / Total Days taken.
- Wages Ratio = Ratio of Work Done (or Ratio of Efficiencies if working for equal days).
- MDH Formula: (M1 × D1 × H1) / W1 = (M2 × D2 × H2) / W2

Q.289 Time & Work

Abhinav and Kabir together can complete a piece of work in 24 days, while Abhinav and Sameer together can complete the same work in 40 days. Kabir is 200% more efficient than Sameer. In how many days can Abhinav, Kabir and Sameer together complete the work?

(A) 15 (B) 20
(C) 18 (D) 24
(E) 12

Q.290 Time & Work

Ramesh is 3 times as efficient as Suresh. The difference between the number of days taken by them individually to complete a job is 24 days. If they work on alternate days starting with Suresh, how many days will they take to complete the whole job?

(A) 14 days (B) 16 days
(C) 20 days (D) 18 days
(E) 22 days

Q.291 Time & Work

A and B together can finish a piece of work in 6 days. A alone can finish the same work in 15 days. If C is 25% more efficient than B, find the number of days C alone will take to finish the work.

(A) 5 days (B) 6 days
(C) 10 days (D) 7 days
(E) 8 days

Q.292 Time & Work

Vikram and Pawan together can finish a project in 15 days. Pawan alone takes 40 days more than Vikram alone to finish the same project. In how many days can Vikram alone finish the project?

(A) 30 (B) 24
(C) 18 (D) 20
(E) 40

Q.293 Time & Work

3 men and 4 apprentices complete a project in 4 days and receive Rs. 1120 for the whole work. If one man alone can complete the same project in 20 days, find the daily wage of one man, assuming wages are proportional to the work done.

(A) Rs. 48 (B) Rs. 60
(C) Rs. 52 (D) Rs. 56
(E) Rs. 64

Q.294 Time & Work

A group of 12 men and 16 women can complete a piece of work in 20 days. A woman takes twice as much time as a man to complete the same work. How many days will 8 men take to complete the work?

(A) 42 days (B) 48 days
(C) 50 days (D) 54 days
(E) 60 days

Q.295 Time & Work

Worker P alone can complete a piece of work in 48 days. If workers P and Q together take 37.5% less time than P alone to complete the same work, then in how many days can Q alone complete the work?

(A) 80 days (B) 90 days
(C) 72 days (D) 96 days
(E) 108 days

Q.296 Time & Work

Jyoti can complete a piece of work in 36 days. Adarsh is twice as efficient as Jyoti. The ratio of the efficiency of Gaurav to that of Adarsh is 3:4. If all three start working together, in how many days will the work be completed?

(A) 6 days (B) 10 days
(C) 12 days (D) 8 days
(E) 9 days

Q.297 Time & Work

Worker P and Worker Q together can complete a piece of work in 18 days. Worker Q and Worker R together can complete the same work in 24 days, while Worker R and Worker P together can complete it in 36 days. Find the time taken by Worker P alone to complete the same work.

(A) 60 days (B) 42 days
(C) 36 days (D) 48 days
(E) 54 days

Q.298 Time & Work

Workers P and Q together can complete a piece of work in 12 days, and Workers Q and R together can complete it in 20 days. P works for 6 days, then Q works for 12 days, and the remaining work is completed by R in 12 days. Find the time taken by Q alone to complete the whole work.

(A) 60 days (B) 72 days
(C) 90 days (D) 80 days
(E) 120 days

Q.299 Time & Work

The average daily income of a group of 12 workers is Rs. 48. The difference between the highest and lowest daily incomes is Rs. 56. If the workers with the highest and lowest incomes are excluded, the average daily income of the remaining workers decreases by Rs. 2. Find the average daily income of the group if, in a separate case, two workers whose combined daily income equals the highest daily income leave the original group of 12 workers.

(A) Rs. 45 (B) Rs. 47
(C) Rs. 49 (D) Rs. 51
(E) Rs. 53

Q.300 Time & Work

Worker P alone can complete a task in 10 days, and worker Q alone can complete the same task in 15 days. If they work on alternate days starting with P, find the number of days required to complete the whole task.

(A) 10 days (B) 11 days
(C) 14 days (D) 12 days
(E) 15 days

Q.301 Time & Work

The average consumption of tea in an office is 210 cups per working day, while the average monthly consumption of tea per employee is 35 cups. If there are 264 working days in a year, find the number of employees in the office. Assume the number of employees remains unchanged throughout the year and all tea is consumed by these employees.

(A) 120 (B) 132
(C) 144 (D) 126
(E) 156

Q.302 Time & Work

Ketan, Lakshya and Nilesh together can complete a piece of work in 15 days, and Lakshya alone can complete the same work in 40 days. Find in how many days Ketan and Nilesh together can complete the whole work.

(A) 20 days (B) 18 days
(C) 30 days (D) 25 days
(E) 24 days

Q.303 Time & Work

Three workers Gopal, Harit and Jatin are hired for a job. Gopal and Harit together can complete the job in 24 days, while Harit and Jatin together can complete the same job in 30 days. To complete the job, Gopal worked for 15 days, Harit worked for 20 days and Jatin worked for 30 days. Find the time taken by Gopal alone to complete the job.

(A) 48 days (B) 72 days
(C) 80 days (D) 60 days
(E) 90 days

Q.304 Time & Work

Vinay can complete a piece of work in 300 days. Hemant is 50% more efficient than Vinay. Vinay and Sameer together can complete the same work in 200 days. If all three work together, in how many days will the work be completed?

(A) 100 days (B) 120 days
(C) 90 days (D) 150 days
(E) 80 days

Q.305 Time & Work

Three workers P, Q and R can complete a job in 16 days, 20 days and 40 days, respectively. They complete the whole job together and receive Rs. 5500 as total wages. Find the difference between the combined wages of P and R and the wages of Q alone. Assume wages are divided in proportion to the work done.

(A) Rs. 1500 (B) Rs. 1200
(C) Rs. 1800 (D) Rs. 2000
(E) Rs. 2500

Q.306 Time & Work

Pawan is 50% less efficient than Raghav, and Raghav is 20% more efficient than Tushar. All three together can complete a piece of work in 42 days. If Raghav and Tushar work together for 36 days and then leave, find the time taken by Pawan alone to complete the remaining work.

(A) 48 days (B) 64 days
(C) 56 days (D) 72 days
(E) 60 days

Q.307 Time & Work

Worker P alone can finish a task in 45 days. Worker Q is 60% more efficient than P, and worker R is 25% less efficient than Q. If all three work together, find the time required to complete the task, rounded to the nearest whole day.

(A) 12 days (B) 15 days
(C) 10 days (D) 18 days
(E) 20 days

Q.308 Time & Work

24 men can complete a piece of work in 20 days, while 20 women can complete the same work in 36 days. X men started the work and worked for 12 days before leaving. If (X + 2) women completed the remaining work in 24 days, then find X.

(A) 12 (B) 16
(C) 20 (D) 14
(E) 18

Q.309 Time & Work

Ravi alone can complete a piece of work in 10 days, and Rajesh alone can complete the same work in 15 days. Ravi and Rajesh work together for 4 days, and then Akash replaces both of them. If Akash completes 60% of the remaining work, find the percentage of the whole work completed by Akash.

(A) 24% (B) 18%
(C) 30% (D) 20%
(E) 15%

Q.310 Time & Work

A certain number of workers can complete a job in 40 days. After working for 10 days, 8 workers left the job. The remaining workers completed the rest of the work in 45 days. Find the number of workers at the beginning.

(A) 18 (B) 20
(C) 28 (D) 24
(E) 30

Q.311 Time & Work

Aniket alone can complete a job in 36 days, and Aman alone can complete the same job in 54 days. They started working together, but Aniket left after some days. Aman then completed the remaining work in 24 days. After how many days did Aniket leave?

(A) 15 days (B) 9 days
(C) 12 days (D) 18 days
(E) 10 days

Q.312 Time & Work

Ananya is 50% more efficient than Divya, and Ananya alone can complete a piece of work in 24 days. Sahil alone can complete the same work in 72 days. Find the time taken by Ananya, Divya and Sahil together to complete the work.

(A) 15 days (B) 18 days
(C) 10 days (D) 16 days
(E) 12 days

Q.313 Time & Work

Kritika and Divya started working together on a project. After n days, Kritika left the work and Divya completed the remaining work in 12 days. Kritika and Divya alone can complete the same work in 30 days and 40 days, respectively. Find the value of n.

(A) 10 days (B) 12 days
(C) 15 days (D) 18 days
(E) 20 days

Q.314 Time & Work

Aniket, Gaurav, and Chirag rented a projector for a workshop for Rs. 4800. They used it for 7 hours, 9 hours, and 14 hours respectively. If the rent is divided according to the usage time, how much should Chirag pay?

(A) Rs. 1840
(B) Rs. 2160
(C) Rs. 2400
(D) Rs. 2240
(E) Rs. 1920

Q.315 Time & Work

Sameer and Ayush together can complete a job in 12 days. Sameer alone can complete the same job in 20 days. Sagar is 50% more efficient than Ayush. Find the number of days Sagar alone will take to complete the same job.

(A) 20 days
(B) 15 days
(C) 18 days
(D) 24 days
(E) 30 days

Q.316 Time & Work

P, Q and R together can complete a piece of work in 8 days, while Q alone can complete the same work in 24 days. P alone takes 10 days less than R to complete the work. In how many days can R complete 60% of the same work?

(A) 12 days
(B) 15 days
(C) 24 days
(D) 18 days
(E) 20 days

Q.317 Time & Work

Machines A, B, and C can complete a job alone in 30 days, 36 days, and 45 days respectively. Machines A and C were operated together for 10 days and then stopped. If the remaining work is completed by machines B and D together in 8 days, find the number of days machine D alone will take to complete 25% of the same job.

(A) 12 days
(B) 15 days
(C) 9 days
(D) 18 days
(E) 6 days

Q.318 Time & Work

Tushar is three times as efficient as Kabir and can complete a piece of work 40 days earlier than Kabir. Find the time taken by Tushar and Kabir together to complete the same work.

(A) 12 days
(B) 18 days
(C) 20 days
(D) 15 days
(E) 24 days

Q.319 Time & Work

Abeer, Lakshya and Chirag can complete a piece of work in 20 days, 30 days and 60 days, respectively. They started the work together, but Chirag left after 5 days. After 3 more days, Lakshya also left the work. Find the total time taken to complete the whole work.

(A) 13 days
(B) 15 days
(C) 12 days
(D) 18 days
(E) 10 days

Q.320 Time & Work

P, Q and R together can complete a piece of work in 80 days. The efficiency of P is 25% more than that of Q, and the efficiency of R is 20% less than that of P. Find in how many days Q and R together can complete 60% of the work.

(A) 78 days
(B) 72 days
(C) 84 days
(D) 90 days
(E) 96 days

Q.321 Time & Work

Eshan alone can complete a piece of work in 30 days, and Nilesh alone can complete the same work in 12 days. If Eshan, Nilesh and Mahesh together can complete the same work in 15/2 days, then by what percentage is Mahesh less efficient than Eshan?

(A) 60%
(B) 25%
(C) 40%
(D) 75%
(E) 50%

Q.322 Time & Work

18 men and 12 women can complete a piece of work in 18 days. A group consisting of 6 fewer men and 12 more women completes the entire same work in 18 days. Find the time taken by 24 women to complete double the same work.

(A) 54 days
(B) 60 days
(C) 72 days
(D) 48 days
(E) 84 days

Q.323 Time & Work

Ayush can complete 3/5 of a piece of work in 12 hours, while Gaurav can complete 2/3 of the same work in 10 hours. Chandan is 25% more efficient than Gaurav. If they work one after another for one hour each repeating the order Ayush, Gaurav and Chandan, then in how many hours will the whole work be completed?

(A) 12 hours
(B) 18 hours
(C) 21 hours
(D) 15 hours
(E) 20 hours

Q.324 Time & Work

Six men can complete a piece of work in 30 days, while nine women can complete the same work in 30 days. Find in how many days three men and three women together can complete the same work.

(A) 24 days
(B) 30 days
(C) 42 days
(D) 40 days
(E) 36 days

Q.325 Time & Work

'N' children can complete a piece of work in 10 days, and 'M' women can complete the same piece of work in 10 days. 6 women and 6 children working together can complete the same work in 8 days. If the efficiency of a woman is 50% more than the efficiency of a child, then find in how many days (N + M) children can complete the same work.

(A) 7 days (B) 5 days
(C) 8 days (D) 6 days
(E) 10 days

📖 Chapter 02 Step-by-Step Smart Solutions — Ikkish Ratio & Shortcut Approaches

Solution Q.289 — **Correct Option: (B)**

• ⚡ Total Work (LCM) = 120 units
• Abhinav and Kabir together complete the work in 24 days.
• So, their combined efficiency = 120 / 24 = 5 units per day
• Abhinav and Sameer together complete the work in 40 days.
• So, their combined efficiency = 120 / 40 = 3 units per day
• Kabir is 200% more efficient than Sameer.
• This means Kabir's efficiency = 3 times Sameer's efficiency.
• Let Sameer's efficiency = x units per day.
• Then Kabir's efficiency = 3x units per day.
• Now,
• Abhinav + Kabir = 5 units
• Abhinav + Sameer = 3 units
• Subtract the second from the first:
• Kabir - Sameer = 5 - 3
• 3x - x = 2
• 2x = 2
• x = 1
• So, Sameer's efficiency = 1 unit per day
• Kabir's efficiency = 3 units per day
• Abhinav's efficiency = 3 - 1 = 2 units per day
• Combined efficiency of all three = 2 + 3 + 1 = 6 units per day
• Required time = 120 / 6 = 20 days

Solution Q.290 — **Correct Option: (D)**

• days
• Solution:
• Ramesh is 3 times as efficient as Suresh.
• So, the efficiency Ratio (Ramesh : Suresh) = 3:1.
• In work and time questions, time taken is inversely proportional to efficiency.
• So, the time Ratio (Ramesh : Suresh) = 1:3.
• The difference between their individual times = 24 days.
• Difference in ratio parts = 3 - 1 = 2 parts.
• So, 2 parts = 24 days.
• Therefore, 1 part = 12 days.
• Time taken by Ramesh alone = 1 part = 12 days.
• Time taken by Suresh alone = 3 parts = 36 days.
• Now take total work as 36 units, because it is easy to use with 12 and 36.
• Ramesh's 1 day work = 3 units.
• Suresh's 1 day work = 1 unit.
• They work on alternate days starting with Suresh.
• So, in 2 days:
• Suresh does 1 unit and Ramesh does 3 units.
• Total work done in 2 days = 1 + 3 = 4 units.
• Total work = 36 units.
• Number of 2-day cycles needed = 36 / 4 = 9 cycles.
• Total time = 9 × 2 = 18 days.

Solution Q.291 — **Correct Option: (E)**

• days
• Solution:
• A and B together can complete the work in 6 days.
• So, one day work of A and B together = 1/6
• A alone can complete the same work in 15 days.
• So, one day work of A = 1/15
• Now, one day work of B = One day work of A and B together - One day work of A
• B's one day work = 1/6 - 1/15
• LCM of 6 and 15 = 30
• So, B's one day work = 5/30 - 2/30
• B's one day work = 3/30 = 1/10
• Therefore, B alone can complete the work in 10 days.
• Now, C is 25% more efficient than B.
• It means, if B's efficiency is 100%, then C's efficiency = 125%.
• Efficiency Ratio (B : C) = 100 : 125 = 4 : 5
• Time is inversely proportional to efficiency.
• So, time Ratio (B : C) = 5 : 4
• B takes 10 days.
• So, 5 parts = 10 days
• 1 part = 10/5 = 2 days
• C's time = 4 parts = 2 × 4 = 8 days

Solution Q.292 — **Correct Option: (D)**

• Let Vikram alone complete the project in x days.
• Pawan takes 40 days more than Vikram.
• So, Pawan alone can complete the project in x + 40 days.
• Vikram's 1-day work = 1/x
• Pawan's 1-day work = 1/(x + 40)
• Together they complete the work in 15 days.
• So, their combined 1-day work = 1/15
• Now:
• 1/x + 1/(x + 40) = 1/15
• (x + 40 + x)/(x(x + 40)) = 1/15
• (2x + 40)/(x(x + 40)) = 1/15
• 15(2x + 40) = x(x + 40)
• 30x + 600 = x^2 + 40x
• x^2 + 40x - 30x - 600 = 0
• x^2 + 10x - 600 = 0
• x^2 + 30x - 20x - 600 = 0
• x(x + 30) - 20(x + 30) = 0
• (x + 30)(x - 20) = 0
• x = 20. The other root is -30 and is rejected because time must be positive.

Solution Q.293 **Correct Option: (D)**

• Rs. 56
• Solution:
• One man alone can complete the project in 20 days.
• So, total work = 20 man-days
• 3 men and 4 apprentices complete the same project in 4 days.
• So, their one-day work = 20/4 = 5 man-units
• This means 3 men + 4 apprentices = 5 men
• So, 4 apprentices = 5 men - 3 men = 2 men
• Therefore, 1 apprentice = 2/4 = 1/2 man
• Total payment for 4 days = Rs. 1120
• Daily payment of the group = 1120/4 = Rs. 280
• Let daily wage of one man = x units
• Then daily wage of one apprentice = x/2
• Daily wage of group = 3x + 4 × x/2
• 280 = 3x + 2x
• 280 = 5x
• x = 280/5 = 56

Solution Q.294 **Correct Option: (C)**

• days
• Solution:
• A woman takes twice as much time as a man to complete the work.
• So, efficiency of 1 woman = 1/2 of 1 man
• 12 men and 16 women can complete the work in 20 days.
• Convert women into men-equivalent.
• 16 women = 16 × 1/2 = 8 men
• So, total efficiency = 12 men + 8 men = 20 men
• This means 20 men can complete the work in 20 days.
• Total work = 20 × 20 = 400 man-days
• Now, 8 men will complete the same work in:
• Time = Total work/Number of men
• Time = 400/8 = 50 days

Solution Q.295 **Correct Option: (A)**

• days
• Solution:
• Worker P alone can complete the work in 48 days.
• Time taken by P and Q together is 37.5% less than 48 days.
• 37.5% = 3/8
• So, time reduced = 48 × 3/8 = 18 days
• Time taken by P and Q together = 48 - 18 = 30 days
• P's 1-day work = 1/48
• P and Q's 1-day work = 1/30
• Q's 1-day work = 1/30 - 1/48
• LCM of 30 and 48 = 240
• Q's 1-day work = 8/240 - 5/240
• Q's 1-day work = 3/240
• Q's 1-day work = 1/80
• Therefore, Q alone can complete the work in 80 days.

Solution Q.296 **Correct Option: (D)**

• days
• Solution:
• Let Jyoti's efficiency be 1 unit per day.
• Jyoti can complete the work in 36 days.
• So, total work = 36 units.
• Adarsh is twice as efficient as Jyoti.
• So, Adarsh's efficiency = 2 units per day.
• Efficiency of Gaurav : Adarsh = 3:4
• So, Gaurav's efficiency = 2 × 3/4 = 3/2 units per day.
• Total efficiency of all three = 1 + 2 + 3/2
• Total efficiency = 3 + 3/2 = 9/2 units per day.
• Time taken = Total work/Total efficiency
• Time taken = 36/(9/2)
• Time taken = 36 × 2/9
• Time taken = 8 days

Solution Q.297 **Correct Option: (D)**

• days
• Solution:
• Take total work as the LCM of 18, 24 and 36.
• LCM = 72 units.
• Efficiency of P and Q together = 72/18 = 4 units per day.
• Efficiency of Q and R together = 72/24 = 3 units per day.
• Efficiency of R and P together = 72/36 = 2 units per day.
• Add all three:
• (P + Q) + (Q + R) + (R + P) = 4 + 3 + 2 = 9 units per day.
• So, 2(P + Q + R) = 9 units per day.
• P + Q + R = 9/2 units per day.
• Now, P alone = (P + Q + R) - (Q + R).
• P alone = 9/2 - 3.
• P alone = 9/2 - 6/2 = 3/2 units per day.
• Total work = 72 units.
• Time taken by P alone = 72/(3/2).
• Time taken = 72 × 2/3 = 48 days.

Solution Q.298 **Correct Option: (A)**

• days
• Solution:
• Let P's 1-day work be p.
• Let Q's 1-day work be q.
• Let R's 1-day work be r.
• P and Q together complete the work in 12 days.
• So:
• p + q = 1/12
• Q and R together complete the work in 20 days.
• So:
• q + r = 1/20
• From p + q = 1/12:
• p = 1/12 - q
• From q + r = 1/20:
• r = 1/20 - q
• Given, P works for 6 days, Q works for 12 days and R works for 12 days to complete
• the work.
• So:
• 6p + 12q + 12r = 1
• Put p = 1/12 - q and r = 1/20 - q.
• 6(1/12 - q) + 12q + 12(1/20 - q) = 1
• 6/12 - 6q + 12q + 12/20 - 12q = 1
• 1/2 + 3/5 - 6q = 1
• 5/10 + 6/10 - 6q = 1
• 11/10 - 6q = 1
• 6q = 11/10 - 1
• 6q = 1/10
• q = 1/60
• So, Q alone can complete the whole work in 60 days.

Solution Q.299 **Correct Option: (C)**

• Rs. 49
• Solution:
• Number of workers = 12
• Average daily income = Rs. 48
• Total daily income of all workers = 12 × 48 = Rs. 576
• After excluding the workers with the highest and lowest incomes, remaining workers
• = 10
• New average decreases by Rs. 2.
• New average = 48 - 2 = Rs. 46
• Total income of remaining 10 workers = 10 × 46 = Rs. 460
• So, sum of highest and lowest incomes = 576 - 460
• Sum = Rs. 116
• Difference between highest and lowest incomes = Rs. 56
• Highest income = (116 + 56)/2
• Highest income = 172/2
• Highest income = Rs. 86
• In the separate case, two workers leave the original group, and their total income
• equals the highest income.
• Total income of two leaving workers = Rs. 86
• Remaining total income = 576 - 86 = Rs. 490
• Remaining number of workers = 12 - 2 = 10
• New average income = 490/10
• New average income = Rs. 49

Solution Q.300 **Correct Option: (D)**

• days
• Solution:
• P alone can complete the work in 10 days.
• So, P's one-day work = 1/10
• Q alone can complete the work in 15 days.
• So, Q's one-day work = 1/15
• They work on alternate days starting with P.
• Work done in 2 days = P's one-day work + Q's one-day work
• Work done in 2 days = 1/10 + 1/15
• LCM of 10 and 15 = 30
• Work done in 2 days = 3/30 + 2/30 = 5/30 = 1/6
• So, in 2 days, they complete 1/6 of the work.
• To complete full work:
• Required time = 6 × 2 = 12 days

Solution Q.301 **Correct Option: (B)**

• Average tea consumption per working day = 210 cups
• Number of working days in a year = 264
• Total tea consumption in a year = 210 × 264
• Total tea consumption in a year = 55440 cups
• Average monthly tea consumption per employee = 35 cups
• Average yearly tea consumption per employee = 35 × 12
• Average yearly tea consumption per employee = 420 cups
• Number of employees = Total yearly tea consumption / Yearly consumption per
• employee
• Number of employees = 55440/420
• Number of employees = 132

Solution Q.302 **Correct Option: (E)**

• days
• Solution:
• Ketan, Lakshya and Nilesh together can complete the work in 15 days.
• Their 1-day work = 1/15
• Lakshya alone can complete the work in 40 days.
• Lakshya's 1-day work = 1/40
• Ketan and Nilesh's 1-day work = 1/15 - 1/40
• LCM of 15 and 40 = 120
• Ketan and Nilesh's 1-day work = 8/120 - 3/120
• Ketan and Nilesh's 1-day work = 5/120
• Ketan and Nilesh's 1-day work = 1/24
• Therefore, Ketan and Nilesh together can complete the whole work in 24 days.

Solution Q.303 **Correct Option: (D)**

• days
• Solution:
• Let Gopal's 1 day work be g.
• Let Harit's 1 day work be h.
• Let Jatin's 1 day work be j.
• Gopal and Harit together complete the job in 24 days.
• So,
• g + h = 1/24
• Harit and Jatin together complete the job in 30 days.
• So,
• h + j = 1/30
• From g + h = 1/24:
• h = 1/24 - g
• From h + j = 1/30:
• j = 1/30 - h
• j = 1/30 - (1/24 - g)
• j = 1/30 - 1/24 + g
• j = g - 1/120
• According to the work done:
• 15g + 20h + 30j = 1
• Put h = 1/24 - g and j = g - 1/120:
• 15g + 20(1/24 - g) + 30(g - 1/120) = 1
• 15g + 20/24 - 20g + 30g - 30/120 = 1
• 25g + 5/6 - 1/4 = 1
• 25g + 10/12 - 3/12 = 1
• 25g + 7/12 = 1
• 25g = 1 - 7/12
• 25g = 5/12
• g = 5/(12 × 25)
• g = 1/60
• Therefore, Gopal alone can complete the job in 60 days.

Solution Q.304 **Correct Option: (A)**

• days
• Solution:
• Take total work as 600 units.
• Vinay can complete the work in 300 days.
• Vinay's 1-day work = 600/300 = 2 units.
• Hemant is 50% more efficient than Vinay.
• Hemant's 1-day work = 2 × 150/100 = 3 units.
• Vinay and Sameer together can complete the work in 200 days.
• Their 1-day work = 600/200 = 3 units.
• Sameer's 1-day work = 3 - 2 = 1 unit.
• Total 1-day work of Vinay, Hemant and Sameer = 2 + 3 + 1 = 6 units.
• Time taken by all three = Total work/1-day work.
• Time taken = 600/6 = 100 days.

Solution Q.305 **Correct Option: (A)**

• Rs. 1500
• Solution:
• Wages are divided in the ratio of efficiency.
• Efficiency ratio of P, Q and R = 1/16 : 1/20 : 1/40.
• LCM of 16, 20 and 40 = 80.
• So, efficiency ratio = 5 : 4 : 2.
• Total parts = 5 + 4 + 2 = 11.
• Value of 1 part = 5500 / 11 = 500.
• P and R together = (5 + 2) parts = 7 parts.
• Wages of P and R together = 7 × 500 = Rs. 3500.
• Wages of Q = 4 × 500 = Rs. 2000.
• Required difference = 3500 - 2000 = Rs. 1500.

Solution Q.306 **Correct Option: (B)**

• days
• Solution:
• Let the efficiency of Tushar = 5 units/day.
• Raghav is 20% more efficient than Tushar.
• Efficiency of Raghav = 5 × 120/100 = 6 units/day
• Pawan is 50% less efficient than Raghav.
• Efficiency of Pawan = 6 × 50/100 = 3 units/day
• So, efficiency ratio of Pawan, Raghav and Tushar = 3:6:5
• Together efficiency = 3 + 6 + 5 = 14 units/day
• They complete the work in 42 days.
• Total work = 14 × 42 = 588 units
• Work done by Raghav and Tushar in 36 days:
• Efficiency of Raghav and Tushar = 6 + 5 = 11 units/day
• Work done = 11 × 36 = 396 units
• Remaining work = 588 - 396 = 192 units
• Time taken by Pawan alone = 192/3 = 64 days

Solution Q.307 **Correct Option: (A)**

• days
• Solution:
• Choose P's work rate as 5 units per day.
• Q is 60% more efficient than P.
• Q's work rate = 5 × 160/100 = 8 units per day.
• R is 25% less efficient than Q.
• R's work rate = 8 × 75/100 = 6 units per day.
• Thus, the work-rate ratio P : Q : R = 5 : 8 : 6.
• P alone completes the task in 45 days.
• Total work = 5 × 45 = 225 units.
• Combined work rate = 5 + 8 + 6 = 19 units per day.
• Exact time = 225/19 days ≈ 11.8421 days.
• Rounded to the nearest whole day, the required time is 12 days.

Solution Q.308 **Correct Option: (B)**

• Total work can be taken according to work rates.
• 24 men complete the work in 20 days.
• So, 1 man's 1 day work = 1/(24 × 20) = 1/480.
• 20 women complete the work in 36 days.
• So, 1 woman's 1 day work = 1/(20 × 36) = 1/720.
• Work done by X men in 12 days = 12X/480 = X/40.
• Work done by (X + 2) women in 24 days = 24(X + 2)/720 = (X + 2)/30.
• Total work = 1.
• So, X/40 + (X + 2)/30 = 1.
• LCM of 40 and 30 = 120.
• 3X + 4(X + 2) = 120.
• 3X + 4X + 8 = 120.
• 7X = 112.
• X = 16.

Solution Q.309 **Correct Option: (D)**

• %
• Solution:
• Ravi can complete the work in 10 days.
• Rajesh can complete the same work in 15 days.
• LCM of 10 and 15 = 30.
• So, take total work = 30 units.
• Ravi's efficiency = 30/10 = 3 units per day.
• Rajesh's efficiency = 30/15 = 2 units per day.
• Together, Ravi and Rajesh can do:
• 3 + 2 = 5 units per day.
• They work together for 4 days.
• Work done by Ravi and Rajesh in 4 days = 5 × 4 = 20 units.
• Remaining work = 30 - 20 = 10 units.
• Akash completes 60% of the remaining work.
• Work done by Akash = 60% of 10 units
• Work done by Akash = 60/100 × 10
• Work done by Akash = 6 units.
• Now, total work = 30 units.
• Percentage of the whole work done by Akash = 6/30 × 100
• Percentage of the whole work done by Akash = 20%.

Solution Q.310 **Correct Option: (D)**

• Let the number of workers at the beginning be N.
• Total work = N × 40 worker-days.
• Work done in the first 10 days = N × 10 = 10N worker-days.
• Remaining work = 40N - 10N = 30N worker-days.
• After 8 workers left, remaining workers = N - 8.
• They completed the remaining work in 45 days.
• So, 45 × (N - 8) = 30N.
• 45N - 360 = 30N.
• 15N = 360.
• N = 360 / 15 = 24.

Solution Q.311 **Correct Option: (C)**

• days
• Solution:
• Aniket's 1 day work = 1/36
• Aman's 1 day work = 1/54
• Together, their 1 day work = 1/36 + 1/54
• LCM of 36 and 54 = 108
• Together, their 1 day work = 3/108 + 2/108 = 5/108
• Let Aniket leave after x days.
• Work done by both together in x days = 5x/108
• After Aniket left, Aman completed the remaining work in 24 days.
• Work done by Aman in 24 days = 24/54 = 4/9
• Total work = 1
• So, 5x/108 + 4/9 = 1
• 5x/108 + 48/108 = 108/108
• 5x/108 = 60/108
• 5x = 60
• x = 12
• So, Aniket left after 12 days.

Solution Q.312 **Correct Option: (E)**

• days
• Solution:
• Ananya is 50% more efficient than Divya.
• So, efficiency ratio of Ananya to Divya = 150 : 100 = 3 : 2.
• Ananya completes the work in 24 days.
• Divya's time = 24 × 3/2 = 36 days.
• Sahil's time = 72 days.
• Work done by Ananya in 1 day = 1/24.
• Work done by Divya in 1 day = 1/36.
• Work done by Sahil in 1 day = 1/72.
• Take LCM of 24, 36 and 72 = 72.
• Ananya's 1 day work = 3/72.
• Divya's 1 day work = 2/72.
• Sahil's 1 day work = 1/72.
• Together 1 day work = 3/72 + 2/72 + 1/72 = 6/72 = 1/12.
• Therefore, all three together can complete the work in 12 days.

Solution Q.313 **Correct Option: (B)**

• days
• Solution:
• Kritika alone can complete the work in 30 days.
• Kritika's 1 day work = 1/30
• Divya alone can complete the work in 40 days.
• Divya's 1 day work = 1/40
• Together, their 1 day work = 1/30 + 1/40
• LCM of 30 and 40 = 120
• Together, their 1 day work = 4/120 + 3/120 = 7/120
• Divya completes the remaining work in 12 days.
• Work done by Divya in 12 days = 12 × 1/40 = 3/10
• So, work completed by both together = 1 - 3/10 = 7/10
• Time taken by both together = (7/10) / (7/120)
• Time = 7/10 × 120/7 = 12 days

Solution Q.314 **Correct Option: (D)**

• Rs. 2240
• Solution:
• Rent should be divided in the ratio of usage time.
• Usage time of Aniket, Gaurav, and Chirag = 7:9:14
• Total parts = 7 + 9 + 14 = 30
• Total rent = Rs. 4800
• Chirag's share = 14 parts
• Chirag's share = 4800 × 14/30
• Chirag's share = 160 × 14 = Rs. 2240

Solution Q.315 **Correct Option: (A)**

• days
• Solution:
• Take total work as 60 units, which is the LCM of 12 and 20.
• Efficiency of Sameer and Ayush together = 60 / 12 = 5 units/day.
• Efficiency of Sameer alone = 60 / 20 = 3 units/day.
• Efficiency of Ayush alone = 5 - 3 = 2 units/day.
• Sagar is 50% more efficient than Ayush.
• Efficiency of Sagar = 2 × 150/100 = 3 units/day.
• Time taken by Sagar alone = 60 / 3 = 20 days.

Solution Q.316 **Correct Option: (D)**

• days
• Solution:
• P, Q and R together complete the work in 8 days.
• So, their combined efficiency = 1/8.
• Q alone completes the work in 24 days.
• So, Q's efficiency = 1/24.
• Therefore, efficiency of P and R together = 1/8 - 1/24.
• 1/8 - 1/24 = 3/24 - 1/24 = 2/24 = 1/12.
• Let R alone complete the work in r days.
• Then P alone completes the work in (r - 10) days.
• So, 1/r + 1/(r - 10) = 1/12.
• (r - 10 + r) / [r(r - 10)] = 1/12.
• (2r - 10) / [r(r - 10)] = 1/12.
• 12(2r - 10) = r(r - 10).
• 24r - 120 = r^2 - 10r.
• r^2 - 34r + 120 = 0.
• r^2 - 30r - 4r + 120 = 0.
• r(r - 30) - 4(r - 30) = 0.
• (r - 30)(r - 4) = 0.
• R cannot take 4 days because then P would take negative time, so R = 30 days.
• Time taken by R to complete 60% work = 60% of 30 = 18 days.

Solution Q.317 **Correct Option: (C)**

• days
• Solution:
• A can complete the work in 30 days.
• A's 1 day work = 1/30
• C can complete the work in 45 days.
• C's 1 day work = 1/45
• A and C together 1 day work = 1/30 + 1/45
• LCM of 30 and 45 = 90
• A and C together 1 day work = 3/90 + 2/90 = 5/90 = 1/18
• Work done by A and C in 10 days = 10 × 1/18 = 5/9
• Remaining work = 1 - 5/9 = 4/9
• B and D complete 4/9 work in 8 days.
• B and D together 1 day work = 4/9 / 8 = 4/72 = 1/18
• B can complete the work in 36 days.
• B's 1 day work = 1/36
• D's 1 day work = 1/18 - 1/36 = 2/36 - 1/36 = 1/36
• So, D alone can complete the full work in 36 days.
• Time taken by D to complete 25% of the work = 36 × 25/100
• Time taken = 36 × 1/4 = 9 days

Solution Q.318 **Correct Option: (D)**

• days
• Solution:
• Let Kabir take x days to complete the work alone.
• Since Tushar is three times as efficient as Kabir, Tushar will take x/3 days to complete
• the same work.
• According to the question, x - x/3 = 40. So, 2x/3 = 40.
• Therefore, x = 60.
• Kabir takes 60 days and Tushar takes 60/3 = 20 days.
• Kabir's 1 day work = 1/60. Tushar's 1 day work = 1/20. Together, their 1 day work =
• 1/60 + 1/20 = 1/60 + 3/60 = 4/60 = 1/15. Hence, they can complete the work together
• in 15 days.

Solution Q.319 **Correct Option: (A)**

• days
• Solution:
• Take total work as 60 units, because LCM of 20, 30 and 60 is 60.
• Abeer's 1 day work = 60 / 20 = 3 units.
• Lakshya's 1 day work = 60 / 30 = 2 units.
• Chirag's 1 day work = 60 / 60 = 1 unit.
• Work done by all three in first 5 days = (3 + 2 + 1) × 5 = 6 × 5 = 30 units. After
• Chirag left, Abeer and Lakshya worked together for 3 days. Work done in these 3 days
• = (3 + 2) × 3 = 15 units. Total work done = 30 + 15 = 45 units.
• Remaining work = 60 - 45 = 15 units.
• Abeer alone completes remaining 15 units in 15 / 3 = 5 days. Total time taken = 5 + 3
• + 5 = 13 days.

Solution Q.320 **Correct Option: (A)**

• days
• Solution:
• Let the efficiency of Q be 4 units/day.
• P is 25% more efficient than Q.
• So, efficiency of P = 4 × 125/100 = 5 units/day.
• R is 20% less efficient than P.
• So, efficiency of R = 5 × 80/100 = 4 units/day.
• Efficiency ratio of P, Q and R = 5 : 4 : 4.
• Total efficiency of P, Q and R = 5 + 4 + 4 = 13 units/day.
• They complete the work in 80 days.
• Total work = 13 × 80 = 1040 units.
• 60% of total work = 1040 × 60/100 = 624 units.
• Efficiency of Q and R together = 4 + 4 = 8 units/day.
• Time taken by Q and R to complete 60% work = 624 / 8 = 78 days.
• Therefore, Q and R together can complete 60% of the work in 78 days.

Solution Q.321 **Correct Option: (E)**

• %
• Solution:
• Take total work as 60 units, because 60 is suitable for 30 days, 12 days and 15/2 days.
• Eshan completes the work in 30 days.
• Eshan's 1-day work = 60/30 = 2 units
• Nilesh completes the work in 12 days.
• Nilesh's 1-day work = 60/12 = 5 units
• Eshan, Nilesh and Mahesh together complete the work in 15/2 days.
• Their 1-day work = 60/(15/2)
• Their 1-day work = 60 × 2/15 = 8 units
• So, Mahesh's 1-day work = 8 - 2 - 5 = 1 unit
• Now compare Mahesh's efficiency with Eshan's efficiency.
• Eshan's efficiency = 2 units per day
• Mahesh's efficiency = 1 unit per day
• Difference = 2 - 1 = 1 unit
• Required percentage less = 1 × 100/2
• Required percentage less = 50%

Solution Q.322 **Correct Option: (C)**

• days
• Solution:
• Let the efficiency of 1 man be M and the efficiency of 1 woman be W.
• 18 men and 12 women complete the work in 18 days.
• In the separate attempt, the group has 18 - 6 = 12 men and 12 + 12 = 24 women.
• Both groups complete the same work in the same number of days.
• So, their one-day work must be equal.
• 18M + 12W = 12M + 24W
• 18M - 12M = 24W - 12W
• 6M = 12W
• M = 2W
• So, 1 man is equal to 2 women in efficiency.
• Take 1 woman's efficiency = 1 unit/day
• Then 1 man's efficiency = 2 units/day
• One-day work of 18 men and 12 women = 18 × 2 + 12 × 1
• One-day work = 36 + 12 = 48 units
• Total work = 48 × 18 = 864 units
• Double the work = 864 × 2 = 1728 units
• One-day work of 24 women = 24 × 1 = 24 units
• Required time = 1728/24 = 72 days

Solution Q.323 **Correct Option: (D)**

• hours
• Solution:
• Ayush can complete 3/5 work in 12 hours
• Ayush's 1-hour work = 3/(5 × 12)
• Ayush's 1-hour work = 3/60
• Ayush's 1-hour work = 1/20
• Gaurav can complete 2/3 work in 10 hours
• Gaurav's 1-hour work = 2/(3 × 10)
• Gaurav's 1-hour work = 2/30
• Gaurav's 1-hour work = 1/15
• Chandan is 25% more efficient than Gaurav
• Chandan's efficiency = 125% of Gaurav's efficiency
• Chandan's 1-hour work = 125/100 × 1/15
• Chandan's 1-hour work = 5/4 × 1/15
• Chandan's 1-hour work = 1/12
• Work done in 3 hours by Ayush, Gaurav and Chandan = 1/20 + 1/15 + 1/12
• LCM of 20, 15 and 12 = 60
• Work done in 3 hours = 3/60 + 4/60 + 5/60
• Work done in 3 hours = 12/60
• Work done in 3 hours = 1/5
• So, 1/5 work is completed in 3 hours
• Whole work will be completed in 15 hours

Solution Q.324 **Correct Option: (E)**

• days
• Solution:
• Six men can complete the work in 30 days.
• So, total work = 6 × 30 = 180 men-days
• Efficiency of one man = 1/180 work per day
• Nine women can complete the work in 30 days.
• So, total work = 9 × 30 = 270 women-days
• Efficiency of one woman = 1/270 work per day
• Efficiency of three men = 3/180 = 1/60 work per day
• Efficiency of three women = 3/270 = 1/90 work per day
• Combined efficiency = 1/60 + 1/90
• Combined efficiency = 3/180 + 2/180 = 5/180 = 1/36
• Time taken = 36 days
• Therefore, three men and three women together can complete the work in 36 days.

Solution Q.325 **Correct Option: (D)**

• days
• Solution:
• Let the efficiency of a child be 2 units/day.
• Efficiency of a woman is 50% more than a child.
• Efficiency of a woman = 2 × 150/100 = 3 units/day
• Efficiency of 6 women and 6 children together = 6 × 3 + 6 × 2
• = 18 + 12 = 30 units/day
• They complete the work in 8 days.
• Total work = 30 × 8 = 240 units
• N children complete the work in 10 days.
• N × 2 × 10 = 240
• 20N = 240
• N = 12
• M women complete the work in 10 days.
• M × 3 × 10 = 240
• 30M = 240
• M = 8
• N + M = 12 + 8 = 20
• Efficiency of 20 children = 20 × 2 = 40 units/day
• Time taken by (N + M) children = 240/40 = 6 days
• Therefore, (N + M) children can complete the same work in 6 days.

CHAPTER 03: PIPES & CISTERNS (INLET, OUTLET & LEAKAGE)

14 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Inlet Pipe = Positive Efficiency (+ units/hour) | Outlet/Leak Pipe = Negative Efficiency (- units/hour)
- Net Efficiency = Sum of Inlets - Sum of Outlets
- Time to fill/empty = Tank Capacity (LCM units) / Net Efficiency

Q.326 Pipes & Cisterns

Pipe A and Pipe B can fill a tank in 20 minutes and 30 minutes respectively. Pipe C can empty the tank at the rate of 4 litres per minute. If all three pipes are opened together, the tank is filled in 24 minutes. Find the capacity of the tank.

(A) 96 litres
(B) 120 litres
(C) 144 litres
(D) 108 litres
(E) 84 litres

Q.327 Pipes & Cisterns

Pipe P alone can fill a tank in 10 minutes. Two outlet pipes Q and R can empty the same tank in 30 minutes and 60 minutes respectively. If all three pipes are opened together and after X minutes pipe Q is closed, the remaining tank is filled by pipes P and R together in 9 minutes. Find X.

(A) 5
(B) 8
(C) 10
(D) 12
(E) 15

Q.328 Pipes & Cisterns

Pipe P can fill 20% of a tank in 4 hours, and Pipe Q can fill 25% of the tank in 5 hours. Pipe R is an outlet pipe and can empty the full tank in 40 hours. If the tank is already 25% filled, find the time taken by all three pipes together to fill the remaining part of the tank.

(A) 8 hours (B) 12 hours
(C) 10 hours (D) 15 hours
(E) 6 hours

Q.329 Pipes & Cisterns

Pipes M and N together can fill 1/3 of a tank in 5 hours, while Pipe S can empty the full tank in 30 hours. Starting with an empty tank, Pipes M and N are opened together for 10 hours and then closed. If Pipe S alone is then opened, find the time taken by Pipe S to empty the filled part of the tank.

(A) 15 hours (B) 18 hours
(C) 24 hours (D) 30 hours
(E) 20 hours

Q.330 Pipes & Cisterns

Pipe P can fill half of a tank in 6 minutes. Pipe Q is 100% more efficient than Pipe P, and Pipe R is 50% less efficient than Pipe Q. Find the time taken by all three pipes together to fill the tank.

(A) 2 minutes (B) 3 minutes
(C) 4 minutes (D) 5 minutes
(E) 6 minutes

Q.331 Pipes & Cisterns

Pipes P, Q and R can fill a tank in 20 minutes, 30 minutes and 60 minutes respectively. Pipes P and Q are opened together. After x minutes, pipe R is also opened, and the remaining tank is filled by all three pipes together in 5 minutes. Find the value of x.

(A) 4 (B) 5
(C) 6 (D) 8
(E) 10

Q.332 Pipes & Cisterns

Four identical inlet pipes can fill 50% of a tank in 5 minutes, while six identical outlet pipes can empty the full tank in 10 minutes. Find the time taken by 2 inlet pipes and 2 outlet pipes together to fill the tank.

(A) 60 min (B) 90 min
(C) 45 min (D) 75 min
(E) 120 min

Q.333 Pipes & Cisterns

Pipe P can fill a tank in 60 hours, Pipe Q is 50% more efficient than Pipe P, and Pipe R can fill the same tank in 10 hours less than Pipe Q. Pipe P and Pipe Q are opened together for X hours and then closed. Pipe R fills the remaining tank in (X + 3) hours. If the ratio of the portion of the tank filled by Pipes P and Q together to the portion filled by Pipe R is 1 : 1, find X.

(A) 8 hr (B) 10 hr
(C) 14 hr (D) 12 hr
(E) 16 hr

Q.334 Pipes & Cisterns

Pipe A can fill a tank in 30 minutes, while pipes A and B together can fill the same tank in 12 minutes. Pipe C is an outlet pipe and its efficiency is 20% less than that of pipe B. Find the time taken by pipe C alone to empty the full tank.

(A) 30 min (B) 20 min
(C) 18 min (D) 24 min
(E) 25 min

Q.335 Pipes & Cisterns

Two inlet pipes M and N together can fill a tank in 20 minutes, while pipes M and O together can fill the same tank in 30 minutes. If all three pipes M, N, and O together can fill the tank in 15 minutes, then pipe N is what percent more efficient than pipe O?

(A) 50% (B) 75%
(C) 125% (D) 100%
(E) 150%

Q.336 Pipes & Cisterns

A tank has two inlet pipes X and Y and one outlet pipe Z. Pipes X and Y alone can fill the tank in 5 hours and 20 hours respectively, while pipe Z can empty the full tank in 8 hours. All three pipes are opened together, and after 2 hours, pipe Z is closed. How much more time is required to fill the tank completely?

(A) 4 hours (B) 5 hours
(C) 2 hours (D) 6 hours
(E) 3 hours

Q.337 Pipes & Cisterns

Pipe A alone can fill a tank in 12 hours, and pipe B alone can fill the same tank in 15 hours. If pipes A, B, and C together can fill the tank in 20/7 hours, then find the time taken by pipe C alone to fill the tank.

(A) 5 hours (B) 6 hours
(C) 8 hours (D) 10 hours
(E) 12 hours

Q.338 Pipes & Cisterns

Pipe P alone can fill a tank in T hours, and pipe Q alone can fill the same tank in (T + 8) hours. If both pipes together fill the tank in 7 1/2 hours, then find the time taken by pipe Q alone to fill the tank.

(A) 16 hours (B) 18 hours
(C) 20 hours (D) 24 hours
(E) 30 hours

Q.339 Pipes & Cisterns

Pipe A can fill 2/3 of a tank in 12 minutes. Pipe B is 100% more efficient than pipe A, and pipe C is 25% less efficient than pipe B. Find the time taken by all three pipes together to fill the tank.

(A) 6 min (B) 5 min
(C) 8 min (D) 4 min
(E) 7 min

Chapter 03 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.326 Correct Option: (A)

• litres
• Solution:
• Let the capacity of the tank = x litres.
• Pipe A fills the tank in 20 minutes.
• So, Pipe A fills x / 20 litres per minute.
• Pipe B fills the tank in 30 minutes.
• So, Pipe B fills x / 30 litres per minute.
• Pipe C empties 4 litres per minute.
• When all three pipes are opened together, the tank is filled in 24 minutes.
• So, net filling rate = x / 24 litres per minute
• Now,
• x / 20 + x / 30 - 4 = x / 24
• Take the filling terms together:
• x / 20 + x / 30 = x / 12
• So,
• x / 12 - 4 = x / 24
• x / 12 - x / 24 = 4
• x / 24 = 4
• x = 96

Solution Q.327 Correct Option: (A)

• Pipe P fills the tank in 10 minutes.
• P's 1-minute work = 1/10
• Pipe Q empties the tank in 30 minutes.
• Q's 1-minute work = -1/30
• Pipe R empties the tank in 60 minutes.
• R's 1-minute work = -1/60
• When all three pipes are opened together:
• Net 1-minute work = 1/10 - 1/30 - 1/60
• LCM of 10, 30 and 60 = 60
• Net 1-minute work = 6/60 - 2/60 - 1/60 = 3/60 = 1/20
• So, work done in X minutes = X/20
• After X minutes, pipe Q is closed.
• Now, pipes P and R work together.
• Net 1-minute work of P and R = 1/10 - 1/60
• Net work = 6/60 - 1/60 = 5/60 = 1/12
• They fill the remaining tank in 9 minutes.
• Work done by P and R in 9 minutes = 9 × 1/12 = 9/12 = 3/4
• So, work done by all three pipes before Q is closed = 1 - 3/4 = 1/4
• Therefore, X/20 = 1/4
• X = 20/4 = 5

Solution Q.328 Correct Option: (C)

• hours
• Solution:
• Pipe P fills 20% of the tank in 4 hours.
• 20% = 20/100 = 1/5
• Work done by Pipe P in 1 hour = 1/5 × 1/4 = 1/20
• Pipe Q fills 25% of the tank in 5 hours.
• 25% = 25/100 = 1/4
• Work done by Pipe Q in 1 hour = 1/4 × 1/5 = 1/20
• Pipe R empties the full tank in 40 hours.
• Work done by Pipe R in 1 hour = -1/40
• Net work done by all three pipes in 1 hour = 1/20 + 1/20 - 1/40
• LCM of 20, 20 and 40 = 40
• Net work = 2/40 + 2/40 - 1/40 = 3/40
• Tank is already 25% filled.
• Remaining part = 75% = 75/100 = 3/4
• Required time = Remaining work/Net one-hour work
• Required time = 3/4 × 40/3 = 10 hours

Solution Q.329 Correct Option: (E)

• hours
• Solution:
• Pipes M and N together can fill 1/3 of the tank in 5 hours.
• So, in 10 hours, Pipes M and N will fill double of 1/3.
• Filled part in 10 hours = 2/3 of the tank.
• Pipe S can empty the full tank in 30 hours.
• Time taken by Pipe S to empty 2/3 of the tank = 2/3 × 30.
• Time taken = 60/3 = 20 hours.

Solution Q.330 Correct Option: (B)

• minutes
• Solution:
• Pipe P fills half of the tank in 6 minutes.
• So, Pipe P fills the full tank in 12 minutes.
• Let the efficiency of Pipe P be 1 unit/minute.
• Pipe Q is 100% more efficient than Pipe P.
• So, efficiency of Pipe Q = 2 units/minute.
• Pipe R is 50% less efficient than Pipe Q.
• So, efficiency of Pipe R = 50% of 2 = 1 unit/minute.
• Efficiency ratio of P, Q and R = 1 : 2 : 1
• Together efficiency = 1 + 2 + 1 = 4 units/minute.
• Pipe P alone takes 12 minutes.
• When all three work together, time taken = 12/4
• Time taken = 3 minutes

Solution Q.331 **Correct Option: (C)**

• Pipe P can fill the tank in 20 minutes.
• So, P's 1-minute work = 1/20
• Pipe Q can fill the tank in 30 minutes.
• So, Q's 1-minute work = 1/30
• Pipe R can fill the tank in 60 minutes.
• So, R's 1-minute work = 1/60
• Work done by all three pipes together in 1 minute:
• 1/20 + 1/30 + 1/60
• LCM of 20, 30 and 60 = 60
• 3/60 + 2/60 + 1/60 = 6/60 = 1/10
• Work done by all three pipes in 5 minutes = 5 × 1/10 = 1/2
• So, work done by P and Q before R is opened = 1 - 1/2 = 1/2
• Work done by P and Q together in 1 minute:
• 1/20 + 1/30
• LCM of 20 and 30 = 60
• 3/60 + 2/60 = 5/60 = 1/12
• Time taken by P and Q to complete 1/2 work = (1/2)/(1/12)
• Time = 1/2 × 12/1 = 6 minutes

Solution Q.332 **Correct Option: (A)**

• min
• Solution:
• Four inlet pipes can fill 50% of the tank in 5 minutes.
• So, four inlet pipes can fill the full tank in 10 minutes.
• Work done by four inlet pipes in 1 minute = 1/10
• Work done by one inlet pipe in 1 minute = 1/40
• Six outlet pipes can empty the full tank in 10 minutes.
• Work done by six outlet pipes in 1 minute = 1/10
• Work done by one outlet pipe in 1 minute = 1/60
• Work done by 2 inlet pipes in 1 minute = 2 × 1/40 = 1/20
• Work done by 2 outlet pipes in 1 minute = 2 × 1/60 = 1/30
• Net work done in 1 minute = 1/20 - 1/30
• Net work done in 1 minute = (3 - 2)/60 = 1/60
• So, the tank will be filled in 60 minutes.

Solution Q.333 **Correct Option: (D)**

• hr
• Solution:
• Pipe P can fill the tank in 60 hours.
• So, efficiency of Pipe P = 1/60 tank per hour.
• Pipe Q is 50% more efficient than Pipe P.
• So, Pipe Q's efficiency = 150/100 × 1/60 = 1/40 tank per hour.
• Therefore, Pipe Q can fill the tank in 40 hours.
• Pipe R can fill the tank in 10 hours less than Pipe Q.
• So, Pipe R can fill the tank in 40 - 10 = 30 hours.
• Efficiency of Pipe R = 1/30 tank per hour.
• Combined efficiency of Pipe P and Pipe Q = 1/60 + 1/40.
• LCM of 60 and 40 = 120.
• Combined efficiency = 2/120 + 3/120 = 5/120 = 1/24 tank per hour.
• The ratio of work done by Pipes P and Q together to Pipe R is 1 : 1.
• So, Pipes P and Q together fill 1/2 of the tank.
• Time taken by Pipes P and Q to fill 1/2 tank = (1/2)/(1/24).
• Time taken = 1/2 × 24 = 12 hours.
• So, X = 12.
• Check: Pipe R takes 30/2 = 15 hours to fill the remaining half. This equals X + 3 = 12
• + 3, so the additional condition is also satisfied.

Solution Q.334 **Correct Option: (E)**

• min
• Solution:
• Pipe A fills the tank in 30 minutes.
• Efficiency of pipe A = 1/30 tank per minute
• Pipes A and B together fill the tank in 12 minutes.
• Combined efficiency of A and B = 1/12 tank per minute
• Efficiency of pipe B = 1/12 - 1/30
• Efficiency of pipe B = (5 - 2)/60 = 3/60 = 1/20
• So, pipe B can fill the tank in 20 minutes.
• Pipe C is 20% less efficient than pipe B.
• Efficiency of pipe C = 80% of pipe B
• Efficiency of pipe C = 80/100 × 1/20 = 1/25
• So, pipe C alone can empty the full tank in 25 minutes.

Solution Q.335 **Correct Option: (D)**

• %
• Solution:
• Let the total work of filling the tank be 60 units because 60 is divisible by 20, 30, and
• 15.
• Efficiency of M and N together = 60 / 20 = 3 units/minute.
• Efficiency of M and O together = 60 / 30 = 2 units/minute.
• Efficiency of M, N, and O together = 60 / 15 = 4 units/minute.
• Efficiency of N = efficiency of all three - efficiency of M and O together.
• So, N = 4 - 2 = 2 units/minute.
• Efficiency of O = efficiency of all three - efficiency of M and N together.
• So, O = 4 - 3 = 1 unit/minute.
• Required percentage = (2 - 1) / 1 × 100 = 100%.
• Therefore, pipe N is 100% more efficient than pipe O.

Solution Q.336 **Correct Option: (E)**

• hours
• Solution:
• Pipe X can fill the tank in 5 hours.
• Work done by pipe X in 1 hour = 1/5
• Pipe Y can fill the tank in 20 hours.
• Work done by pipe Y in 1 hour = 1/20
• Pipe Z can empty the tank in 8 hours.
• Work done by pipe Z in 1 hour = 1/8
• Net work done by all three pipes in 1 hour = 1/5 + 1/20 - 1/8
• 1/5 + 1/20 = 4/20 + 1/20 = 5/20 = 1/4
• Net work = 1/4 - 1/8 = 2/8 - 1/8 = 1/8
• Work done by all three pipes in 2 hours = 2 × 1/8 = 1/4
• Remaining work = 1 - 1/4 = 3/4
• After pipe Z is closed, only X and Y work together.
• Work done by X and Y in 1 hour = 1/5 + 1/20 = 1/4
• Time required to complete remaining work = (3/4) / (1/4) = 3 hours

Solution Q.337 **Correct Option: (A)**

• hours
• Solution:
• Pipe A fills the tank in 12 hours.
• So, A's 1-hour work = 1/12.
• Pipe B fills the tank in 15 hours.
• So, B's 1-hour work = 1/15.
• A, B, and C together fill the tank in 20/7 hours.
• So, their 1-hour work = 1 / (20/7) = 7/20.
• Work done by A and B together in 1 hour = 1/12 + 1/15.
• LCM of 12 and 15 = 60.
• So, 1/12 + 1/15 = 5/60 + 4/60 = 9/60 = 3/20.
• C's 1-hour work = total work of A, B, C - work of A and B.
• C's 1-hour work = 7/20 - 3/20 = 4/20 = 1/5.
• So, pipe C alone can fill the tank in 5 hours.
• Therefore, the time taken by pipe C alone is 5 hours.

Solution Q.338 **Correct Option: (C)**

• hours
• Solution:
• Pipe P alone fills the tank in T hours.
• Pipe Q alone fills the tank in (T + 8) hours.
• Together they fill the tank in 7 1/2 hours = 15/2 hours.
• Combined one-hour work = 1 / (15/2) = 2/15
• So, 1/T + 1/(T + 8) = 2/15
• Shortcut checking from options:
• If pipe Q takes 20 hours, then pipe P takes 20 - 8 = 12 hours.
• Together one-hour work = 1/12 + 1/20
• LCM of 12 and 20 = 60
• Together one-hour work = 5/60 + 3/60 = 8/60 = 2/15
• So, together time = 15/2 hours = 7 1/2 hours
• Therefore, Pipe Q alone takes 20 hours.

Solution Q.339 **Correct Option: (D)**

• min
• Solution:
• Pipe A can fill 2/3 of the tank in 12 minutes.
• So, Pipe A can fill the full tank in 12 × 3/2 = 18 minutes.
• Efficiency of Pipe A = 1/18 tank per minute
• Pipe B is 100% more efficient than Pipe A.
• So, efficiency of Pipe B = 2 × 1/18 = 1/9 tank per minute
• Pipe C is 25% less efficient than Pipe B.
• So, efficiency of Pipe C = 75% of 1/9
• = 3/4 × 1/9
• = 1/12 tank per minute
• Combined efficiency of A, B and C = 1/18 + 1/9 + 1/12
• = 2/36 + 4/36 + 3/36
• = 9/36
• = 1/4 tank per minute
• Time taken by all three pipes together = 4 minutes

CHAPTER 04: SPEED, TIME & DISTANCE (BASICS & MOTION)

20 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Speed = Distance / Time | 1 km/h = 5/18 m/s | 1 m/s = 18/5 km/h
- Average Speed for Equal Distances = (2 × s1 × s2) / (s1 + s2)
- Relative Speed: Opposite Direction = (s1 + s2) | Same Direction = (s1 - s2)

Q.340 Speed

A man travels from Town P to Town Q at 80 km/h in 3 hours. Then he reduces his speed by 40% and travels from Town Q to Town R in 5 hours. Find his average speed for the entire journey.

(A) 54 km/h (B) 60 km/h
(C) 64 km/h (D) 66 km/h
(E) 72 km/h

Q.341 Speed

Sarthak takes 4 hours more than Nilesh to cover a distance of 96 km. If Sarthak doubles his speed, he takes 2 hours less than Nilesh to cover the same distance. Find Sarthak's original speed.

(A) 8 km/h (B) 6 km/h
(C) 12 km/h (D) 10 km/h
(E) 16 km/h

Q.342 Speed

Two vehicles start from the same place at the same time and travel along the same route to the same destination. The speeds of the faster and slower vehicles are 84 km/h and 70 km/h respectively. If the faster vehicle reaches the destination 30 minutes before the slower vehicle, find the distance from the starting point to the destination.

(A) 180 km (B) 240 km
(C) 210 km (D) 280 km
(E) 300 km

Q.343 Speed

Two towns Ajmer and Indore are 240 km apart. Naman leaves Indore for Ajmer at a speed of (x - 10) km/h. After 4 hours, Tushar leaves Indore for Ajmer at a speed of x km/h. If both reach Ajmer at the same time, find the time taken by Naman to reach Ajmer.

(A) 12 hours (B) 10 hours
(C) 15 hours (D) 16 hours
(E) 8 hours

Q.344 Speed

If a cyclist increases his speed from 36 km/h to 48 km/h, he takes 30 minutes less to cover a certain distance D. Find the time taken by the cyclist to cover 150% of D at his original speed.

(A) 150 minutes (B) 120 minutes
(C) 180 minutes (D) 210 minutes
(E) 240 minutes

Q.345 Speed

A van started from city A and reached city B at a speed of X km/h. A car started from city B and reached city A at a speed of (X + 30) km/h. If the car takes 3 hours less than the van to complete the journey, find the value of X + 8. The distance between city A and city B is 540 km.

(A) 56 (B) 70
(C) 65 (D) 75
(E) 68

Q.346 Speed

Two travelers are initially (D + 25) km apart and start moving towards each other at the same time. After 20 minutes, the distance between them becomes (2D − 70) km. If their speeds are 45 km/h and 30 km/h, find the difference between the distances traveled by them when they meet.

(A) 15 km (B) 21 km
(C) 24 km (D) 18 km
(E) 19 km

Q.347 Speed

Two buses start at the same time from two cities toward each other. The distance between the two cities is 360 km. The first bus can cover the whole distance in 6 hours, and the second bus can cover the same distance in 12 hours. After how many hours will they meet?

(A) 4 hours (B) 3 hours
(C) 5 hours (D) 6 hours
(E) 8 hours

Q.348 Speed

Two buses start at the same time from two cities that are 432 km apart and travel towards each other. They meet after 3 hours 36 minutes. If the speed of the first bus is 10 km/h less than the speed of the second bus, find the speed of the second bus.

(A) 60 km/h (B) 65 km/h
(C) 70 km/h (D) 75 km/h
(E) 80 km/h

Q.349 Speed

Three towns P, Q and R are located on the same straight highway, and Q lies between P and R. A bus travels from P to Q at 48 km/h, while a car travels from P to R at 72 km/h. If both vehicles take the same time to reach their destinations and the distance from P to Q is 144 km, find the distance between Q and R.

(A) 60 km (B) 84 km
(C) 96 km (D) 72 km
(E) 48 km

Q.350 Speed

A cab covers a certain distance at a speed of 72 km/h in t minutes. A bus covers the same distance at a speed of 48 km/h in (t + 50) minutes. Find the distance covered.

(A) 96 km (B) 120 km
(C) 144 km (D) 180 km
(E) 108 km

Q.351 Speed

A traveller has to cover a distance of 72 km. His usual speed is 12 km/h. To reach on time, he must increase his speed by X% and reduce his journey time by 4/9 of his usual journey time. Find the required percentage increase in speed.

(A) 70% (B) 90%
(C) 60% (D) 100%
(E) 80%

Q.352 Speed

Two buses P and Q start from two cities towards each other at the same time. The speed of bus Q is 18 km/h more than the speed of bus P, and the distance between the two cities is 648 km. If they meet after 6 hours, then find the speed of bus Q.

(A) 63 km/h (B) 60 km/h
(C) 66 km/h (D) 72 km/h
(E) 54 km/h

Q.353 Speed

Point B lies between points A and C. The distance from A to B is 240 km, and the distance from B to C is 360 km. A car takes the same time to cover A to B and B to C. If the speed of the car from A to B is 40 km/h, then find the difference between the average speed for the whole journey and the speed from B to C.

(A) 8 km/h (B) 12 km/h
(C) 10 km/h (D) 15 km/h
(E) 5 km/h

Q.354 Speed

Ayush covered one part of a trip in 40 minutes at a speed of 54 km/h and the remaining part in 45 minutes at a speed of 32 km/h. Find the total distance covered by him.

(A) 54 km (B) 58 km
(C) 60 km (D) 64 km
(E) 66 km

Q.355 Speed

Aman travels from his home to his office at a speed of 48 km/h and reaches 20 minutes before the scheduled time. If he travels at a speed of 32 km/h, he reaches 25 minutes late. Find the distance between his home and office. He leaves home at the same time and follows the same route in both cases.

(A) 72 km (B) 60 km
(C) 84 km (D) 96 km
(E) 68 km

Q.356 Speed

The length of a rectangular park is 30 meters and its breadth is (Y − 6) meters. The area of the park is 540 sq meters. A man walks along its boundary at a speed of Y m/min. Find the time taken by him to complete one round.

(A) 4 minutes (B) 6 minutes
(C) 8 minutes (D) 5 minutes
(E) 3 minutes

Q.357 Speed

The distances travelled by Ayush, Rhea and Hardik are in the ratio 4 : 3 : 7. The ratio of the speeds of Ayush and Rhea is 2 : 1. If Hardik's speed is 7 km/h and the sum of the time taken by Ayush and Rhea is equal to the time taken by Hardik, then find Ayush's speed.

(A) 10 km/h (B) 8 km/h
(C) 12 km/h (D) 14 km/h
(E) 6 km/h

Q.358 Speed

A delivery van travels from a warehouse to a market at a speed of 75 km/h and returns by the same route at a speed of 50 km/h. Find the average speed of the van for the entire journey.

(A) 52 km/h
(B) 60 km/h
(C) 56 km/h
(D) 65 km/h
(E) 58 km/h

Q.359 Speed

The speed of a bike is 50% more than the speed of a scooter. The scooter can cover X km in 6 hours, while the bike can cover (X + 120) km in 5 hours. If the bike increases its speed by 25%, then find the time taken by the bike to cover (X + 270) km.

(A) 5 hours
(B) 6 hours
(C) 4 hours
(D) 8 hours
(E) 7 hours

Chapter 04 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.340 Correct Option: (B)

• km/h
• Solution:
• Speed from Town P to Town Q = 80 km/h
• Time taken = 3 hours
• Distance from Town P to Town Q = 80 × 3 = 240 km
• Now speed is reduced by 40%.
• So, new speed = 60% of 80
• = 48 km/h
• Time from Town Q to Town R = 5 hours
• Distance from Town Q to Town R = 48 × 5 = 240 km
• Total distance = 240 + 240 = 480 km
• Total time = 3 + 5 = 8 hours
• Average speed = Total distance / Total time
• = 480 / 8
• = 60 km/h

Solution Q.341 Correct Option: (A)

• km/h
• Solution:
• Let the time taken by Sarthak be T hours.
• Sarthak takes 4 hours more than Nilesh.
• So, time taken by Nilesh = T - 4
• If Sarthak doubles his speed, his time becomes T/2.
• Now, he takes 2 hours less than Nilesh.
• So, T/2 = (T - 4) - 2
• T/2 = T - 6
• T = 2T - 12
• T = 12 hours
• Distance covered by Sarthak = 96 km
• Speed of Sarthak = Distance/Time
• Speed = 96/12 = 8 km/h

Solution Q.342 Correct Option: (C)

• km
• Solution:
• Let the distance covered by both vehicles = x km.
• Speed of faster vehicle = 84 km/h
• Speed of slower vehicle = 70 km/h
• Time taken by faster vehicle = x/84 hours
• Time taken by slower vehicle = x/70 hours
• The faster vehicle reaches 30 minutes earlier.
• 30 minutes = 1/2 hour
• So:
• x/70 - x/84 = 1/2
• LCM of 70 and 84 = 420
• 6x/420 - 5x/420 = 1/2
• x/420 = 1/2
• 2x = 420
• x = 210

Solution Q.343 Correct Option: (A)

• hours
• Solution:
• Let the time taken by Naman be t hours.
• Tushar starts 4 hours later, so time taken by Tushar = t - 4 hours.
• Distance = 240 km.
• Speed of Naman = 240/t
• Speed of Tushar = 240/(t - 4)
• According to the question, speed difference = x - (x - 10) = 10 km/h.
• So, 240/(t - 4) - 240/t = 10
• 240 × [t - (t - 4)]/[t(t - 4)] = 10
• 240 × 4/[t(t - 4)] = 10
• 960/[t(t - 4)] = 10
• t(t - 4) = 96
• Now, 12 × (12 - 4) = 12 × 8 = 96
• So, t = 12

Solution Q.344 Correct Option: (C)

• minutes
• Solution:
• Original speed = 36 km/h.
• Increased speed = 48 km/h.
• Time saved = 30 minutes = 1/2 hour.
• Let the distance be D km.
• Time at original speed = D/36.
• Time at increased speed = D/48.
• Difference in time = 1/2 hour.
• So, D/36 - D/48 = 1/2
• Take D common:
• D × (1/36 - 1/48) = 1/2
• 1/36 - 1/48 = 1/144
• So, D × 1/144 = 1/2
• D/144 = 1/2
• D = 144/2 = 72 km.
• Now, 150% of D = 150/100 × 72 = 108 km.
• Time taken at original speed = 108/36 = 3 hours.
• 3 hours = 180 minutes.
• Therefore, the required time is 180 minutes.

Solution Q.345 **Correct Option: (E)**

• Distance between city A and city B = 540 km
• Speed of van = X km/h
• Speed of car = (X + 30) km/h
• Time taken by van = 540/X hours
• Time taken by car = 540/(X + 30) hours
• Car takes 3 hours less than van.
• So:
• 540/X - 540/(X + 30) = 3
• 540[(X + 30) - X] / [X(X + 30)] = 3
• 540 × 30 / [X(X + 30)] = 3
• 16200 = 3X(X + 30)
• 5400 = X(X + 30)
• X² + 30X - 5400 = 0
• X² + 90X - 60X - 5400 = 0
• X(X + 90) - 60(X + 90) = 0
• (X + 90)(X - 60) = 0
• X = 60. The other root is -90 and is rejected because speed must be positive.
• Required value = X + 8
• Required value = 60 + 8 = 68

Solution Q.346 **Correct Option: (E)**

• km
• Solution:
• Initial distance between the two travelers = D + 25 km
• Distance after 20 minutes = 2D - 70 km
• Speed of first traveler = 45 km/h
• Speed of second traveler = 30 km/h
• Relative speed = 45 + 30 = 75 km/h
• Time = 20 minutes = 20/60 = 1/3 hour
• Distance covered together in 20 minutes = 75 × 1/3 = 25 km
• So, decrease in distance = 25 km
• Initial distance - Distance after 20 minutes = 25
• (D + 25) - (2D - 70) = 25
• D + 25 - 2D + 70 = 25
• 95 - D = 25
• D = 70
• Initial distance = D + 25 = 70 + 25 = 95 km
• Time taken to meet = Initial distance/Relative speed
• Time = 95/75 = 19/15 hours
• Difference in their speeds = 45 - 30 = 15 km/h
• Difference in distance traveled by them = 15 × 19/15 = 19 km

Solution Q.347 **Correct Option: (A)**

• hours
• Solution:
• Distance between the two cities = 360 km.
• The first bus covers 360 km in 6 hours.
• Speed of first bus = 360/6 = 60 km/h.
• The second bus covers 360 km in 12 hours.
• Speed of second bus = 360/12 = 30 km/h.
• They move toward each other.
• Relative speed = 60 + 30 = 90 km/h.
• Time taken to meet = Distance/Relative speed
• Time taken = 360/90
• Time taken = 4 hours.

Solution Q.348 **Correct Option: (B)**

• km/h
• Solution:
• Time = 3 hours 36 minutes = 3 + 36/60 = 3.6 hours
• Let the speed of the second bus = x km/h.
• Speed of the first bus = (x - 10) km/h
• Since both buses travel towards each other, their relative speed = x + x - 10 = (2x -
• 10) km/h
• Distance = relative speed × time
• 432 = (2x - 10) × 3.6
• 2x - 10 = 432/3.6
• 2x - 10 = 120
• 2x = 130
• x = 65
• Therefore, the speed of the second bus is 65 km/h.

Solution Q.349 **Correct Option: (D)**

• km
• Solution:
• Time taken by the bus from P to Q = 144 / 48 = 3 hours.
• The car also takes the same time, so time taken by car = 3 hours.
• Distance from P to R = 72 × 3 = 216 km.
• Distance between Q and R = PR - PQ = 216 - 144 = 72 km.

Solution Q.350 **Correct Option: (B)**

• km
• Solution:
• Both cover the same distance.
• Distance = speed × time
• Time of cab = t minutes = t/60 hours
• Time of bus = (t + 50) minutes = (t + 50)/60 hours
• So, 72 × t/60 = 48 × (t + 50)/60
• 72t = 48(t + 50)
• 72t = 48t + 2400
• 24t = 2400
• t = 100 minutes
• Distance covered = 72 × 100/60 = 120 km

Solution Q.351 **Correct Option: (E)**

• %
• Solution:
• Usual speed = 12 km/h.
• Distance = 72 km.
• Usual time = 72 / 12 = 6 hours.
• Journey time is reduced by 4/9.
• So, new time = 1 - 4/9 = 5/9 of the usual time.
• New time = 6 × 5/9 = 10/3 hours.
• New speed = distance / new time.
• New speed = 72 / (10/3) = 72 × 3/10 = 21.6 km/h.
• Increase in speed = 21.6 - 12 = 9.6 km/h.
• Percentage increase = 9.6/12 × 100 = 80%.
• Therefore, X = 80 and the required increase in speed is 80%.

Solution Q.352 **Correct Option: (A)**

• km/h
• Solution:
• Let the speed of bus P = x km/h
• Speed of bus Q = (x + 18) km/h
• They move towards each other, so their relative speed = x + x + 18 = 2x + 18 km/h
• Distance between the two cities = 648 km
• Time taken to meet = 6 hours
• Relative speed = Distance / Time
• Relative speed = 648 / 6 = 108 km/h
• So, 2x + 18 = 108
• 2x = 90
• x = 45
• Speed of bus Q = x + 18 = 45 + 18 = 63 km/h

Solution Q.353 **Correct Option: (C)**

• km/h
• Solution:
• Distance from A to B = 240 km.
• Speed from A to B = 40 km/h.
• Time taken from A to B = 240 / 40 = 6 hours.
• Time taken from B to C is also 6 hours.
• Distance from B to C = 360 km.
• Speed from B to C = 360 / 6 = 60 km/h.
• Total distance = 240 + 360 = 600 km.
• Total time = 6 + 6 = 12 hours.
• Average speed for the whole journey = 600 / 12 = 50 km/h.
• Required difference = 60 - 50 = 10 km/h.

Solution Q.354 **Correct Option: (C)**

• km
• Solution:
• Distance covered in first part = Speed × Time.
• Time = 40 minutes = 40/60 hour.
• Distance = 54 × 40/60 = 36 km.
• Time for second part = 45 minutes = 45/60 hour.
• Distance = 32 × 45/60 = 24 km.
• Total distance = 36 + 24 = 60 km.

Solution Q.355 **Correct Option: (A)**

• km
• Solution:
• At 48 km/h, Aman reaches 20 minutes early.
• At 32 km/h, he reaches 25 minutes late.
• Difference between the two times = 20 + 25 = 45 minutes = 3/4 hour.
• Let the distance be D km.
• Time difference = D/32 - D/48.
• D/32 - D/48 = 3/4.
• Difference of reciprocals = 1/32 - 1/48 = (48 - 32)/(32 × 48) = 16/1536 = 1/96.
• So, D × 1/96 = 3/4.
• D = 3/4 × 96 = 72 km.

Solution Q.356 **Correct Option: (A)**

• minutes
• Solution:
• Length of the park = 30 meters.
• Breadth of the park = (Y - 6) meters.
• Area = 540 sq meters.
• 30 × (Y - 6) = 540
• Y - 6 = 540 / 30 = 18
• Y = 24
• Breadth = 18 meters.
• Perimeter of the park = 2 × (Length + Breadth)
• = 2 × (30 + 18)
• = 2 × 48
• = 96 meters.
• Speed = Y m/min = 24 m/min.
• Time taken = Distance / Speed
• = 96 / 24
• = 4 minutes.

Solution Q.357 **Correct Option: (A)**

• km/h
• Solution:
• Distance travelled by Ayush : Rhea : Hardik = 4 : 3 : 7
• Let their distances be:
• Ayush = 4d
• Rhea = 3d
• Hardik = 7d
• Speed of Ayush : Speed of Rhea = 2 : 1
• Let Ayush's speed = 2x
• Let Rhea's speed = x
• Hardik's speed = 7 km/h
• Time taken by Ayush = 4d / (2x) = 2d/x
• Time taken by Rhea = 3d / x = 3d/x
• Sum of their times = 2d/x + 3d/x = 5d/x
• Time taken by Hardik = 7d / 7 = d
• According to the question:
• 5d/x = d
• x = 5
• Ayush's speed = 2x
• Ayush's speed = 2 × 5 = 10 km/h

Solution Q.358 **Correct Option: (B)**

• km/h
• Solution:
• For equal distances, average speed = 2xy/(x + y)
• Here, x = 75 km/h and y = 50 km/h
• Average speed = 2 × 75 × 50/(75 + 50)
• Average speed = 7500/125
• Average speed = 60 km/h
• Therefore, the average speed of the van is 60 km/h.

Solution Q.359 **Correct Option: (A)**

• hours
• Solution:
• Let distance covered by scooter = X km
• Time taken by scooter = 6 hours
• Speed of scooter = X/6 km/h
• Speed of bike is 50% more than scooter
• Speed of bike = 150% of X/6
• Speed of bike = 3/2 × X/6
• Speed of bike = X/4 km/h
• Bike covers (X + 120) km in 5 hours
• Speed of bike = (X + 120)/5 km/h
• So,
• (X + 120)/5 = X/4
• 4(X + 120) = 5X
• 4X + 480 = 5X
• X = 480
• Initial speed of bike = X/4
• Initial speed of bike = 480/4
• Initial speed of bike = 120 km/h
• Bike increases its speed by 25%
• New speed of bike = 125% of 120
• New speed of bike = 120 × 125/100
• New speed of bike = 150 km/h
• Required distance = X + 270
• Required distance = 480 + 270
• Required distance = 750 km
• Required time = 750/150
• Required time = 5 hours

CHAPTER 05: PROBLEMS ON TRAINS (RELATIVE SPEED & PLATFORMS)

36 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Train crossing a Pole / Man / Tree: Distance = Length of Train (L_T).
- Train crossing a Platform / Bridge / Tunnel: Distance = L_T + Length of Platform (L_P).
- Two trains crossing each other: Distance = L1 + L2.
- Time to cross = (L1 + L2) / Relative Speed

Q.360 Problems on Trains

Train M is 240 meters long and crosses a pole in 24 seconds. Train N crosses a pole in 36 seconds. If the ratio of the speeds of Train M and Train N is 5:3, find the time taken by Train M to overtake Train N.

(A) 96 seconds
(B) 108 seconds
(C) 120 seconds
(D) 126 seconds
(E) 114 seconds

Q.361 Problems on Trains

A train running at 54 km/h crosses a 180-meter-long platform in 22 seconds. Another train running at 72 km/h crosses the same platform in 14 seconds. Find the sum of the lengths of both trains.

(A) 250 meters
(B) 220 meters
(C) 270 meters
(D) 300 meters
(E) 230 meters

Q.362 Problems on Trains

A train is running at 72 km/h. It completely crosses a station platform in 42 seconds. Later, it overtakes a cyclist moving at 36 km/h in the same direction in 26 seconds. Find the length of the platform.

(A) 520 meters
(B) 620 meters
(C) 580 meters
(D) 640 meters
(E) 700 meters

Q.363 Problems on Trains

A train running at a speed of 108 km per hour crosses a platform in 20 seconds. The length of a bridge is three times the length of the platform. If the train crosses the bridge in 28 seconds, find the length of the platform.

(A) 90 meters
(B) 120 meters
(C) 150 meters
(D) 180 meters
(E) 210 meters

Q.364 Problems on Trains

Train P crosses a 180-meter-long platform in 24 seconds, and Train Q crosses a 204-meter-long platform in 24 seconds. Train Q is 420 meters long and completely overtakes Train P in 120 seconds while both run in the same direction. Find the time taken by Train P to cross a 60-meter-long bridge.

(A) 15 seconds
(B) 18 seconds
(C) 21 seconds
(D) 24 seconds
(E) 20 seconds

Q.365 Problems on Trains

A train of length 150 meters is running at a speed of 72 km/h. It crosses a bridge in 35 seconds. Find the length of the bridge.

(A) 520 meters
(B) 450 meters
(C) 600 meters
(D) 500 meters
(E) 550 meters

Q.366 Problems on Trains

A traveller completes a journey of 960 km in 9 hours. He covers the first part by train at 120 km/h and the remaining part by bus at 90 km/h. Find the sum of the terms in the simplified ratio of the time taken by train to the time taken by bus.

(A) 7
(B) 11
(C) 13
(D) 9
(E) 15

Q.367 Problems on Trains

Train L is 240 meters long and crosses a pole in 8 seconds. Train M is 360 meters long and crosses a 180-meter-long platform in 18 seconds. Find the time taken by both trains to cross each other while running in opposite directions.

(A) 12 seconds
(B) 8 seconds
(C) 15 seconds
(D) 9 seconds
(E) 10 seconds

Q.368 Problems on Trains

The length of Train P is (x + 40) meters and the length of Train Q is (x + 20) meters. Train P runs at 72 km/h and Train Q runs 50% faster than Train P. If both trains cross each other in 6 seconds while running in opposite directions, find the time taken by Train Q to cross a 100-meter-long platform.

(A) 6 seconds
(B) 10 seconds
(C) 12 seconds
(D) 9 seconds
(E) 8 seconds

Q.369 Problems on Trains

Train P takes 30 seconds to cross a signal pole and 24 seconds to cross a man walking at 5 m/s towards the train. Train Q travels 6 m/s faster than Train P and takes 20% more time than Train P to cross a signal pole. If the speed of Train Q had been 14 m/s less, it would take p seconds to cross a 204-meter-long platform. Find the value of p − 5.

(A) 84
(B) 90
(C) 88
(D) 96
(E) 92

Q.370 Problems on Trains

Train P, running at 72 km/h, crosses a platform whose length is half of its own length in 45 seconds. Find the time taken by Train P to cross Train Q, whose length is 5/3 times the length of Train P and which is running at 54 km/h in the same direction.

(A) 300 seconds
(B) 280 seconds
(C) 360 seconds
(D) 320 seconds
(E) 400 seconds

Q.371 Problems on Trains

The ratio of the speed of Train X to that of Train Y is 5 : 4. The lengths of Train X and Train Y are 900 meters and 1260 meters respectively. If both trains are running in opposite directions and cross each other in 24 seconds, find the speed of Train X.

(A) 40 m/s
(B) 50 m/s
(C) 60 m/s
(D) 45 m/s
(E) 55 m/s

Q.372 Problems on Trains

Train X, whose length is 60 meters less than the length of Train Y, overtakes Train Y in 100 seconds. The speeds of Train X and Train Y are 72 km/h and 54 km/h, respectively. Find the time taken by Train Y to cross a platform whose length is 100 meters more than the length of Train Y.

(A) 40 seconds
(B) 48 seconds
(C) 52 seconds
(D) 44 seconds
(E) 56 seconds

Q.373 Problems on Trains

The average age of two trainees, a trainer and a supervisor is 32 years. The ratio of the age of the supervisor to the age of the trainer is 7 : 4. If the supervisor is 18 years older than the trainer, find the average age of the two trainees.

(A) 31 years
(B) 28 years
(C) 34 years
(D) 30 years
(E) 32 years

Q.374 Problems on Trains

Two trains M and N cross each other in 60 seconds while running in the same direction. The length of train M is 180 meters, and the speeds of train M and train N are 45 km/h and 63 km/h respectively. Find the length of train N.

(A) 150 m
(B) 120 m
(C) 180 m
(D) 90 m
(E) 140 m

Q.375 Problems on Trains

The lengths of Train P and Train Q are in the ratio of 3:4. Their speeds are 54 km/h and 20 m/s respectively. The two trains cross each other while running in opposite directions in 24 seconds. If the length of a bridge is one-third of the length of Train P, find the time taken by Train Q to cross the bridge.

(A) 24 seconds
(B) 30 seconds
(C) 36 seconds
(D) 42 seconds
(E) 48 seconds

Q.376 Problems on Trains

A passenger train running at 72 km/h crosses a signal pole in 12 seconds. It crosses a goods train running in the opposite direction at 54 km/h in 18 seconds. Find the length of the goods train.

(A) 360 metres
(B) 390 metres
(C) 420 metres
(D) 330 metres
(E) 450 metres

Q.377 Problems on Trains

A 400-meter-long train P crosses a pole in 20 seconds. The speed of train Q is 75% of the speed of train P. If train Q crosses a 240-meter-long bridge in 32 seconds, then the length of train Q is what percentage less than the length of train P?

(A) 20%
(B) 25%
(C) 35%
(D) 40%
(E) 45%

Q.378 Problems on Trains

A train crosses a pole in 18 seconds and a platform of length p meters in 30 seconds. If the train's speed is decreased by 12 km/h, it crosses the same platform in 40 seconds. Find the value of p.

(A) 120
(B) 140
(C) 160
(D) 180
(E) 200

Q.379 Problems on Trains

A 480-metre-long train crosses a bridge whose length is 3/2 times the length of the train in 60 seconds. The same train crosses another train moving in the opposite direction in 32 seconds. If the speed of the second train is 18 km/h more than that of the first train, find the length of the second train.

(A) 900 metres
(B) 960 metres
(C) 840 metres
(D) 1020 metres
(E) 780 metres

Q.380 Problems on Trains

A train covers 960 km at an average running speed of 80 km/h, excluding halts. It halts for 5 minutes after every 160 km, except at the final destination. Find the total time, in minutes, taken by the train to cover the whole distance including halts.

(A) 730 minutes
(B) 750 minutes
(C) 720 minutes
(D) 745 minutes
(E) 765 minutes

Q.381 Problems on Trains

A train is running at 72 km/h and crosses an electric pole in 21 seconds. If the speed of the train is increased by 15 meters/second, find the new time, in seconds, taken to cross the same pole.

(A) 8 seconds
(B) 12 seconds
(C) 15 seconds
(D) 9 seconds
(E) 18 seconds

Q.382 Problems on Trains

Train R and Train S are each 300 meters long. Train R crosses a 300-meter-long platform 5 seconds faster than Train S. If the ratio of the speed of Train S to Train R is 4:5, find the speed of Train R in m/s.

(A) 24 m/s
(B) 36 m/s
(C) 25 m/s
(D) 30 m/s
(E) 20 m/s

Q.383 Problems on Trains

A 420-metre-long train crosses a signal pole in 28 seconds. After increasing its speed by 40%, it crosses a bridge in 44 seconds. Find the ratio of the length of the bridge to the length of the train.

(A) 6:5
(B) 5:6
(C) 4:7
(D) 3:5
(E) 7:6

Q.384 Problems on Trains

A train of length L meters crosses a 300-meter-long platform in 30 seconds. The same train crosses a tunnel whose length is three times the length of the train in 48 seconds. Find the speed of the train in km/h.

(A) 72 km/h
(B) 90 km/h
(C) 60 km/h
(D) 54 km/h
(E) 45 km/h

Q.385 Problems on Trains

Two trains of lengths 180 m and 420 m are running in the same direction at speeds of 72 km/h and 90 km/h respectively. Find the time taken by the faster train, coming from behind, to completely cross the slower train in minutes.

(A) 3
(B) 4
(C) 2
(D) 5
(E) 6

Q.386 Problems on Trains

The distance between two stations P and Q is 630 km. Two trains start at the same time from P and Q towards each other. The speed of Train B is 50% more than the speed of Train A. If Train A runs at its usual speed and Train B reduces its speed by 50%, they take 1.5 hours more to meet than usual. If the length of Train A is 120 metres, find the time taken by Train A to cross a 180-metre-long platform.

(A) 15 seconds
(B) 18 seconds
(C) 20 seconds
(D) 24 seconds
(E) 30 seconds

Q.387 Problems on Trains

Train M runs at a speed of 20 m/s and crosses a signal pole in 18 seconds. It crosses train N, running in the opposite direction, in 20 seconds. If the length of train N is 1.5 times the length of train M, then find the speed of train N.

(A) 15 m/s
(B) 20 m/s
(C) 25 m/s
(D) 30 m/s
(E) 35 m/s

Q.388 Problems on Trains

Train P, running at a speed of s m/s, overtakes train Q, which is running at a speed of 72 km/h, in 18 seconds. The ratio of the length of train P to that of train Q is 5 : 4. If train P crosses an 80-meter-long platform in 7 seconds, find the value of s.

(A) 45
(B) 35
(C) 30
(D) 50
(E) 40

Q.389 Problems on Trains

The length of a platform is 25% of the length of a train. The train crosses the platform at a speed of 30 meters/second in 20 seconds. Find the length of the train.

(A) 420 meters
(B) 450 meters
(C) 480 meters
(D) 500 meters
(E) 540 meters

Q.390 Problems on Trains

A 550-metre-long platform lies beside two parallel tracks. Train R, 250 metres long, is stationary on one track immediately beyond the far end of the platform, with its nearer end aligned with that end of the platform. Train S is 75 metres longer than R and travels on the other track at 90 km/h. Starting when the front of S reaches the near end of the platform, it takes (X + 17) seconds for the rear of S to pass the farthest end of R. Find X.

(A) 32
(B) 28
(C) 35
(D) 25
(E) 40

Q.391 Problems on Trains

A train running at a speed of 126 km/h crosses a platform in 14 seconds. If the length of the platform is 40% of the length of the train, find the length of the platform.

(A) 120 meters
(B) 160 meters
(C) 180 meters
(D) 150 meters
(E) 140 meters

Q.392 Problems on Trains

The ratio between the speed of a train and a bus is 5 : 3, respectively. A car covers 360 km in 6 hours. If the speed of the car is 75% of the speed of the train, then find the speed of the bus.

(A) 60 km/h
(B) 54 km/h
(C) 42 km/h
(D) 45 km/h
(E) 48 km/h

Q.393 Problems on Trains

A 240-meter-long train crosses a 360-meter-long platform in 20 seconds. If the speed of the train is increased by 20%, then find the time taken by the train to cross a man jogging at 14.4 km/h in the opposite direction.

(A) 8 seconds
(B) 10 seconds
(C) 5 seconds
(D) 6 seconds
(E) 12 seconds

Q.394 Problems on Trains

The average weight of a training group is 50 kg. 8 people, each weighing 70 kg, join the group and 8 people, each weighing 30 kg, leave the group. If the initial number of people in the group was 40, then find the percentage increase in the average weight.

(A) 12% (B) 20%
(C) 10% (D) 8%
(E) 16%

Q.395 Problems on Trains

The speeds of two trains M and N are (x + 12) km/h and (2x − 6) km/h, respectively. Train M starts from point A at 7 AM towards point B, and train N starts from point B at 8 AM towards point A. If they meet at 11 AM and the distance between A and B is 390 km, then find the speed of train N.

(A) 54 km/h (B) 60 km/h
(C) 72 km/h (D) 66 km/h
(E) 78 km/h

Chapter 05 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.360 Correct Option: (E)

• seconds
• Solution:
• Length of Train M = 240 meters
• Time taken by Train M to cross a pole = 24 seconds
• So, speed of Train M = 240 / 24
• = 10 m/s
• Given:
• Speed of Train M : Speed of Train N = 5 : 3
• If 5 parts = 10 m/s,
• then 1 part = 2 m/s
• So, speed of Train N = 3 × 2 = 6 m/s
• Train N crosses a pole in 36 seconds.
• So, length of Train N = 6 × 36
• = 216 meters
• When Train M overtakes Train N, both move in the same direction.
• Relative speed = 10 - 6
• = 4 m/s
• Total distance to be covered while overtaking
• = Length of Train M + Length of Train N
• = 240 + 216
• = 456 meters
• Time taken = 456 / 4
• = 114 seconds

Solution Q.361 Correct Option: (A)

• meters
• Solution:
• First train's speed = 54 km/h
• To convert km/h into meter/second, multiply by:
• 5/18
• Speed of first train = 54 × 5/18
• = 15 meters per second
• The first train crosses a 180-meter platform in 22 seconds.
• When a train crosses a platform, it covers:
• Length of train + Length of platform
• Total distance covered by first train = Speed × Time
• Total distance covered by first train = 15 × 22 = 330 meters
• Length of first train = 330 - 180 = 150 meters
• Now, second train's speed = 72 km/h
• Speed of second train = 72 × 5/18
• = 20 meters per second
• The second train crosses the same 180-meter platform in 14 seconds.
• Total distance covered by second train = Speed × Time
• Total distance covered by second train = 20 × 14 = 280 meters
• Length of second train = 280 - 180 = 100 meters
• Required sum of lengths = Length of first train + Length of second train
• Required sum of lengths = 150 + 100 = 250 meters

Solution Q.362 Correct Option: (C)

• meters
• Solution:
• This is a train and platform question. First, find the length of the train by using the
• information about overtaking the cyclist.
• Train speed = 72 km/h.
• Cyclist speed = 36 km/h.
• When the train overtakes the cyclist in the same direction, relative speed = train speed
• - cyclist speed.
• Relative speed = 72 - 36 = 36 km/h.
• Now convert 36 km/h into m/s.
• Since 18 km/h = 5 m/s, therefore 36 km/h = 10 m/s.
• The train overtakes the cyclist in 26 seconds.
• Distance covered while overtaking the cyclist = length of train.
• Length of train = 10 × 26 = 260 meters.
• Now the train crosses the platform in 42 seconds.
• Train speed = 72 km/h = 20 m/s.
• Distance covered in 42 seconds = 20 × 42 = 840 meters.
• When a train crosses a platform, distance covered = length of train + length of
• platform.
• So, length of train + length of platform = 840 meters.
• Length of platform = 840 - 260 = 580 meters.

Solution Q.363 Correct Option: (B)

• meters
• Solution:
• Speed of the train = 108 km per hour.
• Convert the speed into meters per second.
• Speed = 108 × 5/18 = 30 meters per second.
• When a train crosses a platform, it covers the length of train + length of platform.
• Time taken to cross the platform = 20 seconds.
• Distance covered in 20 seconds = 30 × 20 = 600 meters.
• So, length of train + length of platform = 600.
• Let the length of the platform be P meters.
• The length of the bridge is three times the length of the platform.
• So, length of bridge = 3P meters.
• Let the length of the train be T meters.
• From crossing the platform, T + P = 600.
• Now, the train crosses the bridge in 28 seconds.
• Distance covered in 28 seconds = 30 × 28 = 840 meters.
• When a train crosses a bridge, it covers the length of train + length of bridge.
• So, T + 3P = 840.
• Now subtract the first condition from the second condition.
• T + 3P = 840
• T + P = 600
• After subtraction, 2P = 240. P = 240/2 = 120.

Solution Q.364 **Correct Option: (B)**

• seconds
• Solution:
• Let the length of Train P be L meters.
• Let the speed of Train P be S m/s.
• Train P crosses a 180-meter platform in 24 seconds.
• So, L + 180 = 24S.
• Train Q is 420 meters long and crosses a 204-meter platform in 24 seconds.
• Speed of Train Q = (420 + 204)/24
• Speed of Train Q = 624/24 = 26 m/s.
• Train Q crosses Train P in 120 seconds in the same direction.
• Relative speed = 26 - S.
• Distance covered while crossing = L + 420.
• So, (L + 420)/(26 - S) = 120.
• Now, L + 420 = 120(26 - S).
• From the first condition, L = 24S - 180.
• Put L = 24S - 180:
• 24S - 180 + 420 = 120(26 - S)
• 24S + 240 = 3120 - 120S
• 144S = 2880
• S = 2880/144 = 20 m/s.
• Now, L = 24S - 180.
• L = 24 × 20 - 180 = 300 meters.
• Train P has to cross a 60-meter bridge.
• Distance covered = 300 + 60 = 360 meters.
• Time taken = 360/20 = 18 seconds.

Solution Q.365 **Correct Option: (E)**

• meters
• Solution:
• Speed of train = 72 km/h
• Convert km/h into m/s.
• Speed = 72 × 5/18 = 20 m/s
• Time taken to cross the bridge = 35 seconds
• Distance covered while crossing the bridge = Speed × Time
• Distance = 20 × 35 = 700 meters
• When a train crosses a bridge, total distance covered = Length of train + Length of
• bridge
• Length of train = 150 meters
• So, length of bridge = 700 - 150 = 550 meters

Solution Q.366 **Correct Option: (D)**

• Let the time taken by train = x hours.
• Then the time taken by bus = 9 - x hours.
• Distance = Speed × Time.
• Distance covered by train = 120x.
• Distance covered by bus = 90(9 - x).
• Total distance = 960 km.
• So,
• 120x + 90(9 - x) = 960
• 120x + 810 - 90x = 960
• 30x = 960 - 810
• 30x = 150
• x = 150/30
• x = 5
• Time taken by train = 5 hours.
• Time taken by bus = 9 - 5 = 4 hours.
• Ratio of time taken by train to bus = 5 : 4.
• Sum of the terms = 5 + 4 = 9.
• Therefore, the required answer is 9.

Solution Q.367 **Correct Option: (E)**

• seconds
• Solution:
• Train L crosses a pole, so the distance covered is equal to its own length.
• Speed of Train L = 240/8 = 30 m/s.
• Train M crosses a 180-meter platform.
• Distance covered by Train M = 360 + 180 = 540 meters.
• Time taken by Train M = 18 seconds.
• Speed of Train M = 540/18 = 30 m/s.
• When two trains run in opposite directions, relative speed = sum of speeds.
• Relative speed = 30 + 30 = 60 m/s.
• Total distance to be covered = length of Train L + length of Train M
• Total distance = 240 + 360 = 600 meters.
• Time taken = 600/60
• Time taken = 10 seconds

Solution Q.368 **Correct Option: (E)**

• seconds
• Solution:
• Speed of Train P = 72 km/h.
• Speed of Train P in m/s = 72 × 5/18 = 20 m/s.
• Train Q is 50% faster than Train P.
• Speed of Train Q = 150% of 20 = 150/100 × 20 = 30 m/s.
• Both trains are moving in opposite directions.
• Relative speed = 20 + 30 = 50 m/s.
• Time taken to cross each other = 6 seconds.
• Total distance covered while crossing = 50 × 6 = 300 meters.
• Length of Train P + Length of Train Q = 300.
• (x + 40) + (x + 20) = 300
• 2x + 60 = 300
• 2x = 240
• x = 240/2
• x = 120
• Length of Train Q = x + 20 = 120 + 20 = 140 meters.
• Train Q has to cross a 100-meter platform.
• Total distance covered = 140 + 100 = 240 meters.
• Time taken by Train Q = 240/30 = 8 seconds.

Solution Q.369 **Correct Option: (B)**

• Let the speed of Train P be s m/s.
• Train P crosses a signal pole in 30 seconds.
• So, length of Train P = 30s
• Train P crosses a man walking towards it at 5 m/s in 24 seconds.
• Relative speed = s + 5
• So, length of Train P = 24(s + 5)
• Now:
• 30s = 24(s + 5)
• 30s = 24s + 120
• 30s - 24s = 120
• 6s = 120
• s = 120/6
• s = 20 m/s
• Length of Train P = 30 × 20 = 600 meters
• Speed of Train Q = 20 + 6 = 26 m/s
• Train Q takes 20% more time than Train P to cross a signal pole.
• Time taken by Train Q to cross a signal pole = 30 × 120/100
• Time taken by Train Q = 36 seconds
• Length of Train Q = 26 × 36 = 936 meters
• Reduced speed of Train Q = 26 - 14 = 12 m/s
• Platform length = 204 meters
• Total distance covered while crossing the platform = 936 + 204 = 1140 meters
• p = 1140/12
• p = 95 seconds
• Required value = p - 5
• Required value = 95 - 5 = 90

Solution Q.370 **Correct Option: (D)**

• seconds
• Solution:
• Speed of Train P = 72 km/h
• Convert km/h into m/s.
• Speed of Train P = 72 × 5/18 = 20 m/s
• Let the length of Train P be L meters.
• Length of platform = L/2
• Distance covered while crossing the platform = L + L/2 = 3L/2
• Train P crosses the platform in 45 seconds.
• So, distance = Speed × Time
• 3L/2 = 20 × 45
• 3L/2 = 900
• L = 900 × 2/3 = 600 meters
• Length of Train P = 600 meters
• Length of Train Q = 5/3 of Train P
• Length of Train Q = 600 × 5/3 = 1000 meters
• Speed of Train Q = 54 km/h
• Speed of Train Q = 54 × 5/18 = 15 m/s
• Both trains are moving in the same direction.
• Relative speed = 20 - 15 = 5 m/s
• Total distance to cross each other = 600 + 1000 = 1600 meters
• Time taken = Distance/Relative speed
• Time taken = 1600/5 = 320 seconds

Solution Q.371 **Correct Option: (B)**

• m/s
• Solution:
• Ratio (speeds of Train X : Train Y) = 5 : 4.
• Let the speeds of Train X and Train Y be 5x and 4x respectively.
• Length of Train X = 900 meters.
• Length of Train Y = 1260 meters.
• When two trains cross each other in opposite directions, total distance covered = sum
• of their lengths.
• Total distance = 900 + 1260 = 2160 meters.
• Relative speed = 5x + 4x = 9x.
• Time taken to cross each other = 24 seconds.
• So,
• 9x × 24 = 2160
• 216x = 2160
• x = 2160/216
• x = 10.
• Speed of Train X = 5x = 5 × 10 = 50 m/s.

Solution Q.372 **Correct Option: (D)**

• seconds
• Solution:
• Speed of Train X = 72 km/h.
• Convert into m/s:
• 72 × 5/18 = 20 m/s.
• Speed of Train Y = 54 km/h.
• Convert into m/s:
• 54 × 5/18 = 15 m/s.
• Train X overtakes Train Y, so both trains move in the same direction.
• Relative speed = 20 - 15 = 5 m/s.
• Let the length of Train Y be L meters.
• Length of Train X = L - 60 meters.
• Distance covered while overtaking = sum of lengths of both trains.
• Distance = L + (L - 60)
• Distance = 2L - 60.
• Time taken to overtake = 100 seconds.
• So, 2L - 60 = 5 × 100
• 2L - 60 = 500
• 2L = 560
• L = 560/2
• L = 280 meters.
• Length of Train Y = 280 meters.
• Length of platform = 280 + 100 = 380 meters.
• Distance covered by Train Y to cross the platform = 280 + 380
• Distance = 660 meters.
• Speed of Train Y = 15 m/s.
• Time taken = 660/15
• Time taken = 44 seconds.

Solution Q.373 **Correct Option: (A)**

• years
• Solution:
• Average age of four persons = 32 years.
• Total age of all four persons = 32 × 4 = 128 years.
• Ratio of supervisor's age to trainer's age = 7 : 4.
• Difference in ratio parts = 7 - 4 = 3 parts.
• Given difference = 18 years.
• So,
• 3 parts = 18
• 1 part = 18/3 = 6 years.
• Supervisor's age = 7 × 6 = 42 years.
• Trainer's age = 4 × 6 = 24 years.
• Combined age of supervisor and trainer = 42 + 24 = 66 years.
• Combined age of two trainees = 128 - 66 = 62 years.
• Average age of two trainees = 62/2 = 31 years.

Solution Q.374 **Correct Option: (B)**

• m
• Solution:
• Speed of train M = 45 km/h
• Speed of train N = 63 km/h
• Since both trains are running in the same direction, relative speed = 63 - 45 = 18 km/h
• Convert 18 km/h into m/s:
• 18 × 5/18 = 5 m/s
• Time taken to cross each other = 60 seconds
• Total distance covered while crossing = Relative speed × Time
• Total distance = 5 × 60 = 300 meters
• Total distance while crossing = Length of train M + Length of train N
• 300 = 180 + Length of train N
• Length of train N = 300 - 180 = 120 meters

Solution Q.375 **Correct Option: (B)**

• seconds
• Solution:
• Speed of Train P = 54 km/h.
• Convert it into m/s:
• 54 × 5/18 = 15 m/s.
• Speed of Train Q = 20 m/s.
• Since both trains are running in opposite directions, relative speed = 15 + 20 = 35 m/s.
• Time taken to cross each other = 24 seconds.
• So, sum of lengths of both trains = 35 × 24 = 840 meters.
• Ratio (lengths of Train P : Train Q) = 3:4.
• Total parts = 3 + 4 = 7 parts.
• 7 parts = 840 meters.
• 1 part = 840/7 = 120 meters.
• Length of Train P = 3 × 120 = 360 meters.
• Length of Train Q = 4 × 120 = 480 meters.
• Length of bridge = 1/3 of length of Train P.
• Length of bridge = 360 × 1/3 = 120 meters.
• Distance covered by Train Q to cross the bridge = length of Train Q + length of
• bridge.
• Distance = 480 + 120 = 600 meters.
• Time taken by Train Q = 600/20 = 30 seconds.

Solution Q.376 **Correct Option: (B)**

• metres
• Solution:
• Speed of passenger train = 72 km/h
• Convert into m/s:
• 72 × 5/18 = 20 m/s
• Time taken to cross a pole = 12 seconds
• Length of passenger train = 20 × 12 = 240 metres
• Speed of goods train = 54 km/h
• Relative speed in opposite direction = 72 + 54 = 126 km/h
• Convert into m/s:
• 126 × 5/18 = 35 m/s
• Time taken to cross each other = 18 seconds
• Total length of both trains = 35 × 18 = 630 metres
• Length of goods train = 630 - 240 = 390 metres

Solution Q.377 **Correct Option: (D)**

• %
• Solution:
• Train P crosses a pole, so distance covered = length of train P.
• Speed of train P = 400 / 20 = 20 m/s.
• Speed of train Q = 75% of 20 = 15 m/s.
• Train Q crosses a 240-meter bridge in 32 seconds.
• Total distance covered by train Q = 15 × 32 = 480 meters.
• Length of train Q = 480 - 240 = 240 meters.
• Difference in lengths = 400 - 240 = 160 meters.
• Percentage less = 160 × 100 / 400 = 40%.
• So, train Q is 40% shorter than train P.

Solution Q.378 **Correct Option: (C)**

• Let the original speed of the train be v m/s.
• Train crosses a pole in 18 seconds, so length of train = 18v meters.
• Train crosses a platform in 30 seconds, so total distance = 30v meters.
• Platform length = 30v - 18v = 12v meters.
• Speed is decreased by 12 km/h.
• 12 km/h = 12 × 5/18 = 10/3 m/s.
• New speed = v - 10/3 m/s.
• The train crosses the same platform in 40 seconds.
• Distance to cross platform = train length + platform length = 30v.
• So, 40 × (v - 10/3) = 30v.
• 40v - 400/3 = 30v.
• 10v = 400/3.
• v = 40/3 m/s.
• Platform length = 12v = 12 × 40/3 = 160 meters.

Solution Q.379 **Correct Option: (B)**

• metres
• Solution:
• Length of first train = 480 metres
• Length of bridge = 3/2 × 480 = 720 metres
• Total distance covered while crossing the bridge = 480 + 720 = 1200 metres
• Time taken = 60 seconds
• Speed of first train = 1200/60 = 20 m/s
• Speed in km/h = 20 × 18/5 = 72 km/h
• Speed of second train = 72 + 18 = 90 km/h
• Speed of second train in m/s = 90 × 5/18 = 25 m/s
• Since both trains move in opposite directions, relative speed = 20 + 25 = 45 m/s
• Total distance covered in 32 seconds = 45 × 32 = 1440 metres
• Length of second train = 1440 - 480 = 960 metres

Solution Q.380 **Correct Option: (D)**

• minutes
• Solution:
• Running time without halt = Distance / Speed
• Running time = 960 / 80 = 12 hours
• 12 hours = 12 × 60 = 720 minutes
• The train halts after every 160 km.
• Number of 160 km parts in 960 km = 960 / 160 = 6
• Since the train does not halt after reaching the final destination, number of halts = 6 -
• 1 = 5
• Total halt time = 5 × 5 = 25 minutes
• Total time including halts = 720 + 25 = 745 minutes

Solution Q.381 **Correct Option: (B)**

• seconds
• Solution:
• First convert 72 km/h into meters/second.
• 72 km/h = 72 × 5/18 = 20 meters/second
• When a train crosses a pole, it covers its own length.
• Length of train = Speed × Time
• Length of train = 20 × 21 = 420 meters
• New speed = 20 + 15 = 35 meters/second
• New time taken = Length / New speed
• New time taken = 420 / 35 = 12 seconds

Solution Q.382 **Correct Option: (D)**

• m/s
• Solution:
• Distance covered by each train while crossing the platform = length of train + length
• of platform.
• Distance covered by Train R = 300 + 300 = 600 meters.
• Distance covered by Train S = 300 + 300 = 600 meters.
• Let the speeds of Train S and Train R be 4x and 5x m/s respectively.
• Time taken by Train S = 600 / (4x) = 150/x seconds.
• Time taken by Train R = 600 / (5x) = 120/x seconds.
• Train R takes 5 seconds less than Train S.
• So, 150/x - 120/x = 5.
• 30/x = 5.
• x = 6.
• Speed of Train R = 5x = 5 × 6 = 30 m/s.

Solution Q.383 **Correct Option: (A)**

• :5
• Solution:
• Length of train = 420 metres
• Time taken to cross the pole = 28 seconds
• Speed of train = 420 / 28 = 15 m/s
• After increasing speed by 40%:
• New speed = 15 × 140/100 = 21 m/s
• In 44 seconds, distance covered = 21 × 44 = 924 metres
• While crossing a bridge:
• Distance covered = Length of train + Length of bridge
• So, length of bridge = 924 - 420 = 504 metres
• Required ratio = Length of bridge : Length of train
• Required ratio = 504 : 420 = 6 : 5

Solution Q.384 **Correct Option: (C)**

• km/h
• Solution:
• Let the length of the train be L meters and speed be S m/s.
• When the train crosses a 300-meter platform, total distance covered = L + 300.
• So, S = (L + 300) / 30.
• The tunnel length is three times the train length, so tunnel length = 3L.
• When the train crosses the tunnel, total distance covered = L + 3L = 4L.
• So, S = 4L / 48.
• Since speed is the same, (L + 300) / 30 = 4L / 48.
• 48(L + 300) = 120L.
• 48L + 14400 = 120L.
• 72L = 14400.
• L = 200 meters.
• Now, speed = (L + 300) / 30 = (200 + 300) / 30 = 500 / 30 = 50/3 m/s.
• Speed in km/h = 50/3 × 18/5 = 60 km/h.

Solution Q.385 **Correct Option: (C)**

• Since both trains are running in the same direction, relative speed will be the
• difference of their speeds.
• Relative speed = 90 - 72
• = 18 km/h
• Convert 18 km/h into m/s:
• 18 km/h = 5 m/s
• Total distance to be covered while crossing
• = Sum of lengths of both trains
• = 180 + 420
• = 600 metres
• Time taken = Distance / Relative speed
• = 600 / 5
• = 120 seconds
• Now convert seconds into minutes:
• 120 seconds = 2 minutes
• Hence, the faster train will completely cross the slower train in 2 minutes.

Solution Q.386 **Correct Option: (A)**

• seconds
• Solution:
• Let the usual speed of Train A = x kmph.
• Speed of Train B = 50% more than Train A = 1.5x kmph.
• Usual relative speed = x + 1.5x = 2.5x kmph.
• When Train B reduces its speed by 50%, its new speed = 0.75x kmph.
• New relative speed = x + 0.75x = 1.75x kmph.
• Distance between the stations = 630 km.
• Usual meeting time = 630 / (2.5x).
• New meeting time = 630 / (1.75x).
• The new time is 1.5 hours more.
• So,
• 630 / (1.75x) - 630 / (2.5x) = 1.5
• 360 / x - 252 / x = 1.5
• 108 / x = 1.5
• x = 72.
• So speed of Train A = 72 kmph.
• Convert 72 kmph into metres per second:
• 72 × 5 / 18 = 20 metres per second.
• To cross the platform, Train A covers:
• Length of train + length of platform = 120 + 180 = 300 metres.
• Time taken = 300 / 20 = 15 seconds.

Solution Q.387 **Correct Option: (C)**

• m/s
• Solution:
• Speed of train M = 20 m/s.
• Time taken by train M to cross a signal pole = 18 seconds.
• Length of train M = 20 × 18 = 360 meters. Length of train N = 1.5 × 360 = 540
• meters. Total length of both trains = 360 + 540 = 900 meters.
• They cross each other in opposite directions in 20 seconds.
• So, relative speed = 900 / 20 = 45 m/s. Since they are moving in opposite directions,
• relative speed = speed of M + speed of N.
• Speed of train N = 45 - 20 = 25 m/s.

Solution Q.388 **Correct Option: (E)**

• Speed of train Q = 72 km/h
• = 72 × 5/18 = 20 m/s.
• Let the lengths of trains P and Q be 5a meters and 4a meters, respectively.
• When train P overtakes train Q, relative speed = (s - 20) m/s.
• Total distance covered while overtaking = 5a + 4a = 9a.
• So,
• 9a = (s - 20) × 18
• a = 2(s - 20)
• Train P crosses an 80-meter-long platform in 7 seconds.
• Distance covered = Length of train P + Length of platform
• = 5a + 80
• So,
• 5a + 80 = 7s
• Put a = 2(s - 20):
• 5 × 2(s - 20) + 80 = 7s
• 10s - 200 + 80 = 7s
• 10s - 120 = 7s
• 3s = 120
• s = 40

Solution Q.389 **Correct Option: (C)**

• meters
• Solution:
• Let the length of the train = x meters.
• Length of platform = 25% of x = x/4 meters.
• Total distance covered while crossing the platform = Train length + Platform length.
• Total distance = x + x/4 = 5x/4.
• Speed of train = 30 meters/second.
• Time taken = 20 seconds.
• Distance covered = 30 × 20 = 600 meters.
• So, 5x/4 = 600.
• x = 600 × 4/5 = 480.
• Therefore, the length of the train is 480 meters.

Solution Q.390 **Correct Option: (B)**

• Length of Train R = 250 meters.
• Train S is 75 meters longer than Train R.
• Length of Train S = 250 + 75 = 325 meters
• Length of platform = 550 meters
• Train S crosses the platform and Train R completely.
• Total distance covered by Train S = length of platform + length of Train R + length of
• Train S
• Total distance = 550 + 250 + 325 = 1125 meters
• Speed of Train S = 90 km/h
• Convert speed into m/s:
• 90 × 5/18 = 25 m/s
• Time taken = Distance/Speed
• Time taken = 1125/25 = 45 seconds
• Given time = X + 17 seconds
• So, X + 17 = 45
• X = 45 - 17
• X = 28

Solution Q.391 **Correct Option: (E)**

• meters
• Solution:
• Speed of the train = 126 km/h.
• Convert km/h into m/s.
• Speed = 126 × 5/18 = 35 m/s.
• The train crosses the platform in 14 seconds.
• Distance covered while crossing the platform = 35 × 14 = 490 meters.
• When a train crosses a platform, distance covered = length of train + length of
• platform.
• Let the length of the train = x meters.
• Length of platform = 40% of x.
• Length of platform = 40x/100 = 2x/5.
• So,
• x + 2x/5 = 490
• 7x/5 = 490
• x = 490 × 5/7
• x = 350
• Length of the train = 350 meters.
• Length of the platform = 2x/5.
• Length of the platform = 2 × 350/5 = 140 meters.

Solution Q.392 **Correct Option: (E)**

• km/h
• Solution:
• Speed of car = Distance / Time
• = 360 / 6
• = 60 km/h
• Speed of car is 75% of speed of train.
• So, speed of train = 60 × 100/75
• = 80 km/h
• Ratio of speed of train to bus = 5 : 3
• 5 parts = 80
• 1 part = 80/5 = 16
• Speed of bus = 3 × 16
• = 48 km/h

Solution Q.393 **Correct Option: (D)**

• seconds
• Solution:
• Length of train = 240 meters
• Length of platform = 360 meters
• Total distance covered while crossing the platform = 240 + 360 = 600 meters
• Time taken = 20 seconds
• Speed of train = 600/20 = 30 m/s
• Increased speed of train = 30 × 120/100
• = 36 m/s
• Speed of man = 14.4 km/h
• = 14.4 × 5/18
• = 4 m/s
• Since both are moving in opposite directions, relative speed = 36 + 4 = 40 m/s
• Distance covered to cross the man = Length of train = 240 meters
• Required time = 240/40
• = 6 seconds

Solution Q.394 **Correct Option: (E)**

• %
• Solution:
• Initial number of people = 40.
• Initial average weight = 50 kg.
• Initial total weight = 40 × 50 = 2,000 kg.
• Weight added by 8 new people = 8 × 70 = 560 kg.
• Weight removed by 8 people = 8 × 30 = 240 kg.
• New total weight = 2,000 + 560 - 240 = 2,320 kg.
• Number of people remains the same = 40.
• New average weight = 2,320/40 = 58 kg.
• Increase in average weight = 58 - 50 = 8 kg.
• Percentage increase = 8/50 × 100 = 16%.

Solution Q.395 **Correct Option: (D)**

• km/h
• Solution:
• Speed of train M = (x + 12) km/h.
• Speed of train N = (2x - 6) km/h.
• Train M starts at 7 AM and meets train N at 11 AM.
• Time taken by train M = 4 hours.
• Train N starts at 8 AM and meets train M at 11 AM.
• Time taken by train N = 3 hours.
• Distance covered by train M + Distance covered by train N = 390 km.
• 4(x + 12) + 3(2x - 6) = 390
• 4x + 48 + 6x - 18 = 390
• 10x + 30 = 390
• 10x = 360
• x = 36
• Speed of train N = 2x - 6 = 2 × 36 - 6 = 72 - 6 = 66 km/h.

CHAPTER 06: BOATS & STREAMS (UPSTREAM, DOWNSTREAM & RIVER FLOW)

31 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Speed of Boat in Still Water (u) = (Downstream Speed + Upstream Speed) / 2
- Speed of Current/Stream (v) = (Downstream Speed - Upstream Speed) / 2
- Downstream Speed (D) = u + v | Upstream Speed (U) = u - v

Q.396 Boats & Streams

A speedboat covers 84 km downstream in 120 minutes. If the speed of the current is 40% of the speed of the speedboat in still water, find the time taken by the speedboat to cover 126 km upstream.

(A) 7 hours
(B) 5 hours
(C) 8 hours
(D) 6 hours
(E) 9 hours

Q.397 Boats & Streams

A steamer travels 216 km downstream in 180 minutes. The speed of the current is 20% of the speed of the steamer in still water. Find the time taken by the steamer to travel 240 km upstream.

(A) 4 hours
(B) 6 hours
(C) 7 hours
(D) 3 hours
(E) 5 hours

Q.398 Boats & Streams

A boat covers 150 km upstream in 10 hours and can cover X km downstream in 8 hours. If the speed of the boat in still water and the speed of the current are in the ratio of 4:1, respectively, find X.

(A) 180
(B) 200
(C) 240
(D) 160
(E) 220

Q.399 Boats & Streams

A boat covers 135 km upstream in 9 hours and can cover X km downstream in 8 hours. If the speed of the boat in still water and the speed of the current are in the ratio of 5:2, respectively, find X.

(A) 240 km
(B) 300 km
(C) 320 km
(D) 280 km
(E) 260 km

Q.400 Boats & Streams

The ratio of the downstream speed of a boat to the speed of the stream is 6 : 1. The difference between the speed of the boat in still water and its upstream speed is 5 km/h. If the speed of the stream is increased by 4 km/h, find the time taken by the boat to travel 192 km upstream.

(A) 14 hours
(B) 12 hours
(C) 16 hours
(D) 10 hours
(E) 18 hours

Q.401 Boats & Streams

A boat covers 28 km downstream in 40 minutes. The speed of the stream is 40% of the speed of the boat in still water. Find the sum of the terms of the simplest ratio of the time taken by the boat to cover 72 km upstream to the time taken to cover 210 km downstream.

(A) 7
(B) 8
(C) 9
(D) 10
(E) 11

Q.402 Boats & Streams

When the upstream speed of a boat is reduced by 20%, the resulting speed becomes equal to the speed of the current. If the speed of the boat in still water is 90 km/h, find the downstream speed of the boat.

(A) 120 km/h
(B) 130 km/h
(C) 140 km/h
(D) 150 km/h
(E) 160 km/h

Q.403 Boats & Streams

A boat covers 120 km downstream in t hours and 140 km upstream in (t + 3) hours. If the ratio of the speed of the boat in still water to its downstream speed is 5:6, find the time taken by the boat to cover 60 km downstream and 60 km upstream.

(A) 4 hours
(B) 5 hours
(C) 6 hours
(D) 7 hours
(E) 8 hours

Q.404 Boats & Streams

A boat travels 216 km upstream in 18 hours and 300 km downstream in 20 hours. Find the time taken by the boat in still water to cover 81 km.

(A) 4 hours
(B) 5 hours
(C) 6 hours
(D) 8 hours
(E) 9 hours

Q.405 Boats & Streams

A boat covers 160 km downstream in 8 hours. The ratio of the speed of the boat in still water to the speed of the stream is 4:1. If the speed of the stream is increased by 50%, then find the time taken by the boat to cover 90 km upstream.

(A) 9 hours
(B) 10 hours
(C) 12 hours
(D) 8 hours
(E) 15 hours

Q.406 Boats & Streams

The ratio of the downstream speed of a boat to its upstream speed is 5:1. The total time taken by the boat to cover D km downstream and D km upstream together is 18 hours. Find the time taken by the boat to cover 3D km in still water.

(A) 12 hours (B) 15 hours
(C) 18 hours (D) 20 hours
(E) 24 hours

Q.407 Boats & Streams

A boat covers 360 km upstream in a certain time. On the next day, it covers 2/3 of that distance downstream and takes 12 hours less than the previous day. If the speed of the boat in still water is 24 km/h, find the time taken by the boat to cover 450 km downstream.

(A) 12 hours (B) 15 hours
(C) 18 hours (D) 20 hours
(E) 24 hours

Q.408 Boats & Streams

A motorboat covers 240 km downstream in t hours. If its speed in still water is increased by 20%, it covers 420 km downstream in (t + 2) hours. If the ratio of the original speed of the boat in still water to the speed of the stream is 5:1, find the value of t.

(A) 3 (B) 5
(C) 6 (D) 8
(E) 4

Q.409 Boats & Streams

The ratio of the speed of a boat in still water to the speed of the stream is 5:2. The boat travels 210 km downstream from P to Q and returns to P in a total of 20 hours. Find the time taken by the boat to cover 125 km in still water.

(A) 4 hours (B) 6 hours
(C) 8 hours (D) 5 hours
(E) 7 hours

Q.410 Boats & Streams

The speed of a boat in still water is 72 km/h. If the upstream speed of the boat is reduced by 20%, it becomes equal to the speed of the current. Find the original downstream speed of the boat.

(A) 104 km/h (B) 96 km/h
(C) 88 km/h (D) 112 km/h
(E) 120 km/h

Q.411 Boats & Streams

The downstream speed of a boat is 20% more than its speed in still water. If the boat covers 144 km upstream and 216 km downstream in 12 hours, then find the total time taken by the boat to cover 72 km upstream and 72 km downstream.

(A) 5 hours (B) 6 hours
(C) 7 hours (D) 8 hours
(E) 4 hours

Q.412 Boats & Streams

A motorboat takes 3 hours more to cover 180 km upstream than to cover the same distance downstream. If the speed of the boat in still water is 400% of the speed of the stream, find the difference between the speed of the boat in still water and the speed of the stream.

(A) 18 km/h (B) 28 km/h
(C) 20 km/h (D) 30 km/h
(E) 24 km/h

Q.413 Boats & Streams

A boat covers D km upstream in 15 hours. The ratio of the speed of the boat in still water to the speed of the current is 7:3, respectively. If the boat covers 42 km in still water in 3 hours, then find the value of 2D.

(A) 180 (B) 210
(C) 240 (D) 270
(E) 300

Q.414 Boats & Streams

The speed of a boat in still water is 25% more than the speed of the stream. The boat covers 270 km downstream and 90 km upstream in a total of 20 hours. Find the time taken by the boat to cover 324 km downstream.

(A) 8 hours (B) 12 hours
(C) 10 hours (D) 9 hours
(E) 6 hours

Q.415 Boats & Streams

A boat covers 72 km downstream in 3 hours. The ratio of the speed of the boat in still water to the speed of the stream is 5:1. If the ratio of time taken to cover 64 km upstream to the time taken to cover 168 km downstream is 4:n, find the value of n.

(A) 5 (B) 7
(C) 6 (D) 8
(E) 9

Q.416 Boats & Streams

A boat covers 180 km downstream in 6 hours. The ratio of the speed of the boat in still water to the speed of the stream is 4:1. If the boat covers (y − 20) km upstream in 5 hours, find the time taken by the boat to cover (y + 10) km in still water.

(A) 4 hours (B) 6 hours
(C) 5 hours (D) 8 hours
(E) 10 hours

Q.417 Boats & Streams

The speed of a stream is 25% of the speed of a boat in still water. The boat covers 150 km downstream in 5 hours. Find the time taken by the boat to cover 126 km upstream.

(A) 6 hours (B) 9 hours
(C) 7 hours (D) 8 hours
(E) 5 hours

Q.418 Boats & Streams

The speed of the stream is 60% less than the speed of a boat in still water. The boat takes 30 hours to cover X km downstream and return to the starting point. If the speed of the boat in still water is 20 km/h, find the time taken by the boat to cover X/7 km upstream.

(A) 6 hours (B) 4 hours
(C) 3 hours (D) 5 hours
(E) 2 hours

Q.419 Boats & Streams

The ratio of the speed of a motorboat in still water to the speed of the current is 4:1. The boat covers 96 km in still water in 4 hours. If the boat takes 6 hours more to cover x km upstream than to cover x km downstream, then find the time taken by the boat to cover 2x km downstream.

(A) 16 hours (B) 15 hours
(C) 20 hours (D) 12 hours
(E) 18 hours

Q.420 Boats & Streams

The sum of the upstream speed and downstream speed of a boat is 62 km/h. The downstream speed is 18 km/h more than the upstream speed. Find the distance covered by the boat in 4 hours while going upstream.

(A) 76 (B) 84
(C) 96 (D) 80
(E) 88

Q.421 Boats & Streams

The sum of the upstream and downstream speeds of a boat is 90 km/h. The boat covers 140 km upstream in 4 hours 40 minutes. Find the time taken by the boat to cover 180 km downstream.

(A) 2 hours 30 minutes (B) 2 hours 45 minutes
(C) 3 hours 30 minutes (D) 3 hours
(E) 4 hours

Q.422 Boats & Streams

A boat travels 3z/4 km downstream and 5z/4 km upstream. The speed of the boat in still water is 20 km/h and the speed of the stream is 4 km/h. The time taken to travel upstream is 18 hours more than the time taken to travel downstream. Find the distance travelled downstream.

(A) 240 km (B) 320 km
(C) 288 km (D) 360 km
(E) 270 km

Q.423 Boats & Streams

A boat covers 150 km downstream and 108 km upstream in a total of 11 hours. The ratio of the speed of the boat in still water to the speed of the stream is 4 : 1. Find by what percentage the distance covered by the boat in 4 hours upstream is more than the distance covered by it in 2 hours downstream.

(A) 25% (B) 10%
(C) 15% (D) 30%
(E) 20%

Q.424 Boats & Streams

A boat covers 350 km downstream in 7 hours. The ratio of the speed of the stream to the upstream speed of the boat is 3 : 4, respectively. Find the speed of the boat in still water.

(A) 35 km/h (B) 25 km/h
(C) 30 km/h (D) 40 km/h
(E) 45 km/h

Q.425 Boats & Streams

The ratio of the speed of a boat in still water to the speed of the stream is 5 : 2. If the time taken by the boat to cover (K + 210) km downstream is equal to the time taken to cover (K − 150) km upstream, then find K.

(A) 360 (B) 480
(C) 390 (D) 420
(E) 540

Q.426 Boats & Streams

The ratio of the speed of boat P in still water to its downstream speed is 4 : 5. Boat P covers 168 km upstream in 8 hours. If the speed of boat Q in still water is 25% more than that of boat P and the speed of the stream is the same for boat Q, then find the time taken by boat Q to cover 224 km upstream.

(A) 6 hours (B) 8 hours
(C) 10 hours (D) 12 hours
(E) 14 hours

📖 Chapter 06 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.396 — **Correct Option: (A)**

• hours
• Solution:
• First convert 120 minutes into hours.
• 120 minutes = 120/60 = 2 hours.
• The speedboat covers 84 km downstream in 2 hours.
• Downstream speed = Distance/Time = 84/2 = 42 km per hour.
• Let the speed of the speedboat in still water = x km per hour.
• Speed of current = 40% of x = 40x/100.
• Downstream speed = Speed in still water + Speed of current.
• So, downstream speed = x + 40% of x = 140% of x.
• Therefore, 140x/100 = 42.
• Now, x = 42 × 100/140 = 30.
• So, the speed of the speedboat in still water = 30 km per hour.
• Speed of current = 30 × 40/100 = 12 km per hour.
• Upstream speed = Speed in still water - Speed of current.
• Upstream speed = 30 - 12 = 18 km per hour.
• Distance to be covered upstream = 126 km.
• Time taken upstream = Distance/Speed = 126/18 = 7 hours.

Solution Q.397 — **Correct Option: (E)**

• hours
• Solution:
• First convert 180 minutes into hours.
• 180 minutes = 180/60 = 3 hours.
• The steamer covers 216 km downstream in 3 hours.
• Downstream speed = Distance/Time = 216/3 = 72 km per hour.
• Let the speed of the steamer in still water = x km per hour.
• Speed of current = 20% of x = 20x/100.
• Downstream speed = Speed in still water + Speed of current.
• So, downstream speed = x + 20% of x = 120% of x.
• Therefore, 120x/100 = 72.
• Now, x = 72 × 100/120 = 60.
• So, the speed of the steamer in still water = 60 km per hour.
• Speed of current = 60 × 20/100 = 12 km per hour.
• Upstream speed = Speed in still water - Speed of current.
• Upstream speed = 60 - 12 = 48 km per hour.
• Distance to be covered upstream = 240 km.
• Time taken upstream = Distance/Speed = 240/48 = 5 hours.

Solution Q.398 — **Correct Option: (B)**

• Distance covered upstream = 150 km
• Time taken upstream = 10 hours
• Upstream speed = 150/10 = 15 km/h
• Let the speed of the boat in still water be 4a.
• Let the speed of the current be a.
• Upstream speed = Still water speed - Current speed
• Upstream speed = 4a - a = 3a
• So:
• 3a = 15
• a = 5
• Downstream speed = Still water speed + Current speed
• Downstream speed = 4a + a = 5a
• Downstream speed = 5 × 5 = 25 km/h
• The boat covers X km downstream in 8 hours.
• Distance = Speed × Time
• X = 25 × 8
• X = 200

Solution Q.399 — **Correct Option: (D)**

• km
• Solution:
• Distance covered upstream = 135 km
• Time taken upstream = 9 hours
• Upstream speed = 135/9 = 15 km/h
• Let the speed of the boat in still water be 5a.
• Let the speed of the current be 2a.
• Upstream speed = Still water speed - Current speed
• Upstream speed = 5a - 2a = 3a
• So:
• 3a = 15
• a = 5
• Downstream speed = Still water speed + Current speed
• Downstream speed = 5a + 2a = 7a
• Downstream speed = 7 × 5 = 35 km/h
• The boat covers X km downstream in 8 hours.
• Distance = Speed × Time
• X = 35 × 8
• X = 280 km

Solution Q.400 — **Correct Option: (B)**

• hours
• Solution:
• Difference between the speed in still water and upstream speed is equal to the speed of
• the stream.
• Given difference = 5 km/h
• So, speed of stream = 5 km/h
• Ratio of downstream speed to stream speed = 6 : 1
• Downstream speed = 6 × 5 = 30 km/h
• Speed in still water = Downstream speed - Stream speed
• Speed in still water = 30 - 5 = 25 km/h
• Now, speed of stream is increased by 4 km/h.
• New speed of stream = 5 + 4 = 9 km/h
• New upstream speed = Speed in still water - New stream speed
• New upstream speed = 25 - 9 = 16 km/h
• Distance = 192 km
• Time taken = Distance/Speed
• Time taken = 192/16 = 12 hours

Solution Q.401 — **Correct Option: (C)**

• Downstream distance = 28 km.
• Downstream time = 40 minutes = 40/60 hours = 2/3 hours.
• Downstream speed = 28/(2/3)
• Downstream speed = 28 × 3/2 = 42 km/h.
• Let the speed of the boat in still water be s km/h.
• Speed of stream = 40% of s = 40s/100 = 2s/5.
• Downstream speed = s + 2s/5 = 7s/5.
• So, 7s/5 = 42
• s = 42 × 5/7
• s = 30 km/h.
• Speed of stream = 40% of 30 = 12 km/h.
• Upstream speed = 30 - 12 = 18 km/h.
• Time taken to cover 72 km upstream = 72/18 = 4 hours.
• Time taken to cover 210 km downstream = 210/42 = 5 hours.
• Required time ratio = 4:5.
• Sum of the terms of the ratio = 4 + 5 = 9.

Solution Q.402 **Correct Option: (B)**

• km/h
• Solution:
• Let the upstream speed of the boat be U km/h.
• When upstream speed is reduced by 20%, it becomes 80% of U.
• So, speed of current = 80% of U = 80U/100 = 4U/5
• Speed of boat in still water = Upstream speed + Speed of current
• 90 = U + 4U/5
• 90 = 9U/5
• U = 90 × 5/9
• U = 50 km/h
• Speed of current = 4U/5 = 4 × 50/5 = 40 km/h
• Downstream speed = Speed in still water + Speed of current
• Downstream speed = 90 + 40 = 130 km/h

Solution Q.403 **Correct Option: (B)**

• hours
• Solution:
• Let the speed of the boat in still water be 5a.
• Downstream speed = 6a.
• So, speed of stream = 6a - 5a = a.
• Upstream speed = 5a - a = 4a.
• Downstream speed : Upstream speed = 6a : 4a = 3 : 2
• Downstream distance = 120 km and time = t hours.
• Upstream distance = 140 km and time = (t + 3) hours.
• So, downstream speed/upstream speed = 3/2.
• (120/t)/(140/(t + 3)) = 3/2
• 120 × (t + 3)/(140t) = 3/2
• Cross multiply:
• 240(t + 3) = 420t
• 240t + 720 = 420t
• 720 = 180t
• t = 720/180
• t = 4
• Downstream speed = 120/4 = 30 km/h.
• Upstream time = t + 3 = 4 + 3 = 7 hours.
• Upstream speed = 140/7 = 20 km/h.
• Time taken to cover 60 km downstream = 60/30 = 2 hours.
• Time taken to cover 60 km upstream = 60/20 = 3 hours.
• Total time = 2 + 3 = 5 hours.

Solution Q.404 **Correct Option: (C)**

• hours
• Solution:
• Upstream speed = 216/18 = 12 km/h.
• Downstream speed = 300/20 = 15 km/h.
• Speed of boat in still water = (Upstream speed + Downstream speed)/2.
• Speed of boat in still water = (12 + 15)/2.
• Speed of boat in still water = 27/2 km/h.
• Distance to be covered in still water = 81 km.
• Time taken = Distance/Speed.
• Time taken = 81/(27/2).
• Time taken = 81 × 2/27.
• Time taken = 6 hours.

Solution Q.405 **Correct Option: (A)**

• hours
• Solution:
• Downstream distance = 160 km.
• Downstream time = 8 hours.
• Downstream speed = 160/8
• Downstream speed = 20 km/h.
• Ratio of speed of boat in still water to speed of stream = 4:1.
• Let speed of boat in still water = 4x.
• Let speed of stream = x.
• Downstream speed = 4x + x = 5x.
• So, 5x = 20.
• x = 20/5
• x = 4.
• Speed of stream = 4 km/h.
• Speed of boat in still water = 4x = 4 × 4 = 16 km/h.
• Now speed of stream is increased by 50%.
• New speed of stream = 4 × 150/100
• New speed of stream = 6 km/h.
• New upstream speed = Speed in still water - New stream speed
• New upstream speed = 16 - 6
• New upstream speed = 10 km/h.
• Required distance upstream = 90 km.
• Time taken = Distance/Speed
• Time taken = 90/10
• Time taken = 9 hours.

Solution Q.406 **Correct Option: (B)**

• hours
• Solution:
• Let downstream speed = 5x km/h
• Let upstream speed = x km/h
• Time taken to cover D km downstream = D/(5x) hours
• Time taken to cover D km upstream = D/x hours
• Total time = 18 hours
• So:
• D/(5x) + D/x = 18
• D/(5x) + 5D/(5x) = 18
• 6D/(5x) = 18
• D/x = 18 × 5/6
• D/x = 15
• Speed in still water = (Downstream speed + Upstream speed)/2
• Speed in still water = (5x + x)/2
• Speed in still water = 3x km/h
• Time taken to cover 3D km in still water = 3D/(3x)
• Time taken = D/x
• Time taken = 15 hours

Solution Q.407 **Correct Option: (B)**

• hours
• Solution:
• Let the speed of the stream = x km/h.
• Speed of boat in still water = 24 km/h.
• Upstream speed = 24 - x.
• Downstream speed = 24 + x.
• Upstream distance = 360 km.
• Downstream distance on the next day = 2/3 × 360 = 240 km.
• The downstream journey takes 12 hours less than the upstream journey.
• So,
• 360/(24 - x) - 240/(24 + x) = 12
• Multiply by (24 - x)(24 + x):
• 360(24 + x) - 240(24 - x) = 12(24 - x)(24 + x)
• 8640 + 360x - 5760 + 240x = 12(576 - x²)
• 2880 + 600x = 6912 - 12x²
• 12x² + 600x - 4032 = 0
• Divide by 12:
• x² + 50x - 336 = 0
• x² + 56x - 6x - 336 = 0
• x(x + 56) - 6(x + 56) = 0
• (x - 6)(x + 56) = 0
• So, x = 6. The other root, -56, is invalid: the stream speed must satisfy 0 < x < 24.
• Speed of stream = 6 km/h.
• Downstream speed = 24 + 6 = 30 km/h.
• Time taken to cover 450 km downstream = 450/30 = 15 hours.

Solution Q.408 **Correct Option: (E)**

• Let the original speed of the boat in still water = 5a km/h.
• Speed of the stream = a km/h.
• Original downstream speed = 5a + a = 6a km/h
• Time taken to cover 240 km downstream = t hours
• So, 240/(6a) = t
• This gives 40/a = t
• After increasing still water speed by 20%:
• New still water speed = 5a × 120/100 = 6a km/h
• New downstream speed = 6a + a = 7a km/h
• Time taken to cover 420 km downstream = (t + 2) hours
• So, 420/(7a) = t + 2
• This gives 60/a = t + 2
• Now, 40/a = t, so a = 40/t
• Put this in 60/a = t + 2:
• 60/(40/t) = t + 2
• 60t/40 = t + 2
• 3t/2 = t + 2
• 3t = 2t + 4
• t = 4

Solution Q.409 **Correct Option: (D)**

• hours
• Solution:
• Let the speed of the boat in still water = 5x km/h
• Speed of stream = 2x km/h
• Downstream speed = 5x + 2x = 7x km/h
• Upstream speed = 5x - 2x = 3x km/h
• Total time = downstream time + upstream time
• 20 = 210/(7x) + 210/(3x)
• 20 = 30/x + 70/x
• 20 = 100/x
• x = 5
• Speed of boat in still water = 5x = 5 × 5 = 25 km/h
• Time taken to cover 125 km in still water = 125/25 = 5 hours

Solution Q.410 **Correct Option: (A)**

• km/h
• Solution:
• Let the speed of current = x km/h
• Speed of boat in still water = 72 km/h
• Upstream speed = 72 - x
• When upstream speed is reduced by 20%, it becomes equal to the speed of current.
• So, 80% of upstream speed = current speed
• (72 - x) × 80/100 = x
• (72 - x) × 4/5 = x
• 288 - 4x = 5x
• 9x = 288
• x = 32 km/h
• Downstream speed = still water speed + current speed
• Downstream speed = 72 + 32 = 104 km/h

Solution Q.411 **Correct Option: (A)**

• hours
• Solution:
• Let the speed of the boat in still water = 5x km/h.
• Downstream speed is 20% more than still water speed.
• So, downstream speed = 6x km/h.
• Stream speed = 6x - 5x = x km/h.
• Upstream speed = 5x - x = 4x km/h.
• Given total time:
• 144/(4x) + 216/(6x) = 12.
• 36/x + 36/x = 12.
• 72/x = 12.
• x = 6.
• Upstream speed = 4x = 24 km/h.
• Downstream speed = 6x = 36 km/h.
• Time to cover 72 km upstream = 72 / 24 = 3 hours.
• Time to cover 72 km downstream = 72 / 36 = 2 hours.
• Total time = 3 + 2 = 5 hours.

Solution Q.412 **Correct Option: (E)**

• km/h
• Solution:
• Let the speed of stream = x km/h.
• Speed of boat in still water = 400% of x = 4x km/h.
• Upstream speed = 4x - x = 3x km/h.
• Downstream speed = 4x + x = 5x km/h.
• Time taken upstream = 180 / (3x) = 60/x hours.
• Time taken downstream = 180 / (5x) = 36/x hours.
• Difference in time = 60/x - 36/x = 24/x hours.
• Given difference = 3 hours.
• So, 24/x = 3.
• x = 8 km/h.
• Speed of boat in still water = 4 × 8 = 32 km/h.
• Required difference = 32 - 8 = 24 km/h.

Solution Q.413 **Correct Option: (C)**

• Speed of the boat in still water = 42 / 3 = 14 km/h.
• Ratio of still water speed to current speed = 7:3.
• So, 7 parts = 14.
• 1 part = 14 / 7 = 2.
• Current speed = 3 parts = 3 × 2 = 6 km/h.
• Upstream speed = 14 - 6 = 8 km/h.
• Boat covers D km upstream in 15 hours.
• So, D = 8 × 15 = 120 km.
• Required value of 2D = 2 × 120 = 240.

Solution Q.414 **Correct Option: (E)**

• hours
• Solution:
• Let the speed of the stream = 4x km/h.
• Boat's speed in still water is 25% more than stream speed.
• So, speed of boat in still water = 5x km/h.
• Downstream speed = 5x + 4x = 9x km/h
• Upstream speed = 5x - 4x = x km/h
• Total time = 20 hours
• Time taken downstream + Time taken upstream = 20
• 270/(9x) + 90/x = 20
• 30/x + 90/x = 20
• 120/x = 20
• x = 6
• Downstream speed = 9x = 9 × 6 = 54 km/h
• Time taken to cover 324 km downstream = 324 / 54 = 6 hours

Solution Q.415 **Correct Option: (B)**

• Downstream speed = 72 / 3 = 24 km/h.
• Ratio of still water speed to stream speed = 5 : 1.
• So, downstream speed = 5 + 1 = 6 parts.
• 6 parts = 24 km/h.
• 1 part = 24 / 6 = 4 km/h.
• Still water speed = 5 × 4 = 20 km/h.
• Stream speed = 4 km/h.
• Upstream speed = 20 - 4 = 16 km/h.
• Downstream speed = 20 + 4 = 24 km/h.
• Time to cover 64 km upstream = 64 / 16 = 4 hours.
• Time to cover 168 km downstream = 168 / 24 = 7 hours.
• Required ratio = 4 : 7.
• So, n = 7.

Solution Q.416 **Correct Option: (C)**

• hours
• Solution:
• Downstream speed = 180 / 6 = 30 km/h.
• Ratio of speed of boat in still water to speed of stream = 4:1.
• Let still water speed = 4x and stream speed = x.
• Downstream speed = 4x + x = 5x.
• So, 5x = 30.
• x = 6.
• Still water speed = 4x = 4 × 6 = 24 km/h.
• Stream speed = 6 km/h.
• Upstream speed = 24 - 6 = 18 km/h.
• The boat covers (y - 20) km upstream in 5 hours.
• So, y - 20 = 18 × 5 = 90.
• y = 110.
• Required distance in still water = y + 10 = 110 + 10 = 120 km.
• Time taken in still water = 120 / 24 = 5 hours.

Solution Q.417 **Correct Option: (C)**

• hours
• Solution:
• Let the speed of the boat in still water be 100 units.
• Speed of stream = 25% of still water speed = 25 units.
• Downstream speed = 100 + 25 = 125 units.
• Actual downstream speed = 150 / 5 = 30 km/h.
• So, 125 units = 30 km/h.
• 1 unit = 30 / 125 km/h.
• Upstream speed = 100 - 25 = 75 units.
• Upstream speed = 30 × 75 / 125 = 18 km/h.
• Time taken to cover 126 km upstream = 126 / 18 = 7 hours.

Solution Q.418 **Correct Option: (C)**

• hours
• Solution:
• Speed of boat in still water = 20 km/h.
• Speed of stream is 60% less than still water speed.
• Speed of stream = 40% of 20 = 8 km/h.
• Downstream speed = 20 + 8 = 28 km/h.
• Upstream speed = 20 - 8 = 12 km/h.
• Time taken to cover X km downstream and X km upstream = 30 hours.
• So, X/28 + X/12 = 30
• LCM of 28 and 12 = 84
• 3X/84 + 7X/84 = 30
• 10X/84 = 30
• X = 30 × 84 / 10
• X = 252 km.
• Required distance = X/7 = 252/7 = 36 km.
• Time taken upstream = 36 / 12 = 3 hours.

Solution Q.419 **Correct Option: (E)**

• hours
• Solution:
• Speed in still water : speed of current = 4 : 1.
• Boat covers 96 km in still water in 4 hours.
• Speed in still water = 96 / 4 = 24 km/h.
• So, 4 parts = 24.
• 1 part = 6.
• Speed of current = 6 km/h.
• Downstream speed = 24 + 6 = 30 km/h.
• Upstream speed = 24 - 6 = 18 km/h.
• Time difference for x km = 6 hours.
• So, x/18 - x/30 = 6.
• LCM of 18 and 30 = 90.
• 5x/90 - 3x/90 = 6.
• 2x/90 = 6.
• x/45 = 6.
• x = 270 km.
• Required distance = 2x = 2 × 270 = 540 km.
• Time taken downstream = 540 / 30 = 18 hours.

Solution Q.420 **Correct Option: (E)**

• Let the upstream speed = x km/h.
• Then downstream speed = x + 18 km/h.
• Given that the sum of upstream and downstream speeds is 62 km/h.
• So,
• x + (x + 18) = 62
• 2x + 18 = 62
• 2x = 44
• x = 22
• Therefore, upstream speed = 22 km/h.
• Distance covered in 4 hours upstream = 22 × 4 = 88 km.

Solution Q.421 **Correct Option: (D)**

• hours
• Solution:
• Upstream distance = 140 km
• Upstream time = 4 hours 40 minutes = 280 minutes
• Upstream speed = 140 × 60 / 280
• Upstream speed = 30 km/h
• Given:
• Upstream speed + Downstream speed = 90 km/h
• So,
• Downstream speed = 90 - 30
• Downstream speed = 60 km/h
• Required downstream distance = 180 km
• Time taken = 180 / 60
• Time taken = 3 hours

Solution Q.422 **Correct Option: (C)**

• km
• Solution:
• Speed in still water = 20 km/h.
• Speed of stream = 4 km/h.
• Downstream speed = 20 + 4 = 24 km/h.
• Upstream speed = 20 - 4 = 16 km/h.
• Downstream distance = 3z/4 km.
• Upstream distance = 5z/4 km.
• Time taken downstream = (3z/4) / 24
• = 3z/96
• = z/32 hours.
• Time taken upstream = (5z/4) / 16
• = 5z/64 hours.
• Given, upstream time is 18 hours more than downstream time.
• 5z/64 - z/32 = 18
• 5z/64 - 2z/64 = 18
• 3z/64 = 18
• z = 18 × 64/3
• z = 384
• Downstream distance = 3z/4
• = 3 × 384/4
• = 288 km.

Solution Q.423 **Correct Option: (E)**

• %
• Solution:
• Speed of boat in still water : Speed of stream = 4 : 1
• Let speed of boat in still water = 4x
• Let speed of stream = x
• Downstream speed = 4x + x = 5x
• Upstream speed = 4x - x = 3x
• Total time = 11 hours
• According to the question:
• 150 / (5x) + 108 / (3x) = 11
• 30/x + 36/x = 11
• 66/x = 11
• x = 6
• Downstream speed = 5x = 5 × 6 = 30 km/h
• Upstream speed = 3x = 3 × 6 = 18 km/h
• Distance covered in 4 hours upstream:
• 18 × 4 = 72 km
• Distance covered in 2 hours downstream:
• 30 × 2 = 60 km
• Difference = 72 - 60 = 12 km
• Required percentage = 12 × 100 / 60
• Required percentage = 20%

Solution Q.424 **Correct Option: (A)**

• km/h
• Solution:
• Downstream speed of boat = 350 / 7
• = 50 km/h
• Let speed of stream = 3x and upstream speed = 4x.
• Downstream speed = Upstream speed + 2 × Speed of stream
• So, 50 = 4x + 2 × 3x
• 50 = 4x + 6x
• 50 = 10x
• x = 5
• Speed of stream = 3x = 15 km/h
• Upstream speed = 4x = 20 km/h
• Speed of boat in still water = Upstream speed + Speed of stream
• = 20 + 15
• = 35 km/h

Solution Q.425 **Correct Option: (D)**

• Let the speed of the boat in still water = 5x km/h
• Speed of stream = 2x km/h
• Downstream speed = 5x + 2x = 7x km/h
• Upstream speed = 5x - 2x = 3x km/h
• Time taken downstream = Time taken upstream
• (K + 210)/(7x) = (K - 150)/(3x)
• 3(K + 210) = 7(K - 150)
• 3K + 630 = 7K - 1050
• 1680 = 4K
• K = 420
• Therefore, the value of K is 420.

Solution Q.426 **Correct Option: (B)**

• hours
• Solution:
• Let the speed of boat P in still water = 4x km/h.
• Downstream speed of boat P = 5x km/h.
• Speed of stream = 5x - 4x = x km/h.
• Upstream speed of boat P = 4x - x = 3x km/h.
• Boat P covers 168 km upstream in 8 hours.
• Upstream speed of boat P = 168/8 = 21 km/h.
• So, 3x = 21
• x = 7
• Speed of boat P in still water = 4 × 7 = 28 km/h.
• Speed of stream = 7 km/h.
• Speed of boat Q in still water = 28 × 125/100 = 35 km/h.
• Upstream speed of boat Q = 35 - 7 = 28 km/h.
• Required time = 224/28 = 8 hours.

CHAPTER 07: MIXTURES, REPLACEMENTS & ALLIGATIONS

44 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Rule of Alligation: (Cheaper Qty / Dearer Qty) = (Dearer Price - Mean Price) / (Mean Price - Cheaper Price)
- Replacement Formula: Final Pure Qty = Initial Qty × (1 - x / V)^n, where x is quantity replaced each time.

Q.427 Mixtures

A vessel contains 600 litres of a mixture of milk and water, in which water is 50% less than milk. When (x + 50) litres of water is added to the mixture, the quantity of water becomes 70% of the quantity of milk. If x litres of pure milk is added to the original mixture instead, and the ratio of water to milk becomes m:n in its simplest form, find m + n.

(A) 57 (B) 61
(C) 63 (D) 67
(E) 71

Q.428 Mixtures

Container P contains milk and water in the mass ratio 5 : 4. Container Q contains milk, water and sugar in the mass ratio 3 : 7 : 2. When the entire contents of the two containers are combined, the masses of milk and water are 114 kg and 174 kg, respectively. Find the mass of sugar in the final mixture.

(A) 42 kg (B) 30 kg
(C) 24 kg (D) 48 kg
(E) 36 kg

Q.429 Mixtures

A container has 180 liters of a mixture of juice and water in the ratio 5 : 4 respectively. If 36 liters of the mixture is taken out and 24 liters of water is added, find the sum of the numbers in the simplified ratio of water to juice in the new mixture.

(A) 19 (B) 23
(C) 21 (D) 25
(E) 17

Q.430 Mixtures

A tank contains milk and water in the ratio of 7:2. If 18 liters of the mixture are removed and 6 liters of water are added, then the quantity of milk in the final mixture becomes 42 liters. Find the initial quantity of water.

(A) 18 liters (B) 12 liters
(C) 16 liters (D) 20 liters
(E) 24 liters

Q.431 Mixtures

A vessel contains 180 liters of pure milk. When equal quantities of milk and water are added to the vessel, the ratio of milk to water becomes 7:3. Now, 2/3 of the mixture is removed, and X liters of water is added to the remaining mixture such that water becomes 40% of the final mixture. Find X.

(A) 30 (B) 20
(C) 25 (D) 35
(E) 15

Q.432 Mixtures

Two vessels contain mixtures of milk and water. In the first vessel, water is 20%, and in the second vessel, water is 55%. The mixtures are mixed to get a final mixture containing 60% milk. If 96 liters of mixture from the second vessel is used, find the quantity of mixture used from the first vessel.

(A) 84 liters (B) 64 liters
(C) 96 liters (D) 80 liters
(E) 72 liters

Q.433 Mixtures

The ratio of milk to water in a 312-liter mixture is 5 : 7. If 58 liters of water and X liters of milk are added to the mixture, the ratio of milk to water becomes 2 : 3. Find the value of X.

(A) 30 (B) 24
(C) 36 (D) 42
(E) 48

Q.434 Mixtures

A bottle contains 96 ml of a mixture of two liquids P and Q, in which liquid P is 40 ml more than liquid Q. If a separation process removes x ml of pure liquid P without removing any liquid Q, the ratio of liquid P to liquid Q in the new mixture becomes 3:2. Find x.

(A) 26 ml (B) 18 ml
(C) 30 ml (D) 22 ml
(E) 34 ml

Q.435 Mixtures

A mixture of milk and water contains 9 liters of water. When 6 liters of milk and 10 liters of water are added to the mixture, the concentration of milk becomes 75%. Find the total quantity of the initial mixture.

(A) 48 liters (B) 54 liters
(C) 66 liters (D) 60 liters
(E) 72 liters

Q.436 Mixtures

A mixture of milk and water initially contains 12 liters of water. When 6 liters of milk and 18 liters of water are added to it, the concentration of milk in the mixture becomes 75%. Find the total quantity of the initial mixture.

(A) 84 liters (B) 88 liters
(C) 92 liters (D) 96 liters
(E) 104 liters

Q.437 Mixtures

A vendor mixed (x + 40) liters of water with 240 liters of milk. Then 60 liters of the mixture were removed and replaced with 60 liters of pure milk. If the quantity of milk in the final mixture becomes 264 liters, find x.

(A) 100 (B) 140
(C) 120 (D) 160
(E) 180

Q.438 Mixtures

A mixture contains 120 liters of milk and water in the ratio 2 : 3, respectively. If 25% of the mixture is taken out and X liters of milk are added to the remaining mixture, then the ratio of milk to water becomes 4 : 3. Find X.

(A) 24 (B) 30
(C) 36 (D) 42
(E) 48

Q.439 Mixtures

In mixture A, the quantity of milk is 30 liters, which is 150% of the quantity of water. In mixture B, the quantity of milk is 28 liters, which is 70% of the quantity of water. If both mixtures are mixed to form mixture C, then find the difference between the quantities of milk and water in the resulting mixture.

(A) 6 liters (B) 4 liters
(C) 8 liters (D) 2 liters
(E) 10 liters

Q.440 Mixtures

Vessel P contains a mixture of milk and water in which milk is 40% of the total mixture. Vessel Q contains a mixture of milk and water in which the quantity of milk is 36 liters. If the ratio of the quantity of mixture in Vessel P to that in Vessel Q is 3 : 5, and the final mixture contains 30% milk after mixing both vessels, find the total quantity of the final mixture.

(A) 180 liters (B) 200 liters
(C) 240 liters (D) 300 liters
(E) 320 liters

Q.441 Mixtures

A tank contains 150 liters of a mixture of acid and water in the ratio of 2:3, respectively. When X liters of the mixture is taken out and replaced with (X − 10) liters of water, the ratio of acid to water becomes 5:9. Find X.

(A) 15 (B) 20
(C) 25 (D) 30
(E) 35

Q.442 Mixtures

25 kg of rice X and 55 kg of rice Y are mixed and sold at Rs. 102 per kg. If the per kg cost prices of rice X and rice Y are in the ratio 6:5 and the profit earned on the mixture is 20%, find the difference between their per kg cost prices.

(A) Rs. 12 (B) Rs. 20
(C) Rs. 16 (D) Rs. 24
(E) Rs. 18

Q.443 Mixtures

A mixture contains zinc and copper in the ratio of 4 : 7. When 32 kg of zinc and 24 kg of copper are added to the mixture, the ratio of zinc to copper becomes 2 : 3. Find the final quantity of copper.

(A) 168 kg (B) 180 kg
(C) 204 kg (D) 216 kg
(E) 192 kg

Q.444 Mixtures

A grocer mixed 30 kg of wheat costing Rs. 100 per kg with P kg of maize costing Rs. 70 per kg. He sold the mixture at Rs. 68 per kg and incurred a loss of 15%. Find the value of P.

(A) 75 kg (B) 45 kg
(C) 60 kg (D) 90 kg
(E) 50 kg

Q.445 Mixtures

In mixture X, the quantity of milk is 45 liters, which is 150% of the quantity of water. In mixture Y, the quantity of milk is 36 liters, which is 90% of the quantity of water. If both mixtures are combined, find the difference between the total quantity of milk and water.

(A) 9 liters (B) 11 liters
(C) 14 liters (D) 6 liters
(E) 18 liters

Q.446 Mixtures

Mixture P of mango juice and water contains 30% mango juice and the rest water. Then 3x ml of mango juice and x ml of water are added to mixture P, so the ratio of mango juice to water in the final mixture becomes 3:4. If the initial quantity of mixture P is M ml, then find the value of M/x.

(A) 8 (B) 12
(C) 15 (D) 6
(E) 10

Q.447 Mixtures

A vessel contains milk and water in the ratio of 4:1. When 50 liters of water is added to the vessel, water becomes 30% of the final mixture. Find the initial quantity of milk in the vessel.

(A) 240 liters (B) 320 liters
(C) 280 liters (D) 350 liters
(E) 300 liters

Q.448 Mixtures

Two types of pulses costing Rs. 75 per kg and Rs. 125 per kg are mixed in the ratio of 3:2. If this mixture is sold at Rs. 114 per kg, then find the profit percentage.

(A) 15% (B) 25%
(C) 20% (D) 18%
(E) 30%

Q.449 Mixtures

A container has 120 liters of a mixture of milk and water in the ratio of 3:2. If 30 liters of the mixture is removed and x liters of water is added, then the ratio of milk to water in the final mixture becomes 6:7. Find x.

(A) 27 liters (B) 24 liters
(C) 30 liters (D) 36 liters
(E) 21 liters

Q.450 Mixtures

A container has 70 kg of a mixture of spice and salt in the ratio 3:4. Some quantity of the mixture is removed and replaced by twice that quantity of pure salt. If the ratio of spice to salt in the new mixture becomes 2:5, then find the quantity of salt removed with the mixture.

(A) 6 kg (B) 12 kg
(C) 10 kg (D) 8 kg
(E) 14 kg

Q.451 Mixtures

16 men can complete a piece of work in 45 days, and 24 women can complete the same work in 60 days. If 36 women start the work and after 20 days they are replaced by 24 men, then find the total time taken to complete the whole work.

(A) 30 days (B) 35 days
(C) 40 days (D) 45 days
(E) 25 days

Q.452 Mixtures

A vessel contains 80 liters of a mixture of orange juice and water. If a separation process selectively removes 20% of the orange juice and 10% of the water, then 87.5% of the initial mixture remains in the vessel. Find the initial quantity of water in the vessel.

(A) 60 liters (B) 40 liters
(C) 50 liters (D) 45 liters
(E) 55 liters

Q.453 Mixtures

A grocer mixes 48 kg of wheat type P with 72 kg of wheat type Q and sells the whole mixture at Rs. 65 per kg. If the cost price per kg of wheat type P and type Q is in the ratio 7:4, respectively, and the grocer earns a profit of 25%, find the difference between the cost prices per kg of both types of wheat.

(A) 24 (B) 36
(C) 42 (D) 30
(E) 28

Q.454 Mixtures

A 128-litre mixture contains juice and water in the ratio 3:5, respectively. How many litres of juice should be added to the mixture so that the quantities of juice and water become equal?

(A) 20 litres (B) 24 litres
(C) 28 litres (D) 36 litres
(E) 32 litres

Q.455 Mixtures

Container X contains orange juice and water in the ratio of 3:2 respectively. Container Y contains orange juice, water and syrup in the ratio of 5:6:1 respectively. When the mixtures from containers X and Y are mixed, the quantities of orange juice and water in the final mixture become 120 liters and 112 liters respectively. Find the quantity of syrup in the final mixture.

(A) 14 liters (B) 18 liters
(C) 12 liters (D) 16 liters
(E) 10 liters

Q.456 Mixtures

A container has 160 liters of a mixture of milk and water in the ratio 3:5, respectively. If X liters of the mixture are taken out and (X/2 − 5) liters of water are added to the remaining mixture, then the ratio of milk to water becomes 1:2. Find the value of X/4.

(A) 8 (B) 12
(C) 15 (D) 6
(E) 10

Q.457 Mixtures

A tank contains 150 liters of a mixture of milk and water in the ratio of 3:2 respectively. If x liters of the mixture is removed and replaced with the same quantity of pure milk, then the ratio of milk to water in the final mixture becomes 4:1. Find the value of x.

(A) 60 liters (B) 90 liters
(C) 45 liters (D) 75 liters
(E) 80 liters

Q.458 Mixtures

Container A contains a mixture of milk and water in the ratio 2:3, and container B contains a mixture of milk and water in the ratio 3:1. If half of the mixture from each container is taken out and mixed together, the new mixture contains 60 litres of milk and 55 litres of water. Find the initial quantity of mixture in container A.

(A) 160 (B) 140
(C) 150 (D) 170
(E) 180

Q.459 Mixtures

A container of capacity 100 litres is completely filled with a mixture of milk and water. A separation process selectively removes 60% of the milk and 20% of the water. If the container becomes 52% vacant after this, find the initial quantities of water and milk respectively.

(A) 20 litres, 80 litres (B) 25 litres, 75 litres
(C) 30 litres, 70 litres (D) 40 litres, 60 litres
(E) 35 litres, 65 litres

Q.460 Mixtures

A milkman has 75 litres of pure milk. He mixes some water, obtained free of cost, with it and sells the mixture at the cost price per litre of pure milk, earning a profit of 32%. If the cost price of pure milk is Rs. 30 per litre, find the quantity of water mixed.

(A) 18 litres (B) 24 litres
(C) 30 litres (D) 20 litres
(E) 36 litres

Q.461 Mixtures

Vessel A contains milk and water in the ratio 3:1, while vessel B contains milk and water in the ratio 2:1. If 36 litres of mixture from vessel A are poured into vessel B, the new ratio of milk to water in vessel B becomes 5:2. Also, the new quantity of mixture in vessel B is 25% less than the initial quantity of mixture in vessel A. Find the initial quantity of milk in vessel A.

(A) 72 litres (B) 56 litres
(C) 84 litres (D) 60 litres
(E) 63 litres

Q.462 Mixtures

A shopkeeper bought type P rice at Rs. 28 per kg and type Q rice at Rs. 42 per kg. After mixing both types of rice, the cost price of the mixture becomes Rs. 33.60 per kg. If the total quantity of the mixture is 150 kg, then find the quantity of type P rice.

(A) 90 kg (B) 75 kg
(C) 100 kg (D) 84 kg
(E) 105 kg

Q.463 Mixtures

A vessel contains 200 liters of pure milk. 25% of the milk is replaced with water. Further, X% of the resultant mixture is taken out, and 40 liters of water are added to the remaining mixture. If the final ratio of milk to water becomes 3 : 2, find the value of X + 7.

(A) 24
(B) 27
(C) 30
(D) 32
(E) 35

Q.464 Mixtures

In a vessel, the quantity of milk is 25% more than the quantity of water. When 45 liters of the mixture are taken out and 30 liters of water are added to the vessel, the ratio of milk to water in the resultant mixture becomes 5 : 6. Find the initial quantity of milk.

(A) 80 liters
(B) 90 liters
(C) 110 liters
(D) 120 liters
(E) 100 liters

Q.465 Mixtures

A container contains a certain quantity of a milk and water mixture. If 18 liters of the mixture is taken out and replaced with 42 liters of water, then the ratio of the final quantity of mixture to the initial quantity of mixture becomes 6:5. Find the total quantity of the initial mixture.

(A) 100 liters
(B) 140 liters
(C) 120 liters
(D) 90 liters
(E) 150 liters

Q.466 Mixtures

Vessel A contains milk and water in the ratio of 3 : 2, and Vessel B contains milk and water in the ratio of 5 : 4. 50 liters from Vessel A and 45 liters from Vessel B are taken out and mixed in Vessel C, which already contains a mixture of milk and water in the ratio of 2 : 1. If the final ratio of milk to water in Vessel C becomes 5 : 3, find the initial quantity of mixture in Vessel C.

(A) 90 liters
(B) 105 liters
(C) 120 liters
(D) 75 liters
(E) 96 liters

Q.467 Mixtures

A tank contains 140 liters of a mixture of milk and water in the ratio 5 : y. When 25 liters of milk and 15 liters of water are added to the tank, the ratio of milk to water becomes 5 : 7. Find the value of y.

(A) 7
(B) 11
(C) 9
(D) 8
(E) 6

Q.468 Mixtures

A tank contains a mixture of syrup and water in the ratio 5 : 3. When 64 liters of water is added to the mixture, the ratio of syrup to water becomes 3 : 5. Find the original quantity of the mixture.

(A) 96 liters
(B) 84 liters
(C) 108 liters
(D) 120 liters
(E) 72 liters

Q.469 Mixtures

A vessel contains 270 liters of a mixture of liquids P and Q in the ratio 5 : 4, respectively. Some quantity of the mixture is taken out and replaced by an equal volume of liquid R. If the ratio of P, Q and R in the resultant mixture becomes 10 : 8 : 9, respectively, then the quantity of mixture taken out is what percent of the initial quantity of liquid Q?

(A) 60%
(B) 75%
(C) 66.67%
(D) 80%
(E) 72%

Q.470 Mixtures

Vessel A contains a mixture of milk and water in which milk is 40% of the total mixture. Vessel B also contains a mixture of milk and water, and the quantity of milk in vessel B is 36 liters. The ratio of the quantity of mixture in vessel A to that in vessel B is 3 : 7. When both mixtures are mixed together, the resultant mixture contains 30% milk. Find the total quantity of the resultant mixture.

(A) 160 liters
(B) 240 liters
(C) 200 liters
(D) 280 liters
(E) 320 liters

📖 Chapter 07 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.427 — **Correct Option: (C)**

• Total mixture = 600 litres
• Water is 50% less than milk.
• So, if milk = 100 parts, water = 50 parts
• Therefore, milk : water = 2 : 1
• Total parts = 3
• Milk = 600 × 2 / 3 = 400 litres
• Water = 600 × 1 / 3 = 200 litres
• Now (x + 50) litres of water is added.
• Milk remains 400 litres.
• After adding water, quantity of water becomes 70% of milk.
• So, final water = 70% of 400
• = 280 litres
• Therefore,
• Initial water + added water = 280
• 200 + (x + 50) = 280
• x + 250 = 280
• x = 30
• Now, if x litres of pure milk is added to the original mixture:
• Added milk = 30 litres
• Final milk = 400 + 30 = 430 litres
• Water remains = 200 litres
• So, water : milk = 200 : 430
• = 20 : 43
• Therefore, m = 20 and n = 43
• m + n = 20 + 43
• = 63

Solution Q.428 — **Correct Option: (E)**

• kg
• Solution:
• Let the common multiplier of Container P be a.
• In Container P, milk : water = 5 : 4
• So, milk in Container P = 5a and water in Container P = 4a
• Let the common multiplier of Container Q be b.
• In Container Q, milk : water : sugar = 3 : 7 : 2
• So, milk in Container Q = 3b, water in Container Q = 7b and sugar in Container Q =
• 2b
• After mixing both containers, total milk = 114 kg.
• So, 5a + 3b = 114
• After mixing both containers, total water = 174 kg.
• So, 4a + 7b = 174
• Now, solve both equations by elimination.
• First equation:
• 5a + 3b = 114
• Multiply it by 4:
• 20a + 12b = 456
• Second equation:
• 4a + 7b = 174
• Multiply it by 5:
• 20a + 35b = 870
• Now, subtract the first new equation from the second new equation:
• 20a + 35b - 20a - 12b = 870 - 456
• 23b = 414
• Value of b: 414/23
• = 18
• Sugar in Container Q = 2b
• Sugar = 2 × 18 = 36 kg
• Since Container P has no sugar, the sugar in the final mixture is 36 kg.

Solution Q.429 — **Correct Option: (C)**

• The container has 180 liters of mixture.
• The ratio of juice and water is 5 : 4.
• Total ratio parts = 5 + 4 = 9.
• Quantity of juice in the original mixture = 180 × 5/9 = 100 liters.
• Quantity of water in the original mixture = 180 × 4/9 = 80 liters.
• Now, 36 liters of the mixture is taken out.
• Since the mixture is taken out, juice and water are removed in the same ratio of 5 : 4.
• Quantity of juice removed = 36 × 5/9 = 20 liters.
• Quantity of water removed = 36 × 4/9 = 16 liters.
• Remaining juice = 100 - 20 = 80 liters.
• Remaining water = 80 - 16 = 64 liters.
• Now, 24 liters of water is added.
• New quantity of water = 64 + 24 = 88 liters.
• New quantity of juice = 80 liters.
• So, the new ratio of water to juice = 88 : 80.
• Now simplify the ratio.
• 88 and 80 have common factor 8.
• 88/8 = 11 and 80/8 = 10.
• So, simplified ratio of water to juice = 11 : 10.
• Sum of the numbers in the ratio = 11 + 10 = 21.

Solution Q.430 — **Correct Option: (C)**

• liters
• Solution:
• Let the initial quantity of the mixture = x liters.
• Milk : Water = 7:2
• Total parts = 7 + 2 = 9
• Initial milk = 7x/9
• Initial water = 2x/9
• 18 liters of mixture are removed.
• Milk removed from 18 liters = 18 × 7/9 = 14 liters
• Adding 6 liters of water does not change the quantity of milk.
• Final milk = 42 liters
• So, 7x/9 - 14 = 42
• 7x/9 = 42 + 14
• 7x/9 = 56
• x = 56 × 9/7
• x = 72 liters
• Initial water = 72 × 2/9
• Initial water = 16 liters

Solution Q.431 **Correct Option: (C)**

• Let the equal quantity of milk and water added = y liters each.
• Initially, milk = 180 liters and water = 0 liters.
• After adding y liters of milk and y liters of water:
• Milk = 180 + y
• Water = y
• Given ratio of milk to water = 7:3
• So, (180 + y)/y = 7/3
• Cross multiply:
• 3 × (180 + y) = 7y
• 540 + 3y = 7y
• 540 = 4y
• y = 540/4 = 135
• So, milk added = 135 liters and water added = 135 liters.
• Milk in the mixture = 180 + 135 = 315 liters
• Water in the mixture = 135 liters
• Total mixture = 315 + 135 = 450 liters
• Now, 2/3 of the mixture is removed.
• So, remaining mixture = 1/3 of 450 = 150 liters
• Milk remaining = 315 × 1/3 = 105 liters
• Water remaining = 135 × 1/3 = 45 liters
• Now, X liters of water is added.
• Final water = 45 + X
• Final mixture = 150 + X
• Given, water becomes 40% of the final mixture.
• So, 45 + X = 40/100 × (150 + X)
• 45 + X = 2/5 × (150 + X)
• Cross multiply:
• 5 × (45 + X) = 2 × (150 + X)
• 225 + 5X = 300 + 2X
• 5X - 2X = 300 - 225
• 3X = 75
• X = 75/3 = 25

Solution Q.432 **Correct Option: (E)**

• liters
• Solution:
• In the first vessel, water is 20%.
• So, milk in the first vessel = 100% - 20% = 80%
• In the second vessel, water is 55%.
• So, milk in the second vessel = 100% - 55% = 45%
• Required milk percentage in the final mixture = 60%
• Using alligation:
• Ratio of first mixture to second mixture = (60 - 45) : (80 - 60)
• Ratio = 15 : 20
• Ratio = 3 : 4
• So, first vessel mixture : second vessel mixture = 3 : 4
• Given mixture used from the second vessel = 96 liters
• Here, 4 parts = 96 liters
• 1 part = 96/4 = 24 liters
• Mixture used from the first vessel = 3 parts
• Mixture used from the first vessel = 3 × 24 = 72 liters

Solution Q.433 **Correct Option: (A)**

• Total quantity of mixture = 312 liters
• Ratio of milk to water = 5 : 7
• Total parts = 5 + 7 = 12
• Initial quantity of milk = 312 × 5/12 = 130 liters
• Initial quantity of water = 312 × 7/12 = 182 liters
• Now, 58 liters of water is added.
• Final quantity of water = 182 + 58 = 240 liters
• X liters of milk is added.
• Final quantity of milk = 130 + X
• Final ratio of milk to water = 2 : 3
• So, (130 + X)/240 = 2/3
• Cross multiply:
• 3 × (130 + X) = 2 × 240
• 390 + 3X = 480
• 3X = 480 - 390
• 3X = 90
• X = 90/3 = 30

Solution Q.434 **Correct Option: (A)**

• ml
• Solution:
• Total quantity of mixture = 96 ml.
• Liquid P is 40 ml more than liquid Q.
• Let liquid P = p ml and liquid Q = q ml.
• So, p + q = 96
• Also, p - q = 40
• Add both equations:
• 2p = 136
• p = 136/2 = 68 ml
• So, liquid P = 68 ml.
• Liquid Q = 96 - 68 = 28 ml.
• Now x ml of liquid P is removed.
• New quantity of liquid P = 68 - x.
• Quantity of liquid Q remains 28 ml.
• New ratio of P to Q = 3:2.
• So, (68 - x)/28 = 3/2
• Cross multiply:
• 2 × (68 - x) = 3 × 28
• 136 - 2x = 84
• 2x = 136 - 84
• 2x = 52
• x = 52/2 = 26
• Therefore, x = 26 ml.

Solution Q.435 **Correct Option: (D)**

• liters
• Solution:
• Let the initial quantity of the mixture = x liters.
• Initial water = 9 liters.
• Initial milk = x - 9 liters.
• Now, 6 liters of milk and 10 liters of water are added.
• Final milk = x - 9 + 6 = x - 3 liters.
• Final total mixture = x + 6 + 10 = x + 16 liters.
• Milk concentration in the final mixture = 75%.
• So,
• (x - 3)/(x + 16) = 75/100
• (x - 3)/(x + 16) = 3/4
• 4(x - 3) = 3(x + 16)
• 4x - 12 = 3x + 48
• 4x - 3x = 48 + 12
• x = 60

Solution Q.436 **Correct Option: (D)**

• liters
• Solution:
• Let the initial quantity of milk = x liters.
• Initial quantity of water = 12 liters
• Initial total mixture = x + 12 liters
• Milk added = 6 liters
• Water added = 18 liters
• Final quantity of milk = x + 6
• Final total mixture = x + 12 + 6 + 18 = x + 36
• Final concentration of milk = 75% = 75/100 = 3/4
• So, (x + 6)/(x + 36) = 3/4
• Cross multiply:
• 4(x + 6) = 3(x + 36)
• 4x + 24 = 3x + 108
• 4x - 3x = 108 - 24
• x = 84
• Initial total mixture = 84 + 12 = 96 liters

Solution Q.437 **Correct Option: (C)**

• Quantity of milk = 240 liters.
• Quantity of water added = (x + 40) liters.
• Total mixture = 240 + x + 40 = (x + 280) liters.
• 60 liters of mixture is removed.
• Milk removed = 60 × 240/(x + 280)
• After this, 60 liters of pure milk is added.
• Final milk = Initial milk - Milk removed + Milk added
• Final milk = 240 - 60 × 240/(x + 280) + 60
• Given final milk = 264 liters.
• So, 240 + 60 - 60 × 240/(x + 280) = 264
• 300 - 60 × 240/(x + 280) = 264
• 60 × 240/(x + 280) = 36
• 14400/(x + 280) = 36
• x + 280 = 14400/36
• x + 280 = 400
• x = 400 - 280
• x = 120

Solution Q.438 **Correct Option: (C)**

• Total mixture = 120 liters.
• Ratio of milk to water = 2 : 3.
• Total ratio parts = 2 + 3 = 5.
• Initial milk = 120 × 2/5 = 48 liters.
• Initial water = 120 × 3/5 = 72 liters.
• 25% of the mixture is taken out.
• Quantity taken out = 120 × 25/100 = 30 liters.
• Milk removed = 30 × 2/5 = 12 liters.
• Water removed = 30 × 3/5 = 18 liters.
• Remaining milk = 48 - 12 = 36 liters.
• Remaining water = 72 - 18 = 54 liters.
• Now X liters of milk are added.
• New milk = 36 + X.
• Water remains = 54 liters.
• New ratio of milk to water = 4 : 3.
• So,
• (36 + X)/54 = 4/3
• 3(36 + X) = 54 × 4
• 108 + 3X = 216
• 3X = 216 - 108
• 3X = 108
• X = 108/3
• X = 36

Solution Q.439 **Correct Option: (D)**

• liters
• Solution:
• In mixture A, milk = 30 liters.
• Milk is 150% of water.
• So, 30 = 150/100 × water
• 30 = 3/2 × water
• Water = 30 × 2/3
• Water = 20 liters.
• In mixture B, milk = 28 liters.
• Milk is 70% of water.
• So, 28 = 70/100 × water
• 28 = 7/10 × water
• Water = 28 × 10/7
• Water = 40 liters.
• Now both mixtures are mixed.
• Total milk = 30 + 28 = 58 liters.
• Total water = 20 + 40 = 60 liters.
• Required difference = 60 - 58
• Required difference = 2 liters.

Solution Q.440 **Correct Option: (C)**

• liters
• Solution:
• Ratio of quantity of mixture in Vessel P to Vessel Q = 3 : 5
• Let quantity in Vessel P = 3x liters.
• Let quantity in Vessel Q = 5x liters.
• Milk in Vessel P = 40% of 3x
• Milk in Vessel P = 3x × 40/100 = 6x/5
• Milk in Vessel Q = 36 liters
• Total final mixture = 3x + 5x = 8x liters
• Final mixture contains 30% milk.
• So, total milk in final mixture = 30% of 8x
• Total milk = 8x × 30/100 = 12x/5
• Now,
• Milk from P + Milk from Q = Total milk in final mixture
• 6x/5 + 36 = 12x/5
• 36 = 12x/5 - 6x/5
• 36 = 6x/5
• x = 36 × 5/6
• x = 30
• Total final mixture = 8x
• Total final mixture = 8 × 30 = 240 liters

Solution Q.441 **Correct Option: (C)**

• Total mixture = 150 liters
• Ratio of acid to water = 2:3
• Acid = 150 × 2/5 = 60 liters
• Water = 150 × 3/5 = 90 liters
• X liters of mixture is removed.
• Acid removed = 2X/5
• Water removed = 3X/5
• After removing X liters:
• Acid left = 60 - 2X/5
• Water left = 90 - 3X/5
• Now, (X - 10) liters of water is added.
• Final water = 90 - 3X/5 + X - 10
• Final water = 80 + 2X/5
• Final ratio of acid to water = 5:9
• So:
• (60 - 2X/5) : (80 + 2X/5) = 5 : 9
• 9(60 - 2X/5) = 5(80 + 2X/5)
• 540 - 18X/5 = 400 + 10X/5
• 540 - 400 = 18X/5 + 10X/5
• 140 = 28X/5
• X = 140 × 5/28
• X = 25

Solution Q.442 **Correct Option: (C)**

• Rs. 16
• Solution:
• Selling price of the mixture = Rs. 102 per kg.
• Profit = 20%.
• So, selling price = 120% of cost price.
• Average cost price of the mixture = 102 × 100/120
• Average cost price of the mixture = Rs. 85 per kg.
• Let the per kg cost prices of rice X and rice Y be Rs. 6x and Rs. 5x.
• Quantity of rice X = 25 kg.
• Quantity of rice Y = 55 kg.
• Total quantity = 25 + 55 = 80 kg.
• Total cost = 25 × 6x + 55 × 5x
• Total cost = 150x + 275x
• Total cost = 425x.
• Average cost price = Total cost/Total quantity.
• So, 425x/80 = 85
• 425x = 85 × 80
• 425x = 6800
• x = 6800/425
• x = 16.
• Difference between per kg cost prices = 6x - 5x
• Difference = x = Rs. 16.

Solution Q.443 **Correct Option: (E)**

• kg
• Solution:
• Let the initial quantities of zinc and copper be 4x kg and 7x kg respectively.
• After adding 32 kg zinc, final zinc = 4x + 32
• After adding 24 kg copper, final copper = 7x + 24
• Final ratio of zinc to copper = 2 : 3
• So, (4x + 32)/(7x + 24) = 2/3
• Cross multiply:
• 3(4x + 32) = 2(7x + 24)
• 12x + 96 = 14x + 48
• 14x - 12x = 96 - 48
• 2x = 48
• x = 48/2 = 24
• Initial quantity of copper = 7 × 24 = 168 kg
• Final quantity of copper = 168 + 24 = 192 kg

Solution Q.444 **Correct Option: (C)**

• kg
• Solution:
• Selling price of mixture = Rs. 68 per kg
• Loss = 15%
• So, selling price = 85% of cost price
• Cost price per kg of mixture = 68 × 100/85 = Rs. 80
• Now, wheat costs Rs. 100 per kg and maize costs Rs. 70 per kg.
• Average cost price of mixture = Rs. 80 per kg
• Using alligation:
• Quantity ratio of wheat to maize = (80 - 70):(100 - 80)
• Quantity ratio of wheat to maize = 10:20 = 1:2
• Wheat quantity = 30 kg
• So, maize quantity = 30 × 2 = 60 kg

Solution Q.445 **Correct Option: (B)**

• liters
• Solution:
• In mixture X:
• Milk = 45 liters
• Milk is 150% of water.
• So, 150% of water = 45
• Water = 45 × 100/150 = 30 liters
• In mixture Y:
• Milk = 36 liters
• Milk is 90% of water.
• So, 90% of water = 36
• Water = 36 × 100/90 = 40 liters
• Total milk after mixing = 45 + 36 = 81 liters
• Total water after mixing = 30 + 40 = 70 liters
• Difference = 81 - 70 = 11 liters

Solution Q.446 **Correct Option: (E)**

• Let the initial quantity of mixture P = M ml.
• Mango juice in mixture P = 30% of M = 3M/10.
• Water in mixture P = 70% of M = 7M/10.
• After adding 3x ml mango juice and x ml water:
• Mango juice = 3M/10 + 3x.
• Water = 7M/10 + x.
• Final ratio = 3:4.
• So, (3M/10 + 3x)/(7M/10 + x) = 3/4.
• By cross multiplication:
• 4 × (3M/10 + 3x) = 3 × (7M/10 + x).
• 12M/10 + 12x = 21M/10 + 3x.
• 12x - 3x = 21M/10 - 12M/10.
• 9x = 9M/10.
• x = M/10.
• So, M/x = 10.

Solution Q.447 **Correct Option: (C)**

• liters
• Solution:
• Let the initial quantity of milk and water be 4x liters and x liters.
• Initial total mixture = 5x liters
• After adding 50 liters of water:
• Quantity of water = x + 50
• Total mixture = 5x + 50
• Water becomes 30% of the final mixture.
• x + 50 = 30/100 × (5x + 50)
• x + 50 = 3/10 × (5x + 50)
• 10x + 500 = 15x + 150
• 5x = 350
• x = 70
• Initial quantity of milk = 4x = 4 × 70 = 280 liters

Solution Q.448 **Correct Option: (C)**

• %
• Solution:
• Cost price of first type = Rs. 75 per kg
• Cost price of second type = Rs. 125 per kg
• Mixing ratio = 3:2
• Average cost price of mixture = (75 × 3 + 125 × 2)/(3 + 2)
• Average cost price = (225 + 250)/5
• Average cost price = 475/5 = Rs. 95 per kg
• Selling price = Rs. 114 per kg
• Profit = 114 - 95 = Rs. 19
• Profit percentage = 19/95 × 100 = 20%

Solution Q.449 **Correct Option: (A)**

• liters
• Solution:
• Total mixture = 120 liters
• Milk : Water = 3:2
• Milk = 3/5 × 120 = 72 liters
• Water = 2/5 × 120 = 48 liters
• 30 liters of mixture is removed in the same ratio 3:2.
• Milk removed = 3/5 × 30 = 18 liters
• Water removed = 2/5 × 30 = 12 liters
• Milk left = 72 - 18 = 54 liters
• Water left = 48 - 12 = 36 liters
• Now x liters of water is added.
• Final milk = 54 liters
• Final water = 36 + x liters
• Final ratio of milk to water = 6:7
• 54/(36 + x) = 6/7
• 54 × 7 = 6 × (36 + x)
• 378 = 216 + 6x
• 6x = 162
• x = 27 liters

Solution Q.450 **Correct Option: (D)**

• kg
• Solution:
• Total mixture = 70 kg
• Ratio of spice to salt = 3:4
• Spice = 70 × 3/7 = 30 kg
• Salt = 70 × 4/7 = 40 kg
• Let x kg mixture be removed.
• Spice removed = 3x/7
• Salt removed = 4x/7
• Pure salt added = 2x kg
• New spice = 30 - 3x/7
• New salt = 40 - 4x/7 + 2x = 40 + 10x/7
• New ratio of spice to salt = 2:5
• So, 5 × (30 - 3x/7) = 2 × (40 + 10x/7)
• 150 - 15x/7 = 80 + 20x/7
• 70 = 35x/7
• 70 = 5x
• x = 14 kg
• Quantity of salt removed = 4x/7 = 4 × 14/7 = 8 kg

Solution Q.451 **Correct Option: (B)**

• days
• Solution:
• Take total work as 16 × 45 = 720 man-days.
• So, 1 man's 1 day work = 1/720.
• Also, 24 women complete the work in 60 days.
• Total work = 24 × 60 = 1440 woman-days.
• So, 1 woman's 1 day work = 1/1440.
• Work done by 36 women in 20 days = 36 × 20 / 1440 = 720 / 1440 = 1/2 of the work.
• Remaining work = 1 - 1/2 = 1/2.
• 24 men's 1 day work = 24 / 720 = 1/30.
• Time taken by 24 men to complete 1/2 work = (1/2) / (1/30) = 15 days.
• Total time taken = 20 + 15 = 35 days.

Solution Q.452 **Correct Option: (A)**

• liters
• Solution:
• Let the initial quantity of orange juice be J liters and water be W liters.
• Total mixture = 80 liters.
• So, J + W = 80.
• 87.5% of the initial mixture remains, so 12.5% of the mixture is removed.
• Quantity removed = 12.5% of 80 = 10 liters.
• 20% of orange juice and 10% of water are removed.
• So, 20% of J + 10% of W = 10.
• This means 0.20J + 0.10W = 10.
• Multiply by 10:
• 2J + W = 100.
• We already have J + W = 80.
• Subtracting, J = 20 liters.
• So, W = 80 - 20 = 60 liters.
• Therefore, the initial quantity of water is 60 liters.

Solution Q.453 **Correct Option: (D)**

• Selling price of mixture per kg = Rs. 65.
• Profit = 25%.
• Cost price of mixture per kg = 65 × 100 / 125 = Rs. 52.
• Let the cost price per kg of wheat type P and type Q be 7x and 4x respectively.
• Total cost of 48 kg wheat type P = 48 × 7x = 336x.
• Total cost of 72 kg wheat type Q = 72 × 4x = 288x.
• Total cost of mixture = 336x + 288x = 624x.
• Total quantity = 48 + 72 = 120 kg.
• Cost price per kg of mixture = 624x / 120 = 5.2x.
• So, 5.2x = 52.
• x = 52 / 5.2 = 10.
• Cost price per kg of wheat type P = 7 × 10 = Rs. 70.
• Cost price per kg of wheat type Q = 4 × 10 = Rs. 40.
• Required difference = 70 - 40 = Rs. 30.

Solution Q.454 **Correct Option: (E)**

• litres
• Solution:
• Total mixture = 128 litres
• Ratio of juice to water = 3:5
• Total ratio units = 3 + 5 = 8
• Quantity of juice = 128 × 3/8 = 48 litres
• Quantity of water = 128 × 5/8 = 80 litres
• To make juice and water equal, juice should become 80 litres.
• Required juice to be added = 80 - 48 = 32 litres

Solution Q.455 **Correct Option: (C)**

• liters
• Solution:
• Let the quantity of mixture in container X be 5a liters.
• Orange juice in X = 3a and water in X = 2a.
• Let the quantity of mixture in container Y be 12b liters.
• Orange juice in Y = 5b, water in Y = 6b and syrup in Y = b.
• Total orange juice = 120 liters.
• So, 3a + 5b = 120.
• Total water = 112 liters.
• So, 2a + 6b = 112.
• Multiply 3a + 5b = 120 by 2.
• 6a + 10b = 240.
• Multiply 2a + 6b = 112 by 3.
• 6a + 18b = 336.
• Now subtract:
• 8b = 96.
• b = 12.
• Quantity of syrup in the final mixture = b = 12 liters.

Solution Q.456 **Correct Option: (E)**

• Total mixture = 160 liters
• Ratio of milk to water = 3:5
• Total parts = 3 + 5 = 8
• Milk = 160 × 3/8 = 60 liters
• Water = 160 × 5/8 = 100 liters
• X liters of mixture are removed.
• Milk removed = 3X/8
• Water removed = 5X/8
• Milk left = 60 - 3X/8
• Water left = 100 - 5X/8
• Water added = X/2 - 5
• Final water = 100 - 5X/8 + X/2 - 5
• Final water = 95 - X/8
• Final ratio of milk to water = 1:2
• So, (60 - 3X/8) : (95 - X/8) = 1 : 2
• 2 × (60 - 3X/8) = 95 - X/8
• 120 - 6X/8 = 95 - X/8
• 120 - 95 = 6X/8 - X/8
• 25 = 5X/8
• X = 25 × 8 / 5 = 40
• Required value = X/4 = 40/4 = 10

Solution Q.457 **Correct Option: (D)**

• liters
• Solution:
• Total mixture = 150 liters.
• Milk : water = 3 : 2.
• Total parts = 3 + 2 = 5.
• Milk = 150 × 3/5 = 90 liters.
• Water = 150 × 2/5 = 60 liters.
• When x liters of mixture is removed, milk and water are removed in the ratio 3:2.
• Water removed = 2x/5.
• Milk removed = 3x/5.
• After adding x liters of pure milk:
• Final milk = 90 - 3x/5 + x = 90 + 2x/5.
• Final water = 60 - 2x/5.
• Final ratio of milk to water = 4 : 1.
• So, (90 + 2x/5) : (60 - 2x/5) = 4 : 1.
• 90 + 2x/5 = 4(60 - 2x/5).
• 90 + 2x/5 = 240 - 8x/5.
• 2x/5 + 8x/5 = 240 - 90.
• 10x/5 = 150.
• 2x = 150.
• x = 75 liters.

Solution Q.458 **Correct Option: (C)**

• Let the initial quantity in container A be 10a litres.
• Since milk and water are in the ratio 2:3, half of container A's mixture is 5a litres.
• In this 5a litres, milk = 2a litres and water = 3a litres.
• Let the initial quantity in container B be 8b litres.
• Since milk and water are in the ratio 3:1, half of container B's mixture is 4b litres.
• In this 4b litres, milk = 3b litres and water = b litres.
• After mixing the taken-out quantities, total milk = 60 litres.
• So,
• 2a + 3b = 60
• Total water = 55 litres.
• So,
• 3a + b = 55
• From the second equation:
• b = 55 - 3a
• Put this value in the first equation:
• 2a + 3(55 - 3a) = 60
• 2a + 165 - 9a = 60
• 165 - 7a = 60
• 7a = 105
• a = 15
• Initial quantity in container A = 10a = 10 × 15 = 150 litres.

Solution Q.459 **Correct Option: (A)**

• litres, 80 litres
• Solution:
• Let the initial quantity of milk be M litres.
• Let the initial quantity of water be W litres.
• The container is completely filled and its capacity is 100 litres.
• So,
• M + W = 100
• After selective removal, the container becomes 52% vacant.
• So, quantity removed = 52% of 100
• Quantity removed = 52 litres
• According to the question:
• 60% of milk and 20% of water are removed.
• So,
• 60% of M + 20% of W = 52
• 0.60M + 0.20W = 52
• Multiply by 10:
• 6M + 2W = 520
• Divide by 2:
• 3M + W = 260
• Now subtract M + W = 100 from 3M + W = 260:
• 2M = 160
• M = 80
• So,
• W = 100 - 80
• W = 20

Solution Q.460 **Correct Option: (B)**

• litres
• Solution:
• Quantity of pure milk = 75 litres.
• Cost price of milk = Rs. 30 per litre.
• Total cost price = 75 × 30 = Rs. 2250.
• Let the quantity of water mixed = x litres.
• Since water has no cost, the total cost price remains Rs. 2250.
• Total quantity of mixture = 75 + x litres.
• The mixture is sold at Rs. 30 per litre.
• So selling price = (75 + x) × 30.
• Profit = Selling price - Cost price
• Profit = 30(75 + x) - 2250
• Profit = 2250 + 30x - 2250
• Profit = 30x.
• Given profit percentage = 32%.
• So profit = 32% of 2250
• = 2250 × 32 / 100
• = Rs. 720.
• Therefore,
• 30x = 720
• x = 24.

Solution Q.461 **Correct Option: (E)**

• litres
• Solution:
• Vessel A has milk and water in the ratio 3:1.
• So, in 36 litres taken from vessel A:
• Milk = 36 × 3 / 4 = 27 litres.
• Water = 36 × 1 / 4 = 9 litres.
• Let the initial quantity of mixture in vessel B = y litres.
• Vessel B has milk and water in the ratio 2:1.
• So initial milk in B = 2y / 3.
• Initial water in B = y / 3.
• After adding 36 litres from vessel A:
• Milk in B = 2y / 3 + 27.
• Water in B = y / 3 + 9.
• New ratio in B = 5:2.
• So,
• (2y / 3 + 27) : (y / 3 + 9) = 5 : 2
• 2(2y / 3 + 27) = 5(y / 3 + 9)
• 4y / 3 + 54 = 5y / 3 + 45
• 9 = y / 3
• y = 27.
• So initial quantity in vessel B = 27 litres.
• New quantity in vessel B = 27 + 36 = 63 litres.
• This is 25% less than the initial quantity in vessel A.
• So 63 is 75% of the initial quantity in vessel A.
• Initial quantity in vessel A = 63 × 100 / 75 = 84 litres.
• Milk in vessel A = 75% of 84 = 63 litres.

Solution Q.462 **Correct Option: (A)**

• kg
• Solution:
• Price of type P rice = Rs. 28 per kg.
• Price of type Q rice = Rs. 42 per kg.
• Average price of mixture = Rs. 33.60 per kg.
• By alligation,
• Quantity of type P : Quantity of type Q = (42 - 33.60) : (33.60 - 28).
• = 8.40 : 5.60.
• = 3 : 2.
• Total ratio = 3 + 2 = 5.
• Quantity of type P rice = 150 × 3/5 = 90 kg.

Solution Q.463 **Correct Option: (B)**

• Initial quantity of pure milk = 200 liters.
• 25% of milk is replaced with water.
• Milk left = 75% of 200 = 150 liters.
• Water in the mixture = 25% of 200 = 50 liters.
• Let the remaining part after taking out X% mixture be k.
• Milk left = 150k.
• Water left = 50k.
• After adding 40 liters of water, water = 50k + 40.
• Final ratio of milk to water = 3 : 2.
• 150k/(50k + 40) = 3/2.
• 300k = 150k + 120.
• 150k = 120.
• k = 120/150 = 4/5.
• So, remaining mixture = 4/5 = 80%.
• Mixture taken out = 100% - 80% = 20%.
• X = 20.
• Required value = X + 7 = 20 + 7 = 27.

Solution Q.464 **Correct Option: (E)**

• liters
• Solution:
• Milk is 25% more than water.
• So, milk : water = 125 : 100 = 5 : 4.
• Let the initial quantities of milk and water be 5x liters and 4x liters respectively.
• Total mixture = 9x liters.
• When 45 liters of mixture is removed, milk and water are removed in the ratio 5 : 4.
• Milk removed = 45 × 5/9 = 25 liters.
• Water removed = 45 × 4/9 = 20 liters.
• Milk left = 5x - 25.
• Water left = 4x - 20.
• After adding 30 liters of water:
• Water = 4x - 20 + 30 = 4x + 10.
• New ratio of milk to water = 5 : 6.
• So, (5x - 25)/(4x + 10) = 5/6.
• 6 × (5x - 25) = 5 × (4x + 10).
• 30x - 150 = 20x + 50.
• 10x = 200.
• x = 20.
• Initial quantity of milk = 5x = 5 × 20 = 100 liters.
• Therefore, the initial quantity of milk is 100 liters.

Solution Q.465 **Correct Option: (C)**

• liters
• Solution:
• Let the initial quantity of mixture = x liters.
• 18 liters of mixture is taken out.
• 42 liters of water is added.
• So, final quantity of mixture = x - 18 + 42
• Final quantity = x + 24
• Given, final quantity : initial quantity = 6:5
• So, (x + 24)/x = 6/5
• Cross multiply:
• 5 × (x + 24) = 6x
• 5x + 120 = 6x
• 6x - 5x = 120
• x = 120

Solution Q.466 **Correct Option: (B)**

• liters
• Solution:
• From Vessel A, 50 liters mixture is taken out.
• Ratio (milk : water in Vessel A) = 3 : 2
• Total parts = 3 + 2 = 5
• Milk from Vessel A = 50 × 3/5 = 30 liters
• Water from Vessel A = 50 × 2/5 = 20 liters
• From Vessel B, 45 liters mixture is taken out.
• Ratio (milk : water in Vessel B) = 5 : 4
• Total parts = 5 + 4 = 9
• Milk from Vessel B = 45 × 5/9 = 25 liters
• Water from Vessel B = 45 × 4/9 = 20 liters
• Total milk added to Vessel C = 30 + 25 = 55 liters
• Total water added to Vessel C = 20 + 20 = 40 liters
• Let the initial quantity of mixture in Vessel C = x liters.
• In Vessel C, milk and water are in the ratio of 2 : 1.
• Initial milk in Vessel C = 2x/3
• Initial water in Vessel C = x/3
• Final milk in Vessel C = 55 + 2x/3
• Final water in Vessel C = 40 + x/3
• Final ratio of milk to water = 5 : 3
• So, (55 + 2x/3)/(40 + x/3) = 5/3
• Cross multiply:
• 3 × (55 + 2x/3) = 5 × (40 + x/3)
• 165 + 2x = 200 + 5x/3
• Multiply by 3:
• 495 + 6x = 600 + 5x
• 6x - 5x = 600 - 495
• x = 105

Solution Q.467 **Correct Option: (C)**

• Initial quantity of mixture = 140 liters
• Ratio of milk to water = 5 : y
• Initial milk = 140 × 5/(5 + y)
• Initial water = 140 × y/(5 + y)
• 25 liters of milk and 15 liters of water are added
• New milk = 140 × 5/(5 + y) + 25
• New water = 140 × y/(5 + y) + 15
• Final ratio of milk to water = 5 : 7
• [140 × 5/(5 + y) + 25] / [140 × y/(5 + y) + 15] = 5/7
• 7 × [700/(5 + y) + 25] = 5 × [140y/(5 + y) + 15]
• 4900/(5 + y) + 175 = 700y/(5 + y) + 75
• Multiply by (5 + y)
• 4900 + 175(5 + y) = 700y + 75(5 + y)
• 4900 + 875 + 175y = 700y + 375 + 75y
• 5775 + 175y = 775y + 375
• 5400 = 600y
• y = 9

Solution Q.468 **Correct Option: (A)**

• liters
• Solution:
• Let the initial quantities of syrup and water be 5x liters and 3x liters respectively.
• After adding 64 liters of water,
• Quantity of syrup = 5x liters
• Quantity of water = 3x + 64 liters
• New ratio of syrup to water = 3 : 5
• So, 5x/(3x + 64) = 3/5
• 25x = 9x + 192
• 16x = 192
• x = 12
• Original quantity of mixture = 5x + 3x = 8x
• Original quantity = 8 × 12 = 96 liters
• Therefore, the original quantity of the mixture is 96 liters.

Solution Q.469 **Correct Option: (B)**

• %
• Solution:
• Initial Ratio (P : Q) = 5 : 4
• Total initial mixture = 270 liters
• Initial quantity of Q = 270 × 4/9 = 120 liters
• After some mixture is taken out, the ratio of P and Q remains the same because P and
• Q are taken out in the same ratio.
• Resultant ratio of P, Q and R = 10 : 8 : 9
• Here, P : Q = 10 : 8 = 5 : 4, so the ratio is consistent.
• Total parts = 10 + 8 + 9 = 27
• Quantity of R added = 270 × 9/27 = 90 liters
• So, quantity of mixture taken out = 90 liters
• Required percentage = 90/120 × 100 = 75%
• Therefore, the quantity of mixture taken out is 75% of the initial quantity of liquid Q.

Solution Q.470 **Correct Option: (C)**

• liters
• Solution:
• Let the quantities of mixture in vessels A and B be 3x and 7x liters respectively.
• Total resultant mixture = 3x + 7x = 10x liters
• Milk in vessel A = 40% of 3x
• = 40/100 × 3x
• = 6x/5 liters
• Milk in vessel B = 36 liters
• Total milk in resultant mixture = 6x/5 + 36
• Resultant mixture contains 30% milk.
• So, milk in resultant mixture = 30% of 10x
• = 30/100 × 10x
• = 3x
• According to the question,
• 6x/5 + 36 = 3x
• 36 = 3x - 6x/5
• 36 = 9x/5
• x = 36 × 5/9
• x = 20
• Total quantity of resultant mixture = 10x = 10 × 20 = 200 liters

CHAPTER 08: MENSURATION (2D AREAS & 3D VOLUMES)

26 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Circle: Area = πr^2 | Perimeter = 2πr | Semi-circle Perimeter = πr + 2r
- Cylinder: Volume = πr^2h | Curved Surface Area (CSA) = 2πrh | Total SA = 2πr(r + h)
- Cone: Volume = 1/3 πr^2h | Slant height l = √(r^2 + h^2) | CSA = πrl
- Cube: Volume = a^3 | SA = 6a^2 | Cuboid: Volume = l × b × h | SA = 2(lb + bh + hl)

Q.471 Mensuration

The total surface area of a solid sphere is 22176 m². The sphere is melted and recast into a cylinder of height 14 meters. Find the curved surface area of the cylinder. Take pi as 22 / 7.

(A) 6864 (B) 7056
(C) 7216 (D) 7392
(E) 7560

Q.472 Mensuration

The volume of a cube is 125 cm³. The side of the cube is equal to the breadth of a cuboid. The length of the cuboid is 3 cm more than its height. If the volume of the cuboid is 350 cm³, find the height of the cuboid.

(A) 6 cm (B) 7 cm
(C) 9 cm (D) 11 cm
(E) 14 cm

Q.473 Mensuration

The area of a rectangular ground having length 140 meters and breadth 35 meters is equal to the area of an isosceles right-angled triangle. If the radius of a sphere is 15% of the hypotenuse of that triangle, find the total surface area of the sphere. Use pi = 22/7.

(A) 4312 sq. meters (B) 4928 sq. meters
(C) 6160 sq. meters (D) 5544 sq. meters
(E) 7056 sq. meters

Q.474 Mensuration

The base of a right-angled triangle is equal to the length of a rectangle whose breadth is 54 cm. If the height of the triangle is 26 cm and the area of the rectangle is 3240 square cm, find the area of the right-angled triangle.

(A) 650 square cm (B) 720 square cm
(C) 780 square cm (D) 840 square cm
(E) 900 square cm

Q.475 Mensuration

The circumference of a circular field is 220 meters. The length of a rectangular field is 160% of the radius of the circular field. If the area of the rectangular field is 1568 square meters, find the percentage by which the radius of the circular field is more than the breadth of the rectangular field. Use π = 22/7.

(A) 20% (B) 25%
(C) 30% (D) 35%
(E) 40%

Q.476 Mensuration

The circumference of a circular garden is 440 meters. The length of a rectangular garden is 160% of the radius of the circular garden. If the area of the rectangular garden is 6272 square meters, find by what percentage the radius of the circular garden is more than the breadth of the rectangular garden. Use π = 22/7.

(A) 30% (B) 40%
(C) 20% (D) 50%
(E) 25%

Q.477 Mensuration

The sum of two numbers is equal to the sum of the square of 12 and the cube of

Q.478 Mensuration

The perimeter of a right-angled triangle is 84 meters and its hypotenuse is 35 meters. If the base is the smallest side, find the length of the base.

(A) 15 meters (B) 18 meters
(C) 24 meters (D) 21 meters
(E) 28 meters

Q.479 Mensuration

A solid sphere of radius 15 cm is melted and recast into a cylinder whose radius is the same as that of the sphere. Find the height of the cylinder.

(A) 15 cm (B) 18 cm
(C) 20 cm (D) 24 cm
(E) 30 cm

Q.480 Mensuration

The sum of the breadth of a rectangle and the side of a square is 39 meters. If the perimeter of the rectangle is 84 meters and the length of the rectangle is 26 meters, find the side of the square.

(A) 19 meters (B) 21 meters
(C) 25 meters (D) 18 meters
(E) 23 meters

Q.481 Mensuration

The area of a square is 784 sq. cm, and its side is equal to the side of a cube. Find the total surface area of the cube.

(A) 4200 cm^2 (B) 4480 cm^2
(C) 4704 cm^2 (D) 4900 cm^2
(E) 5040 cm^2

Q.482 Mensuration

The circumference of a circle with radius 14 cm is 16 cm less than the perimeter of a square. Find the area of the square. Use pi = 22/7.

(A) 625 sq. cm (B) 576 sq. cm
(C) 676 sq. cm (D) 729 sq. cm
(E) 784 sq. cm

Q.483 Mensuration

The total cost of fencing a circular park of radius 10.5 metres is Rs. 5280. If the fencing cost per metre for a square park is the same and the total cost of fencing the square park is Rs. 5760, then find the side of the square park in metres. Take π = 22/7.

(A) 14 metres (B) 20 metres
(C) 16 metres (D) 15 metres
(E) 18 metres

Q.484 Mensuration

The perimeter of a square is 24 m more than the perimeter of a rectangle. The area of the square is 225 m^2 more than the area of the rectangle. If the breadth of the rectangle is 6 m less than its length, then find the length of the rectangle.

(A) 15 m (B) 18 m
(C) 20 m (D) 22 m
(E) 24 m

Q.485 Mensuration

The diameter of a circle is equal to the side of a square whose perimeter is 168 cm. Find the difference between the area of the square and the area of the circle. Use π = 22/7.

(A) 378 cm^2 (B) 392 cm^2
(C) 336 cm^2 (D) 420 cm^2
(E) 350 cm^2

Q.486 Mensuration

The circumference of the first circle is 176 cm and the circumference of the second circle is 132 cm. Find the difference between the areas of both circles. Use π = 22/7.

(A) 1008 cm^2 (B) 968 cm^2
(C) 1078 cm^2 (D) 1155 cm^2
(E) 1022 cm^2

Q.487 Mensuration

The breadth of a rectangle is 32 cm. When its length is increased by 5 cm, the area of the rectangle becomes equal to the area of a circle whose diameter is 56 cm. Find the original length of the rectangle. Use pi = 22/7.

(A) 68 cm (B) 74 cm
(C) 72 cm (D) 76 cm
(E) 80 cm

Q.488 Mensuration

The height of a cone is equal to the side of a square. If the radius of the cone is 7 cm and its volume is 924 cm^3, then find the perimeter of the square. Use pi = 22/7.

(A) 64 cm (B) 84 cm
(C) 56 cm (D) 80 cm
(E) 72 cm

Q.489 Mensuration

The perimeter of a square is 168 cm, and the height of a cylinder is 33 1/3% less than the side of the square. If the volume of the cylinder is 38808 cubic cm, then find the total surface area of the cylinder. Use pi = 22/7.

(A) 6248 sq cm (B) 6864 sq cm
(C) 6720 sq cm (D) 6468 sq cm
(E) 6048 sq cm

Q.490 Mensuration

The perimeter of a rectangle is y cm and its breadth is 10 cm. The radius of a circle is 6 cm less than the length of the rectangle. If the circumference of the circle is 28 cm more than the perimeter of the rectangle, find the area of the circle. Take π = 22/7.

(A) 616 cm²
(B) 554 cm²
(C) 484 cm²
(D) 770 cm²
(E) 660 cm²

Q.491 Mensuration

The length of a rectangular notice board is 12 cm more than its breadth. If its length is increased by 6 cm and its breadth is decreased by 4 cm, its area remains unchanged. Find the original area of the notice board.

(A) 1644 cm²
(B) 1728 cm²
(C) 1800 cm²
(D) 1896 cm²
(E) 1600 cm²

Q.492 Mensuration

The length of a cuboid is doubled, its breadth becomes four times the original breadth, and its height is reduced to half. The new volume of the cuboid is how many times the original volume?

(A) 4
(B) 3
(C) 5
(D) 6
(E) 2

Q.493 Mensuration

A circular piece is cut out from a square metal sheet in such a way that the side of the square is equal to the diameter of the circle. If the area of the remaining part of the sheet is 378 cm², then find the radius of the circle. Use pi = 22/7.

(A) 14 cm
(B) 21 cm
(C) 28 cm
(D) 35 cm
(E) 7 cm

Q.494 Mensuration

The length of a rectangular garden is 40% more than its breadth. If the area of the garden is 3500 sq. meters, find its perimeter.

(A) 220 meters
(B) 260 meters
(C) 240 meters
(D) 280 meters
(E) 200 meters

Q.495 Mensuration

The total surface area of a cube is 384 cm². If the length of a rectangle is 10 cm more than the edge of the cube and the perimeter of the rectangle is 60 cm, then find the area of the rectangle.

(A) 180 cm²
(B) 216 cm²
(C) 240 cm²
(D) 192 cm²
(E) 252 cm²

Q.496 Mensuration

The length of a rectangle is 28 cm, and the side of a square is 30 cm less than the sum of the length and breadth of the rectangle. If the perimeter of the rectangle is 20 cm more than the perimeter of the square, then find the area of the square.

(A) 324 cm²
(B) 484 cm²
(C) 256 cm²
(D) 361 cm²
(E) 400 cm²

Chapter 08 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.471 — **Correct Option: (D)**

• Total surface area of sphere = 22176 m²
• Formula for total surface area of sphere = 4 × π × r × r
• Using π = 22 / 7:
• 4 × 22 / 7 × r × r = 22176
• 88 × r × r / 7 = 22176
• r × r = 22176 × 7 / 88
• r × r = 1764
• So, radius of sphere = 42 meters
• Now the sphere is melted and recast into a cylinder.
• So, volume of sphere = volume of cylinder
• Volume of sphere = 4 × π × r × r × r / 3
• = 4 × π × 42 × 42 × 42 / 3
• Height of cylinder = 14 meters
• Let radius of cylinder = R
• Volume of cylinder = π × R × R × 14
• So,
• π × R × R × 14 = 4 × pi × 42 × 42 × 42 / 3
• Cancel pi from both sides:
• R × R × 14 = 4 × 42 × 42 × 42 / 3
• R × R = 4 × 42 × 42 × 42 / (3 × 14)
• R × R = 7056
• R = 84 meters
• Curved surface area of cylinder = 2 × π × R × height
• = 2 × 22 / 7 × 84 × 14
• = 7392 m²

Solution Q.472 — **Correct Option: (B)**

• cm
• Solution:
• Volume of the cube = 125 cm³.
• Side of cube means cube root of 125.
• Since 5 × 5 × 5 = 125, side of the cube = 5 cm.
• According to the question, breadth of the cuboid = side of the cube.
• So, breadth of the cuboid = 5 cm.
• Let the height of the cuboid be h cm.
• Then the length of the cuboid = h + 3 cm.
• Volume of cuboid = length × breadth × height.
• So, 350 = (h + 3) × 5 × h.
• Now divide both sides by 5.
• h × (h + 3) = 70.
• Now use the banking exam shortcut by checking the factor pair of 70 whose difference
• is 3.
• The factor pair is 7 and 10.
• Here, height = 7 and length = 10.
• Length is 3 cm more than height, so this satisfies the condition.

Solution Q.473 — **Correct Option: (D)**

• sq. meters
• Solution:
• Area of rectangular ground = length × breadth.
• Area = 140 × 35 = 4900 sq. meters.
• This area is equal to the area of an isosceles right-angled triangle.
• For an isosceles right-angled triangle:
• Area = hypotenuse²/4.
• So, hypotenuse²/4 = 4900.
• Hypotenuse² = 4900 × 4 = 19600.
• Hypotenuse = 140 meters.
• Radius of the sphere = 15% of hypotenuse.
• Radius = 15/100 × 140 = 21 meters.
• Total surface area of sphere = 4 × pi × r × r.
• Using pi = 22/7:
• Total surface area = 4 × 22/7 × 21 × 21.
• Total surface area = 5544 sq. meters.

Solution Q.474 — **Correct Option: (C)**

• square cm
• Solution:
• Area of rectangle = 3240 square cm
• Breadth of rectangle = 54 cm
• Length of rectangle = Area/Breadth
• Length = 3240/54 = 60 cm
• The base of the right-angled triangle is equal to the length of the rectangle.
• So, base of triangle = 60 cm
• Height of triangle = 26 cm
• Area of right-angled triangle = 1/2 × Base × Height
• Area = 1/2 × 60 × 26
• Area = 30 × 26 = 780 square cm

Solution Q.475 — **Correct Option: (B)**

• %
• Solution:
• Circumference of circular field = 220 meters.
• Using π = 22/7,
• Circumference = 2πr
• So, 220 = 2 × 22/7 × r
• r = 220 × 7/44
• r = 35 meters.
• Radius of circular field = 35 meters.
• Length of rectangular field = 160% of radius
• Length = 160/100 × 35
• Length = 56 meters.
• Area of rectangular field = 1568 square meters.
• Breadth = Area/Length
• Breadth = 1568/56
• Breadth = 28 meters.
• Required percentage more = (Radius - Breadth) × 100/Breadth
• Required percentage more = (35 - 28) × 100/28
• Required percentage more = 7 × 100/28
• Required percentage more = 25%
• Therefore, the radius of the circular field is 25% more than the breadth of the
• rectangular field.

Solution Q.476 — **Correct Option: (E)**

• %
• Solution:
• Circumference of circular garden = 440 meters.
• Using π = 22/7,
• Circumference = 2 × π × radius
• 440 = 2 × 22/7 × radius
• Radius = 440 × 7/44
• Radius = 70 meters.
• Length of rectangular garden = 160% of radius.
• Length = 160/100 × 70
• Length = 112 meters.
• Area of rectangular garden = 6272 square meters.
• Breadth = Area/Length
• Breadth = 6272/112
• Breadth = 56 meters.
• Required percentage more = (Radius - Breadth) × 100/Breadth
• Required percentage more = (70 - 56) × 100/56
• Required percentage more = 14 × 100/56
• Required percentage more = 25%

Solution Q.477 **Correct Option: (B)**

• Sum of the two numbers = square of 12 + cube of 8.
• Square of 12 = 12 × 12 = 144.
• Cube of 8 = 8 × 8 × 8 = 512.
• Sum of two numbers = 144 + 512 = 656.
• Larger number is 20 less than the square of 24.
• Square of 24 = 24 × 24 = 576.
• Larger number = 576 - 20 = 556.
• Smaller number = 656 - 556 = 100.
• Now find twice 40% of the smaller number.
• 40% of smaller number = 40/100 × 100 = 40.
• Twice 40% of smaller number = 2 × 40 = 80.
• Half of larger number = 556/2 = 278.
• Required sum = 80 + 278 = 358.

Solution Q.478 **Correct Option: (D)**

• meters
• Solution:
• Perimeter of the right-angled triangle = 84 meters
• Hypotenuse = 35 meters
• Sum of the other two sides = 84 - 35
• Sum of the other two sides = 49 meters
• Let the two smaller sides = x and 49 - x.
• According to Pythagoras theorem:
• $x^2 + (49 - x)^2 = 35^2$
• $x^2 + 2401 - 98x + x^2 = 1225$
• $2x^2 - 98x + 2401 = 1225$
• $2x^2 - 98x + 1176 = 0$
• $x^2 - 49x + 588 = 0$
• $x^2 - 21x - 28x + 588 = 0$
• $x(x - 21) - 28(x - 21) = 0$
• $(x - 21)(x - 28) = 0$
• So, the two sides are 21 meters and 28 meters.
• The base is the smallest side.
• Therefore, the base is 21 meters.

Solution Q.479 **Correct Option: (C)**

• cm
• Solution:
• When a sphere is melted and recast into a cylinder, their volumes are equal.
• Radius of sphere = 15 cm
• Radius of cylinder = 15 cm
• Volume of sphere = 4/3 × π × r × r × r
• Volume of cylinder = π × r × r × h
• Since both volumes are equal:
• 4/3 × π × 15 × 15 × 15 = π × 15 × 15 × h
• Cancel π × 15 × 15 from both sides.
• h = 4/3 × 15
• h = 60/3 = 20 cm

Solution Q.480 **Correct Option: (E)**

• meters
• Solution:
• Perimeter of rectangle = 84 meters
• Formula of perimeter = 2 × (Length + Breadth)
• So, 2 × (Length + Breadth) = 84
• Length + Breadth = 84/2 = 42 meters
• Length of rectangle = 26 meters
• Breadth of rectangle = 42 - 26 = 16 meters
• Given, breadth of rectangle + side of square = 39 meters
• Side of square = 39 - 16 = 23 meters

Solution Q.481 **Correct Option: (C)**

• cm^2
• Solution:
• Area of square = 784 sq. cm.
• Side of square = square root of 784 = 28 cm.
• Side of cube = side of square = 28 cm.
• Total surface area of cube = 6 × side × side.
• Total surface area = 6 × 28 × 28.
• Total surface area = 6 × 784.
• Total surface area = 4704 sq. cm.

Solution Q.482 **Correct Option: (C)**

• sq. cm
• Solution:
• Radius of the circle = 14 cm.
• Circumference of circle = 2 × pi × radius.
• Using pi = 22/7:
• Circumference = 2 × 22/7 × 14.
• Circumference = 88 cm.
• The circumference of the circle is 16 cm less than the perimeter of the square.
• So, perimeter of square = 88 + 16 = 104 cm.
• Side of square = Perimeter/4.
• Side of square = 104/4 = 26 cm.
• Area of square = side × side.
• Area = 26 × 26 = 676 sq. cm.

Solution Q.483 **Correct Option: (E)**

• metres
• Solution:
• Radius of circular park = 10.5 metres
• Circumference of circular park = 2πr
• Circumference = 2 × 22/7 × 10.5
• Circumference = 66 metres
• Total cost of fencing circular park = Rs. 5280
• Cost per metre = 5280/66 = Rs. 80
• Total cost of fencing square park = Rs. 5760
• Perimeter of square park = 5760/80 = 72 metres
• Side of square park = 72/4 = 18 metres

Solution Q.484 **Correct Option: (B)**

• m
• Solution:
• Let the length of the rectangle = x m.
• Breadth of the rectangle = x - 6 m.
• Perimeter of rectangle = 2 × (x + x - 6) = 4x - 12.
• Perimeter of square = 24 m more than rectangle.
• So, perimeter of square = 4x - 12 + 24 = 4x + 12.
• Side of square = (4x + 12) / 4 = x + 3.
• Area of square = (x + 3) × (x + 3).
• Area of rectangle = x × (x - 6).
• Given difference in areas = 225.
• So, (x + 3) × (x + 3) - x × (x - 6) = 225.
• $x^2 + 6x + 9 - x^2 + 6x = 225$.
• 12x + 9 = 225.
• 12x = 216.
• x = 216 / 12 = 18.
• Length of rectangle = 18 m.

Solution Q.485 **Correct Option: (A)**

• cm^2
• Solution:
• Perimeter of square = 168 cm.
• Side of square = 168 / 4 = 42 cm.
• Diameter of circle = side of square = 42 cm.
• Radius of circle = 42 / 2 = 21 cm.
• Area of square = 42 × 42 = 1764 cm^2.
• Area of circle = 22/7 × 21 × 21 = 1386 cm^2.
• Required difference = 1764 - 1386 = 378 cm^2.

Solution Q.486 **Correct Option: (C)**

• cm²
• Solution:
• Use circumference = 2 × π × radius.
• Take π = 22/7.
• For the first circle:
• 176 = 2 × 22/7 × radius.
• Radius = 176 × 7 / 44 = 28 cm.
• For the second circle:
• 132 = 2 × 22/7 × radius.
• Radius = 132 × 7 / 44 = 21 cm.
• Difference between areas = π × (28 × 28 - 21 × 21).
• = 22/7 × (784 - 441).
• = 22/7 × 343.
• = 22 × 49 = 1078 cm².

Solution Q.487 **Correct Option: (C)**

• cm
• Solution:
• Diameter of circle = 56 cm
• Radius of circle = 56 / 2 = 28 cm
• Area of circle = 22/7 × 28 × 28 = 2464 square cm
• Let the original length of the rectangle be L cm.
• New length of rectangle = L + 5 cm
• Breadth of rectangle = 32 cm
• According to the question:
• 32 × (L + 5) = 2464
• L + 5 = 2464 / 32 = 77
• L = 77 - 5 = 72 cm
• So, the original length of the rectangle is 72 cm.

Solution Q.488 **Correct Option: (E)**

• cm
• Solution:
• Volume of cone = 1/3 × pi × r × r × h.
• Given, radius = 7 cm and volume = 924 cm³.
• Using pi = 22/7:
• 924 = 1/3 × 22/7 × 7 × 7 × h.
• 924 = 154h / 3.
• h = 924 × 3 / 154.
• h = 18 cm.
• The side of the square is equal to the height of the cone.
• So, side of square = 18 cm.
• Perimeter of square = 4 × side = 4 × 18 = 72 cm.

Solution Q.489 **Correct Option: (D)**

• sq cm
• Solution:
• Perimeter of square = 168 cm
• Side of square = 168 / 4 = 42 cm
• Height of cylinder is 33 1/3% less than the side.
• 33 1/3% = 1/3
• Height of cylinder = 42 × 2/3 = 28 cm
• Volume of cylinder = pi × r × r × h
• 38808 = 22/7 × r × r × 28
• 22/7 × 28 = 88
• r × r = 38808 / 88 = 441
• Radius = 21 cm
• Total surface area of cylinder = 2 × pi × r × (r + h)
• Total surface area = 2 × 22/7 × 21 × (21 + 28)
• Total surface area = 2 × 22/7 × 21 × 49
• Total surface area = 6468 sq cm

Solution Q.490 **Correct Option: (A)**

• cm²
• Solution:
• Let the length of the rectangle be L cm.
• Breadth = 10 cm.
• Perimeter of rectangle = 2(L + 10).
• Radius of circle = L - 6.
• Circumference of circle = 2 × 22/7 × (L - 6).
• Given, circumference of circle is 28 cm more than the perimeter of rectangle.
• So, 2 × 22/7 × (L - 6) = 2(L + 10) + 28.
• 44/7 × (L - 6) = 2L + 20 + 28.
• 44/7 × (L - 6) = 2L + 48.
• Multiply by 7:
• 44(L - 6) = 14L + 336.
• 44L - 264 = 14L + 336.
• 44L - 14L = 336 + 264.
• 30L = 600.
• L = 20 cm.
• Radius of circle = L - 6 = 20 - 6 = 14 cm.
• Area of circle = 22/7 × 14 × 14 = 616 cm².

Solution Q.491 **Correct Option: (B)**

• cm²
• Solution:
• Let the breadth of the notice board = x cm.
• Then length = x + 12 cm
• Original area = x × (x + 12)
• After change:
• New length = x + 12 + 6 = x + 18
• New breadth = x - 4
• Area remains unchanged.
• So,
• x(x + 12) = (x + 18)(x - 4)
• x² + 12x = x² + 14x - 72
• 12x = 14x - 72
• 2x = 72
• x = 36
• Breadth = 36 cm
• Length = 36 + 12 = 48 cm
• Original area = 48 × 36
• Original area = 1728 cm²

Solution Q.492 **Correct Option: (A)**

• Let the original volume of the cuboid be V.
• Volume of cuboid = Length × Breadth × Height
• New length = 2 times the original length
• New breadth = 4 times the original breadth
• New height = half of the original height
• So, new volume factor
• = 2 × 4 / 2
• = 4
• Therefore, new volume = 4V.
• Hence, the new volume is 4 times the original volume.

Solution Q.493 **Correct Option: (B)**

• cm
• Solution:
• Let the radius of the circle = r cm
• Diameter of the circle = 2r cm
• Side of the square = 2r cm
• Area of square = 2r × 2r = 4r²
• Area of circle = 22/7 × r²
• Remaining area = Area of square - Area of circle
• Remaining area = 4r² - 22/7 × r²
• Remaining area = 28/7 × r² - 22/7 × r²
• Remaining area = 6/7 × r²
• According to the question:
• 6/7 × r² = 378
• r² = 378 × 7 / 6
• r² = 441
• r = 21
• Radius of the circle = 21 cm

Solution Q.494 **Correct Option: (C)**

• meters
• Solution:
• Let the breadth of the rectangular garden = x meters.
• Length is 40% more than breadth.
• So, length = 140% of x.
• Length = 140x/100 = 7x/5.
• Area of rectangle = length × breadth.
• Area = 3500 sq. meters.
• So,
• 7x/5 × x = 3500
• 7x²/5 = 3500
• x² = 3500 × 5/7
• x² = 2500
• x = 50.
• Breadth = 50 meters.
• Length = 7x/5 = 7 × 50/5 = 70 meters.
• Perimeter of rectangle = 2 × (length + breadth).
• Perimeter = 2 × (70 + 50).
• Perimeter = 2 × 120 = 240 meters.

Solution Q.495 **Correct Option: (B)**

• cm²
• Solution:
• Total surface area of cube = 6 × side × side
• 384 = 6 × side × side
• side × side = 384/6 = 64
• Side of cube = 8 cm
• Length of rectangle = 8 + 10 = 18 cm
• Perimeter of rectangle = 60 cm
• 2 × (Length + Breadth) = 60
• Length + Breadth = 60/2 = 30
• Breadth = 30 - 18 = 12 cm
• Area of rectangle = 18 × 12
• = 216 cm²

Solution Q.496 **Correct Option: (E)**

• cm²
• Solution:
• Let the breadth of the rectangle = b cm
• Length of the rectangle = 28 cm
• Sum of length and breadth = 28 + b
• Side of the square = (28 + b) - 30
• Side of the square = b - 2
• Perimeter of rectangle = 2(28 + b)
• Perimeter of square = 4(b - 2)
• According to the question,
• Perimeter of rectangle = Perimeter of square + 20
• 2(28 + b) = 4(b - 2) + 20
• 56 + 2b = 4b - 8 + 20
• 56 + 2b = 4b + 12
• 56 - 12 = 4b - 2b
• 44 = 2b
• b = 22
• Side of the square = b - 2
• Side of the square = 22 - 2
• Side of the square = 20 cm
• Area of the square = 20 × 20
• Area of the square = 400 cm²

CHAPTER 09: PERMUTATIONS, COMBINATIONS & PROBABILITY

4 Questions

ESSENTIAL FORMULAS & CONCEPT SHORTCUTS

- Combination (Selection): nCr = n! / [r! × (n - r)!]
- Probability P(E) = Number of Favourable Outcomes / Total Number of Outcomes
- Coins, Dice & Pack of 52 Cards Standard Outcomes.

Q.497 Permutations

Box A contains nine cards, one numbered with each integer from 1 to 9. Box B contains seven cards, one numbered with each integer from 1 to 7. One card is selected independently and at random from each box. Find the denominator of the probability that the product of the selected numbers is odd, when expressed in its lowest form.

(A) 56 (B) 49
(C) 63 (D) 72
(E) 81

Q.498 Permutations

A cupboard contains 8 blue scarves and 7 white scarves. If two scarves are selected randomly without replacement, then the probability of selecting two scarves of the same colour is p/q, where p and q are co-prime. Find p + q.

(A) 22 (B) 18
(C) 19 (D) 20
(E) 24

Q.499 Permutations

There are 6 red, 5 green, and 4 yellow balls in a bag. If 3 balls are chosen randomly without replacement, find the numerator of the probability of getting at least one yellow ball when the probability is written in its lowest form.

(A) 44 (B) 58
(C) 66 (D) 72
(E) 86

Q.500 Permutations

A bag contains 6 red balls, 5 green balls and 4 yellow balls. Four balls are picked at random without replacement. If the probability of getting exactly 2 red balls, 1 green ball and 1 yellow ball is written in lowest terms as p/q, find p + q.

(A) 99 (B) 101
(C) 111 (D) 121
(E) 131

Chapter 09 Step-by-Step Smart Solutions

Ikkish Ratio & Shortcut Approaches

Solution Q.497 Correct Option: (C)

• Product of two numbers will be odd only when both selected numbers are odd.
• In Box A, numbers are from 1 to 9.
• Odd numbers in Box A = 1, 3, 5, 7, 9.
• Number of odd numbers in Box A = 5.
• In Box B, numbers are from 1 to 7.
• Odd numbers in Box B = 1, 3, 5, 7.
• Number of odd numbers in Box B = 4.
• Total outcomes = 9 × 7 = 63.
• Favourable outcomes = 5 × 4 = 20.
• Required probability = 20/63.
• 20/63 is already in lowest form.
• So, denominator = 63.

Solution Q.498 Correct Option: (A)

• Total scarves = 8 + 7 = 15
• Total ways of selecting 2 scarves = 15 × 14/2 = 105
• Ways of selecting 2 blue scarves = 8 × 7/2 = 28
• Ways of selecting 2 white scarves = 7 × 6/2 = 21
• Favourable ways = 28 + 21 = 49
• Probability = 49/105 = 7/15
• So, p = 7 and q = 15
• p + q = 7 + 15 = 22

Solution Q.499 Correct Option: (B)

• Total balls = 6 + 5 + 4 = 15.
• Total ways to choose 3 balls = 15 × 14 × 13 / (3 × 2 × 1) = 455.
• To get at least one yellow ball, use the shortcut:
• Favourable ways = Total ways - Ways of choosing no yellow ball.
• Non-yellow balls = 6 red + 5 green = 11.
• Ways of choosing 3 non-yellow balls = 11 × 10 × 9 / (3 × 2 × 1) = 165.
• Favourable ways = 455 - 165 = 290.
• Probability = 290/455.
• Divide numerator and denominator by 5.
• 290/455 = 58/91.
• Numerator in the lowest form = 58.

Solution Q.500 Correct Option: (C)

• Total balls = 6 + 5 + 4 = 15.
• We need exactly 2 red balls, 1 green ball and 1 yellow ball.
• Ways to select 2 red balls = 6 × 5/2 = 15.
• Ways to select 1 green ball = 5.
• Ways to select 1 yellow ball = 4.
• Favourable selections = 15 × 5 × 4 = 300.
• Total selections of 4 balls = (15 × 14 × 13 × 12)/(4 × 3 × 2 × 1) = 1365.
• Required probability = 300/1365 = 20/91.
• Thus, p = 20 and q = 91.
• p + q = 20 + 91 = 111.

🎉 Congratulations on Completing All 500 Questions!

You have mastered the complete 500 Arithmetic Questions curriculum across all 15 core chapters for SBI Clerk, IBPS Clerk & RRB exams!

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